

ASTE-331: Spacecraft Systems Engineering

Attitude Control System (ACS)

Instructors: Jim Chase, Danielle Marsh

TAs: Daejah Dandrich, Luis Diaz, Liam Kerrigan, Emerson Lo

Sept 26th, 2025

- **Overview**
- **Design**
 - Disturbances (aerodynamic, gravity gradient, magnetic, solar radiation, mass expulsion, internal)
 - Sensors (sun, earth, magnetometer, star tracker, gyroscope/IMU, GPS)
 - Actuators & Effectors (reaction wheels, momentum wheels, CMGs, torque rods, RCS thrusters)
 - Block Diagrams (physical & logical architectures)
 - Architectures (eg, 3-axis, spinning)
 - Mission Geometry & Pointing Concerns
 - Sizing
- **Example: ISS**
- **Control & Logic**
 - Rotational Kinematics
 - Moments of Inertial & Angular Acceleration
 - Angular Momentum & Torque
 - Euler's Equations
 - Modeling, Slew Rates, & Spacecraft Motion
- **Example: Osiris Rex Sample Collection**
- **Case Study: Phoenix EDL**
- **Case Study: Deep Impact Star Tracker Anomaly**

Notes

- I'm likely to move the order around a bit..
- And, I haven't had a chance to update the overall format (eg, film-strip, wide-screen, etc.)

- **Attitude Determination and Control**
 - Spacecraft subsystem that provides attitude control, knowledge, and stability, including the actuators, sensors, and onboard logic/control
- **Also Known As...**
 - AACCS: Attitude and Articulation Control System
 - ACS: Attitude Control System
 - ADCS: Attitude Determination Control System
 - ADACS: Attitude Determination and Control System
 - AOCS: Attitude and Orbit Control System
 - G&C: Guidance and Control System
 - GN&C: Guidance, Navigation, and Control
 - PCS: Pointing Control System
 - ...

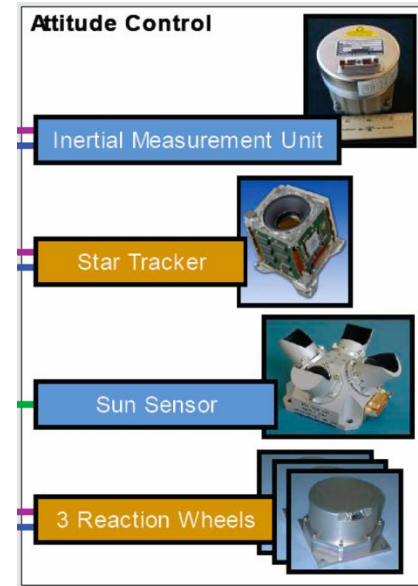
Attitude Control Subsystem

• Function

- Provide attitude control, knowledge, and stability to support spacecraft pointing
- Primary drivers for pointing are:
 - Science observations (instrument FOVs)
 - Trajectory course maneuvers (thrust)
 - Telecom (uplink/downlink to Earth)
 - Thermal (wrt sun, Earth, etc.)
- Also called ACS, ADCS, AOCS, etc.

• GRAIL Example

- 1 star tracker
- 3 reaction wheels
- 1 inertial measurement unit (IMU)
- 1 sun sensor (x4)
- Total = ~5 kg



• Common Components

- Sensors
 - Sun sensors, magnetometers
 - Gyros, GPS receivers
 - Star trackers
- Control Effectors
 - Reaction or momentum wheels
 - Control moment gyros
 - Magnetic torquers
 - Reaction control thrusters (see propulsion)

• Key Trades & Analyses

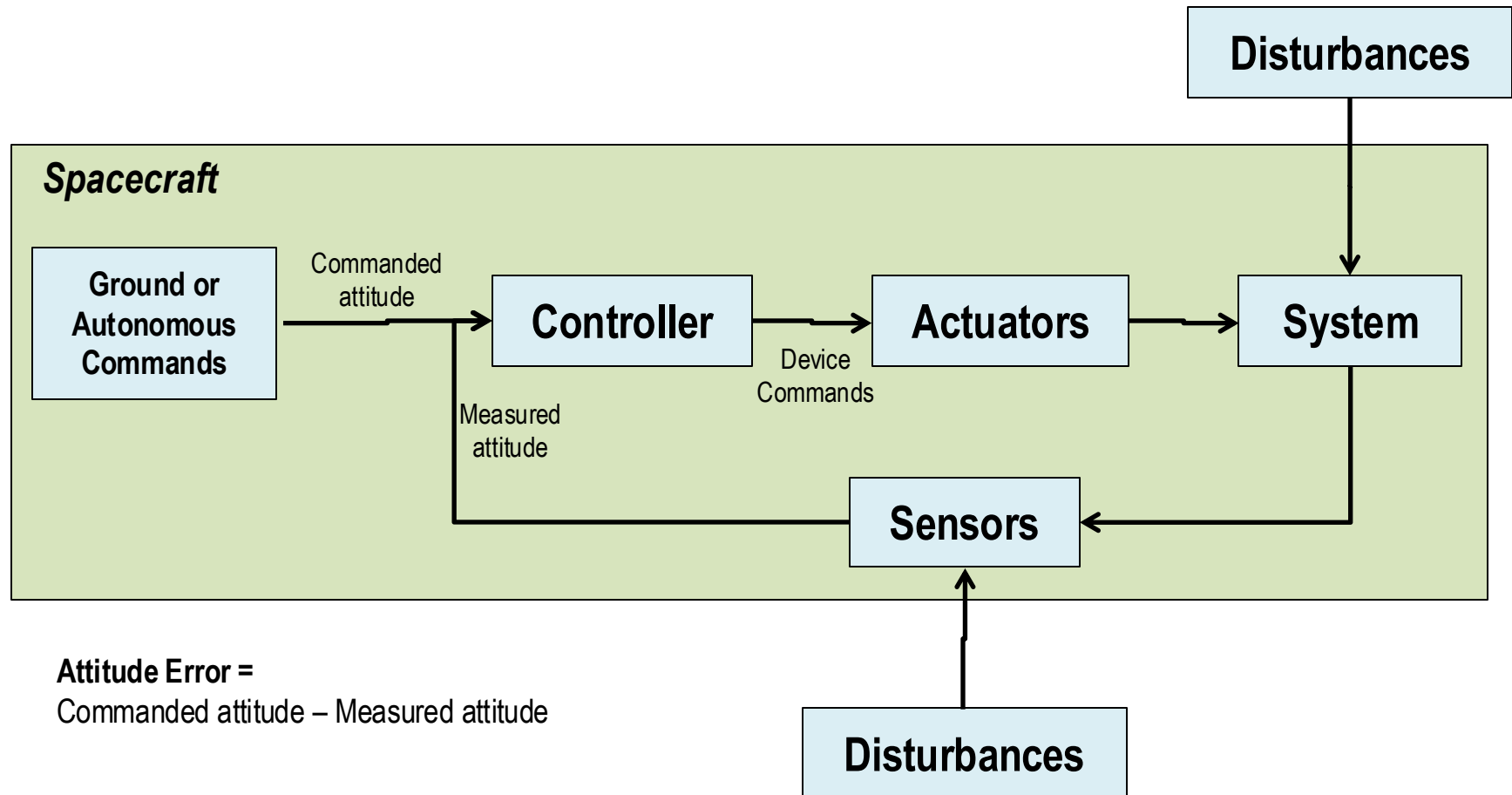
- 3-axis control, spin, gravity gradient...
 - Reaction wheels vs. thrusters, etc.
- Control/knowledge/stability analyses and error budgets
- Heritage from prior systems

• Key Parameters

- Mass, power, and cost
- Pointing control, knowledge, stability/jitter

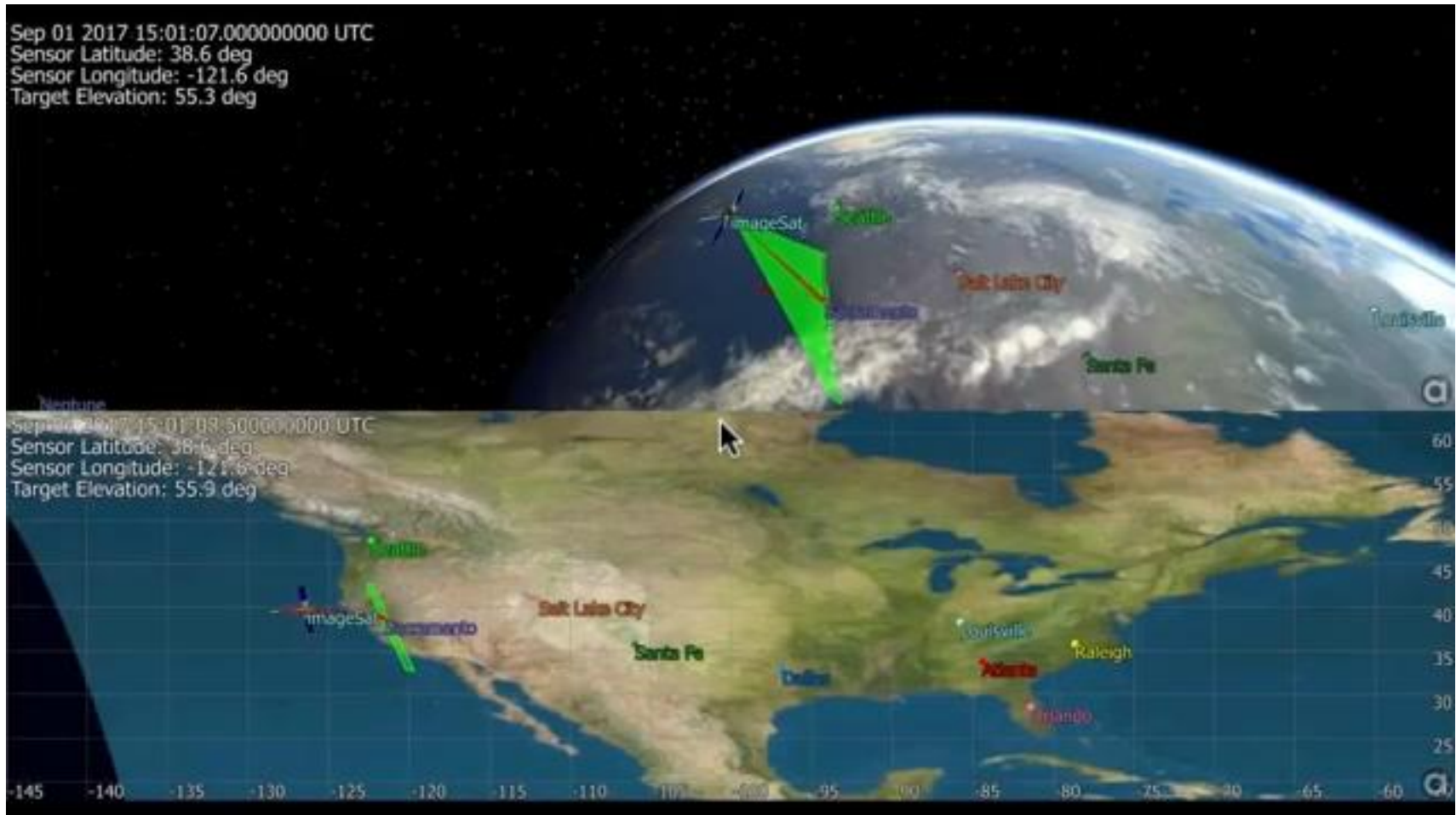
- **Attitude Determination and Control**

- Spacecraft subsystem that provides attitude control, knowledge, and stability, including the actuators, sensors, and onboard logic/control

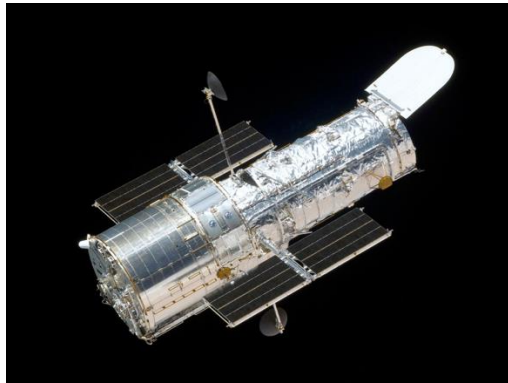


Slew Rates & Spacecraft Motion

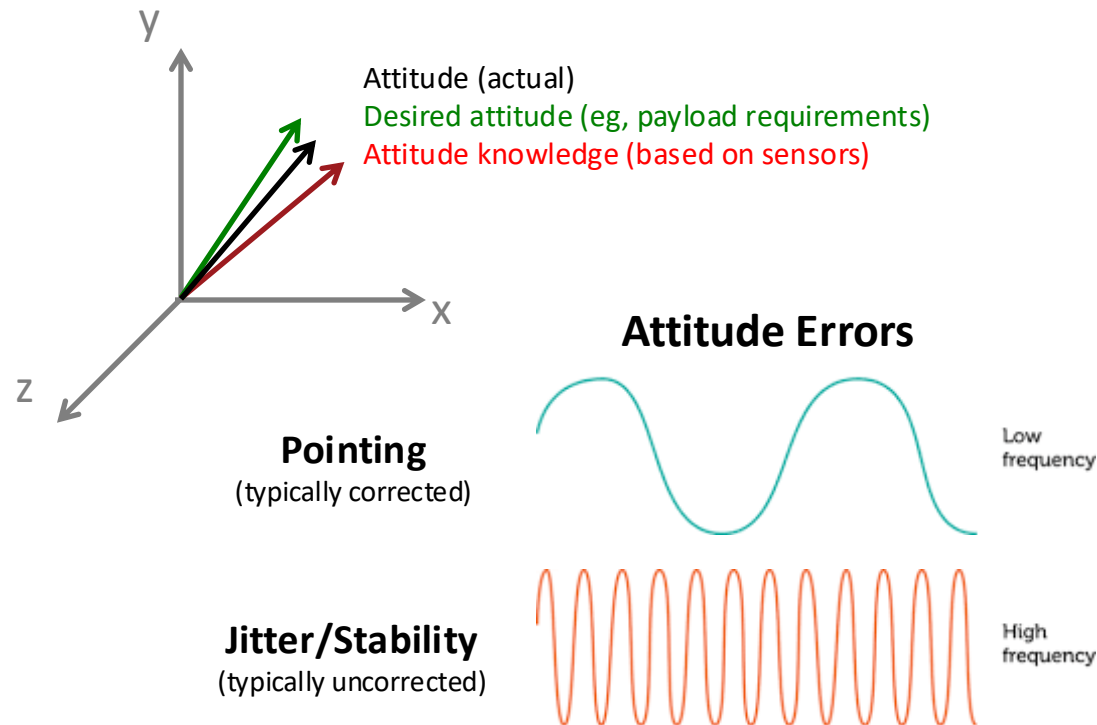
- What do typical spacecraft motions look like?



Some Terminology



For example, imagine the boresight of a telescope is represented by the vectors shown below:

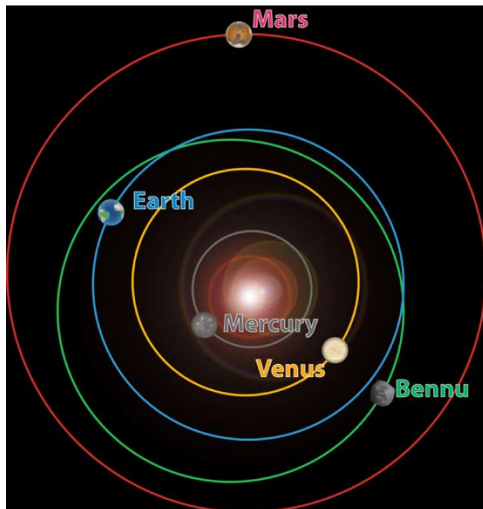
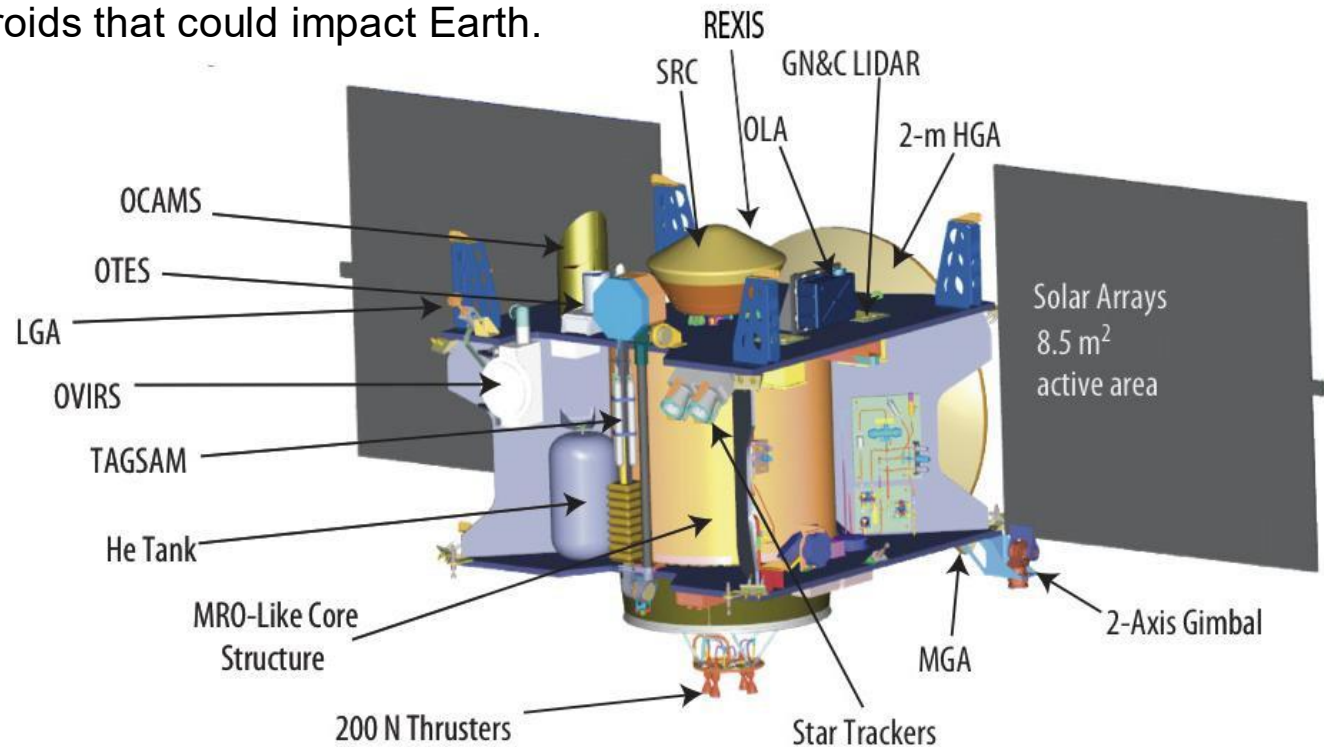


- **Attitude:** Orientation of a defined reference system attached to the spacecraft body
- **Determination:** Knowledge within a specified tolerance (realtime or post-facto)
- **Control:** Maintenance of specified attitude within a given tolerance
- **Pointing Error:** Low frequency component of attitude misalignment
- **Jitter:** Spacecraft stability, consisting of the high frequency component of attitude error (minimized by design)

OSIRIS-REX Example

Mission Objectives

- Launched on Sept. 8, 2016, the spacecraft traveled to a near-Earth asteroid called Bennu
- It has collected a sample of rocks and material from the surface that it returned to Earth **on Sept. 24, 2023**
- The mission will help scientists investigate how planets formed and how life began, as well as improve our understanding of asteroids that could impact Earth.





Instrument Deck

Instruments

OCAMS: Camera Suite (UofA)

- PolyCam: Narrow angle, high resolution (8-inch telescope)
- MapCam: Wide angle, high resolution (4-color, asteroid mapping)
- SamCam: Monitors TAG and sample acquisition

OLA: LIDAR for high-resolution topography (CSA contribution)

OTES: Thermal Spectrometer, 5.7-100 microns (ASU)

OVIRS: Visible & IR Spectrometer, 0.4-4.3 microns (GSFC)

REXIS: X-ray Spectrometer (student experiment)

GN&C Components

4 Sun Sensor Assemblies (SSA)

- 4 heads & 2 heads (x2 = 12)

4 Reaction Wheels

2 Star Trackers

2 MIMUs (Miniature IMUs)

2 LIDARs (40 cm @ 3 km)

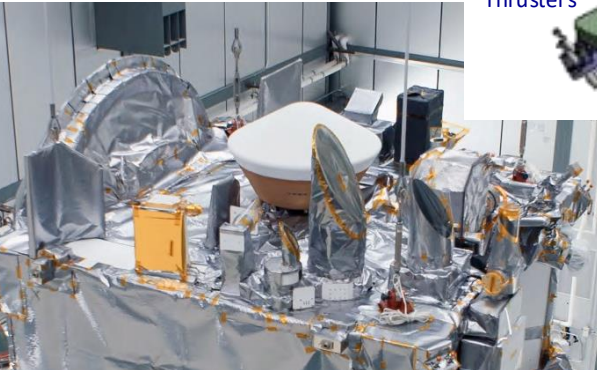
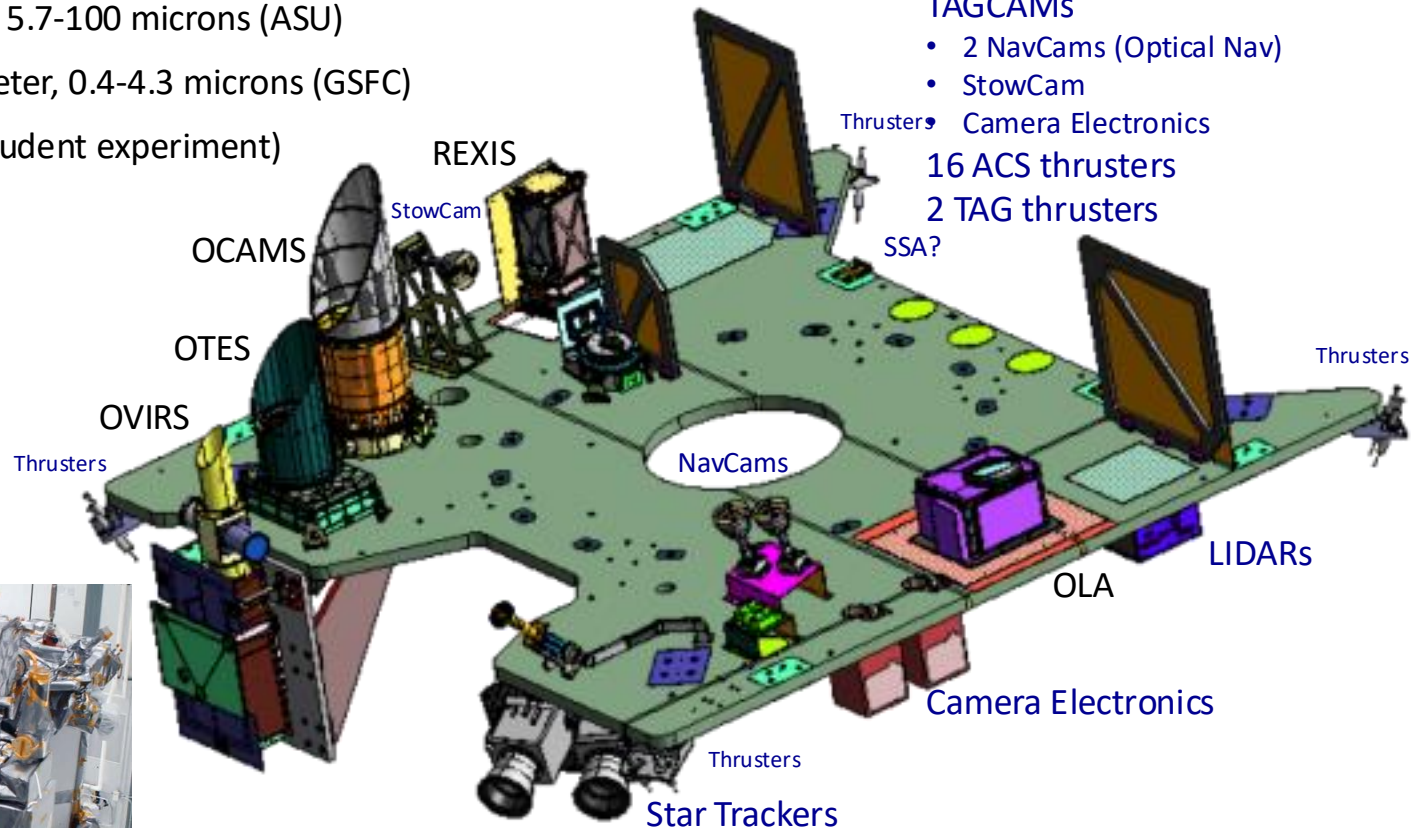
TAGCAMS

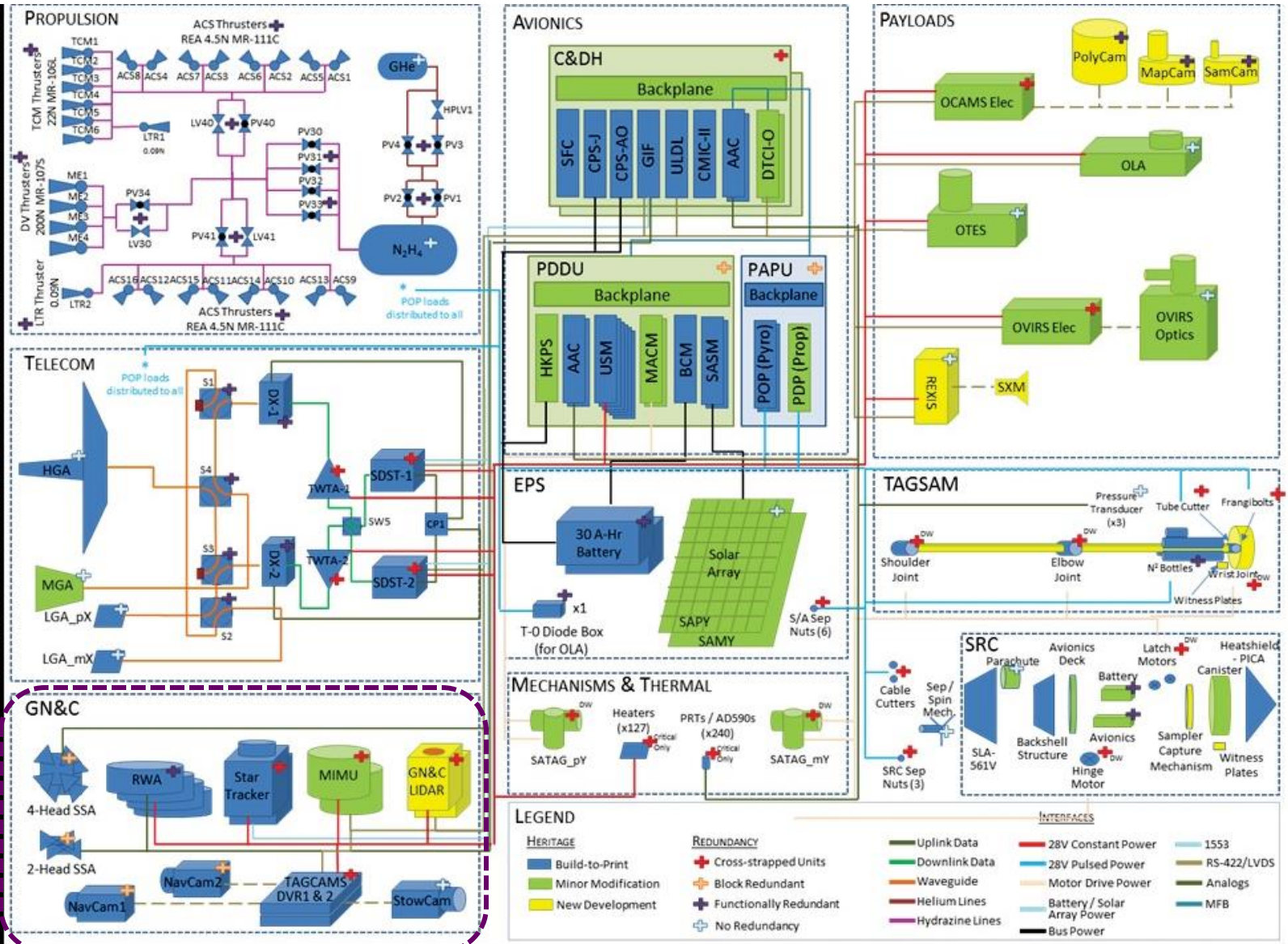
- 2 NavCams (Optical Nav)
- StowCam

• Camera Electronics

16 ACS thrusters

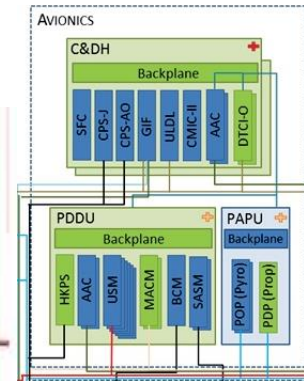
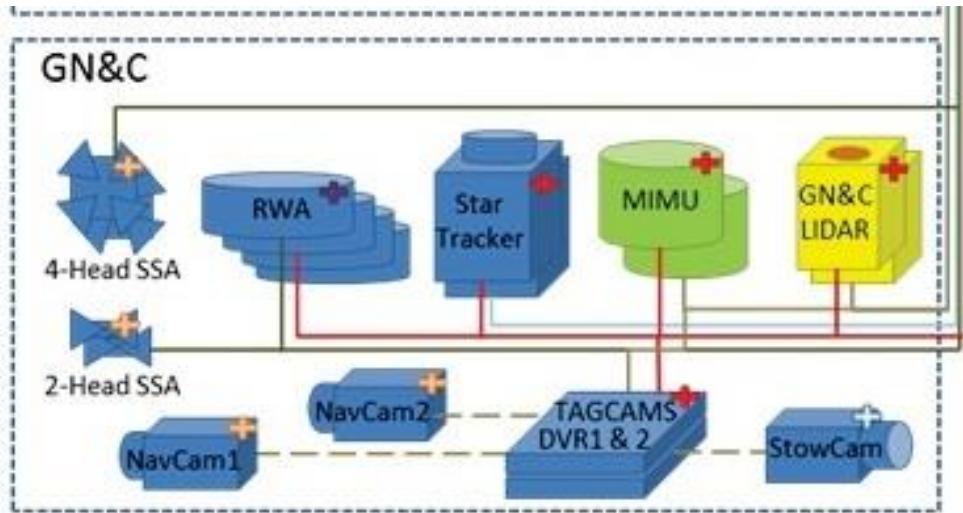
2 TAG thrusters





Discussion Topics

- Sensors, Actuators, Electronics, and software?
- Redundancy?
- Heritage?



C&DH

Power Electronics

Mil-Spec 1553 Interfaces



Analog/Etc. Connectors



LVDS Connector



LEGEND

HERITAGE

- Build-to-Print
- Minor Modification
- New Development

REDUNDANCY

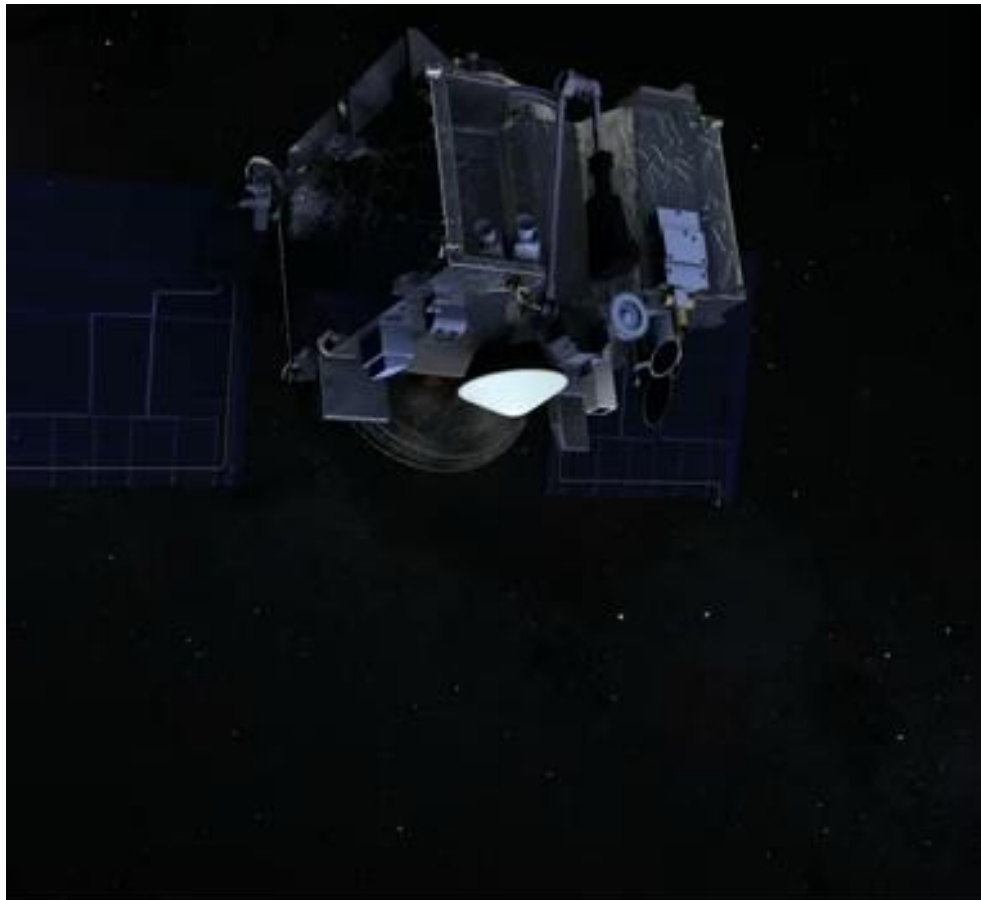
- Cross-strapped Units
- Block Redundant
- Functionally Redundant
- No Redundancy

INTERFACES

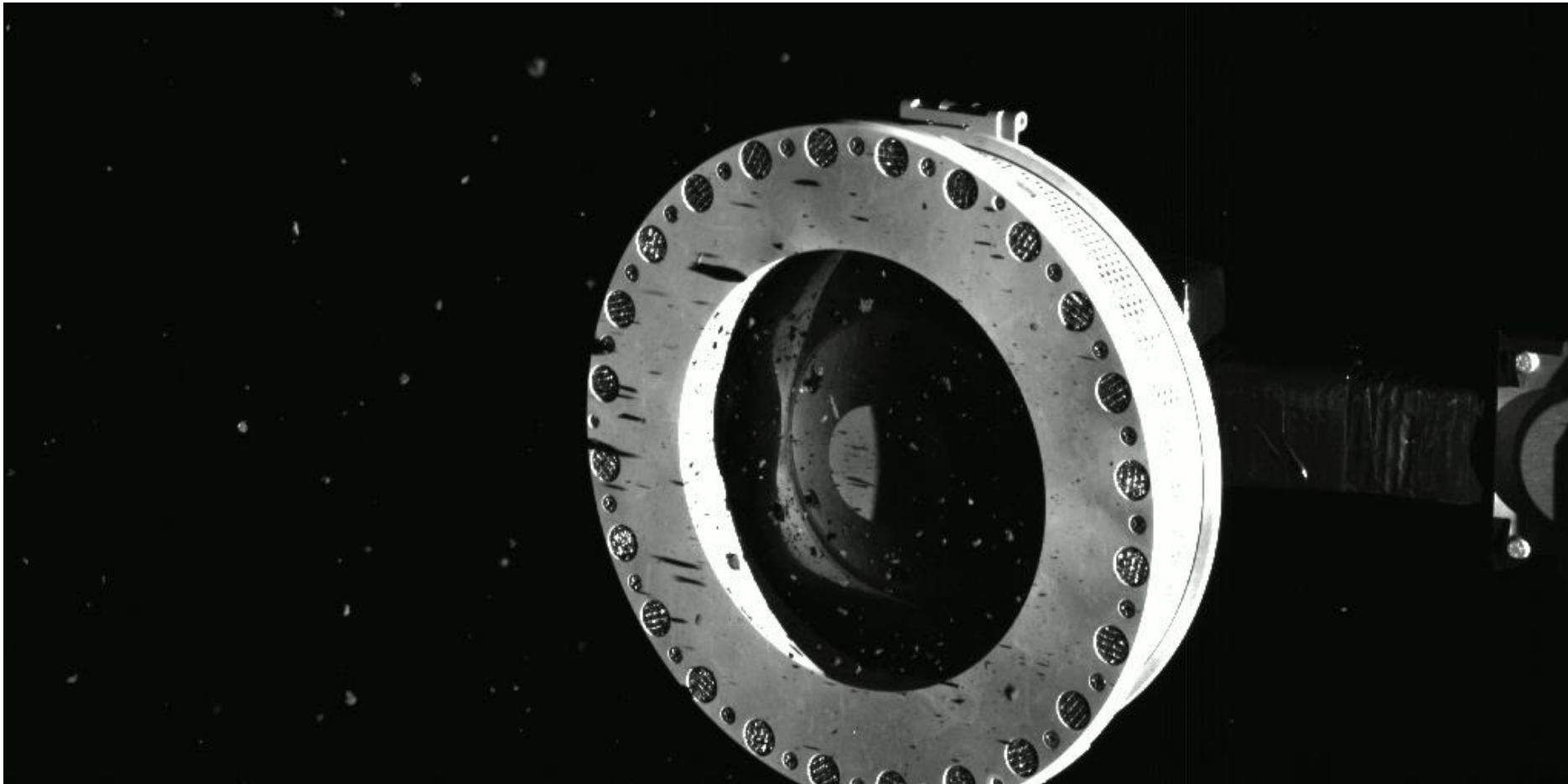
- Uplink Data
- Downlink Data
- Waveguide
- Helium Lines
- Hydrazine Lines
- 28V Constant Power
- 28V Pulsed Power
- Motor Drive Power
- Battery / Solar Array Power
- Bus Power
- 1553
- RS-422/LVDS
- Analogs
- MFB

- **Questions from Last Year**

- Why was there no realtime video? (What was the downlink rate?)
- How does the prediction compare to the sample?

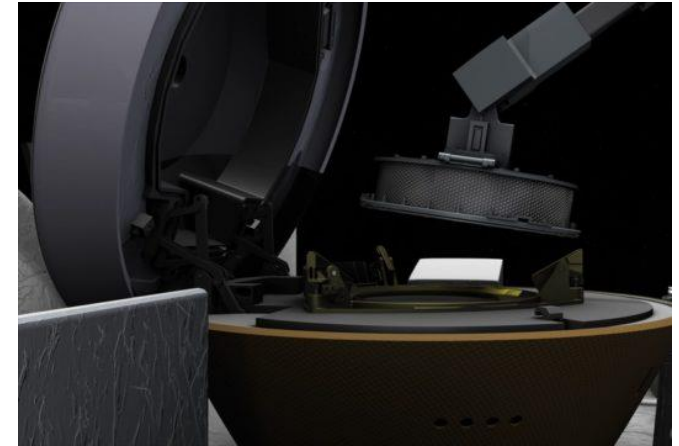


- **1-2 Days Later**
 - Collector lid likely not fully closing due to larger rocks, thus leaking...
 - Forgoing sample measurement in favor of earlier stowing (significant sample observed)



Sample Stowage & Earth Return

- **Stow successful and ready for its return**
- **Delivery achieved on 9/24/2023**
- **Entry, descent, and landing**
 - Heatshield to slow high-speed entry
 - Parachute to reduce speed further
 - Landing area is the Utah test range

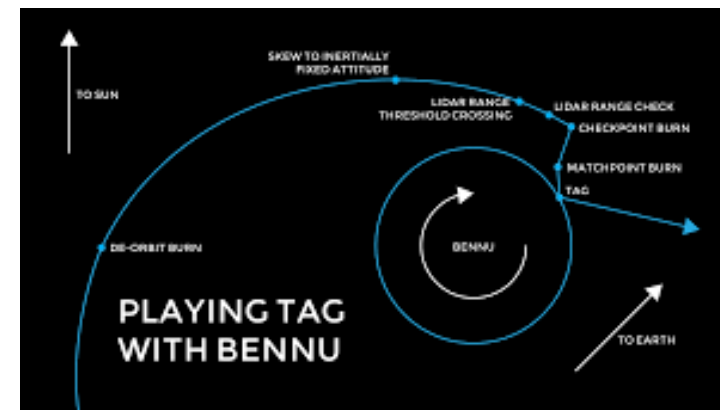


Questions from Last Year (1 of 2)

- **From 1 to 10, how bad was the slight error the OSIRIS-REX Touch and Go? Are these kinds of things usually expected or was it a complete surprise?**
 - Likely well within the modeled capability (~3)
 - Typically, videos are based on the 'nominal' scenario, but analysis is far more encompassing
 - That said, there are still surprises... (both MER rovers were barely within their performance envelopes)
- **How long is each step in the Osiris Rex mission take? As in, how long until NASA decides to return to the asteroid for a second sample if needed?**
 - Scientific consensus & technology development: Varies greatly (many years possible)
 - Proposal Process: 1-3 years
 - Development: 4-5 years
 - Operations and Sample Return: 5-10 years
 - Total: ≥ 10 -20 years
- **Is there a way to change the downlink or uplink in the middle of a mission?**
 - Options for increasing the uplink/downlink rates:
 - Greater use of data compression
 - Increase downlink at the expense of the bit error rate (BER)
 - Adjust trajectory to reduce distance (mission dependent)
 - Going to look at this in detail when we talk about Telecom

Questions from Last Year (2 of 2)

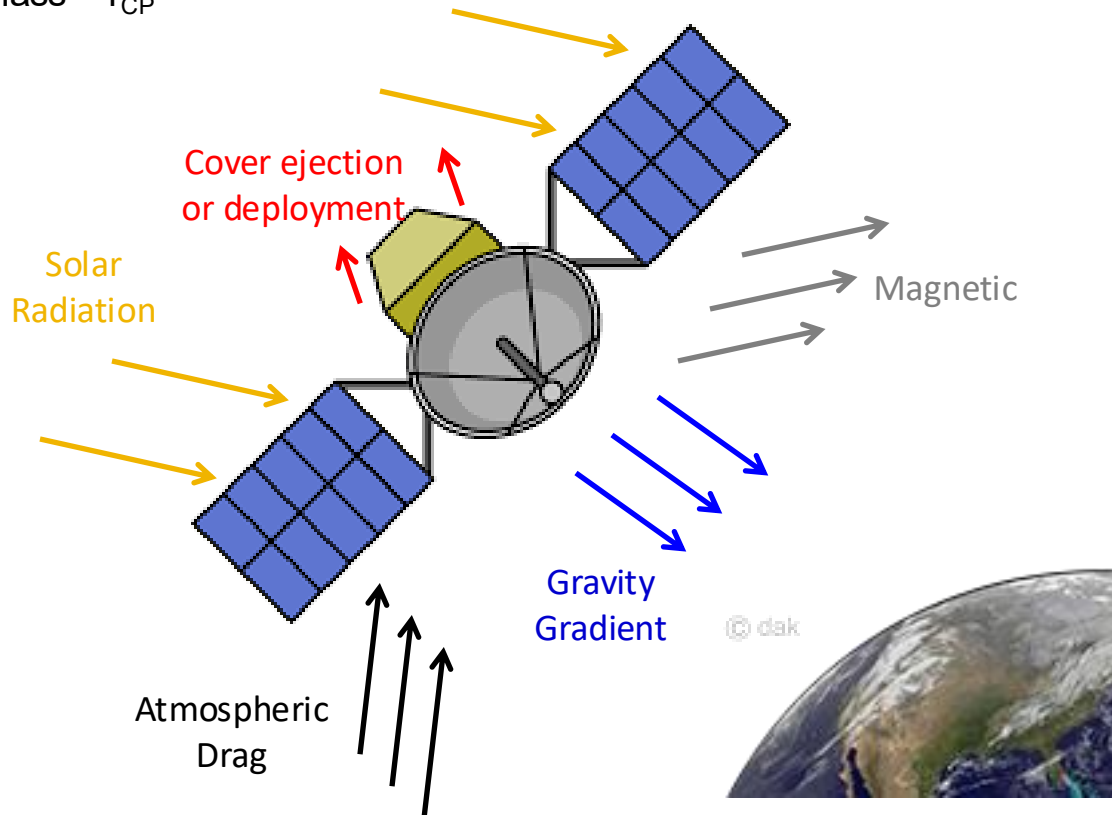
- **In situations like the Osiris Rex touch and go, ...what is a reasonable approach velocity to expect?**
 - At touch-down, expected 0.4 m/s
 - Comparison: Phoenix lander touched down around 1.5 m/s
 - MER Rovers (w/airbags): < 26 m/s
- **How is propellant estimated versus the uncertainty of the asteroid size/shape?**
 - Bounding estimates are developed for the asteroid size, shape, gravity map
 - Sufficient propellant margin is carried to encompass these estimates
 - Prior to approach, significant observations (at 2-3 km) are undertaken (along with fly by's) to improve fidelity of the gravity field.
- **Is the asteroid massive enough that the SC is actually orbiting the asteroid?**
 - Yes, 62 hrs/orbit and more or less circular
 - Using features on Bennu for attitude knowledge
 - Simple fault protection:
 - If something goes wrong, burn towards the sun



- **Disturbances are torques that are typically induced on the spacecraft by external forces.**
 - Typically modeled as a force (representing a center of pressure) acting at on the spacecraft body at some radius from the center of mass.
 - Torque (τ) = $r_{CP} \times F$
 - Radius from center of mass = r_{CP}
 - Force = F

- **Examples**

- Launch vehicle tip-off
- Aerodynamic
- Gravity Gradient
- Magnetic
- Solar Radiation
- Mass Expulsion
- Internal



Aerodynamic Torque

“Weathervane” effect of Center of Mass/Center of Pressure (CM/CP) offset.

Aerodynamic torque: $\tau_a = r_{CP} \times F_a$

Where: $F_a = \frac{1}{2} \rho S C_D v v$

- r_{CP} = Center of pressure position vector in body fixed frame
- F_a = Aerodynamic force
- ρ = Atmospheric density
- v = Spacecraft velocity
- S = Projected area (perpendicular to v)
- C_D = Drag coefficient (use 2 as for a worst case analysis)

From “Attitude Determination and Control”, Joel Sercel, 2003

For example, at 400 km altitude, there might be a 3.11×10^{-5} N torque, Which would result in an attitude error of 82-deg after one orbit

Gravity gradient creates a torque tending to rotate the spacecraft to align its minimum inertia axis with the local vertical.

The torque is given by: $\vec{\tau} = 3n^2 \hat{r} \times I \cdot \hat{r}$ $n^2 = \frac{\mu}{a^3} = \frac{\mu}{R^3}$ = Orbital Rate

$\hat{r}$ = Earth - to - vehicle unit vector

θ = Pitch angle

In body frame: $\hat{r} = (-\sin \theta, \sin \phi, 1 - \sin^2 \theta - \sin^2 \phi)$
 $\approx (-\theta, \phi, 1)$

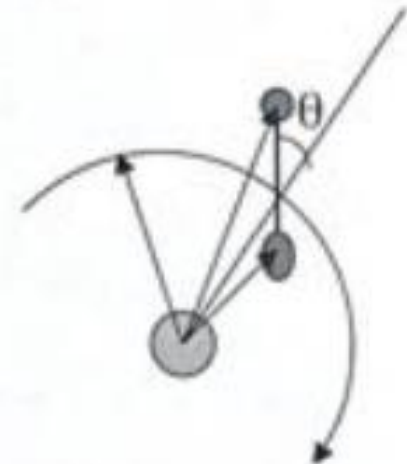
$$\therefore \vec{\tau} \approx 3n^2 \begin{bmatrix} [(I_3 - I_2)^2] \\ [(I_3 - I_1)^2] \\ 0 \end{bmatrix}$$

Typical values:

$$\Delta I = 1000 \text{ kg m}^2$$

$$n = 0.001 \text{ s}^{-1}$$

$$\tau = 6.7 \times 10^{-5} \text{ N m/deg}$$



From "Attitude Determination and Control", Joel Sercel, 2003

Magnetic Torque

Present when in orbit around a planet with a substantial magnetic field, like Earth and Jupiter. The torque is given by:

$$\boldsymbol{\tau}_m = \mathbf{M} \times \mathbf{B}$$

where:

- $\mathbf{M}$ is the spacecraft magnetic dipole moment due to current loops and residual magnetization in the spacecraft. It is expressed in Am^2 : *Ampere-m²* (SI) or *pole-cm* (CGS). $1 \text{ Am}^2 = 1000 \text{ pole cm}$.
- $\mathbf{B}$ is the planet magnetic field vector expressed in spacecraft coordinates; it is proportional to $1/r^3$, where r is the radius vector to the spacecraft. It is measured in *Tesla* (SI), or *Gauss* (CGS). $1 \text{ Tesla} = 10^4 \text{ Gauss}$.

Typical values for a small spacecraft in LEO:

$$B = 3 \times 10^{-5} \text{ Tesla}$$

$$M = 0.1 \text{ Am}^2$$

$$\tau_m = 3 \times 10^{-6} \text{ N m}$$

From "Attitude Determination and Control", Joel Sercel, 2003

Solar Radiation

In body coordinates, the solar radiation pressure torque is given by:

$$\tau_s = \mathbf{r} \times \mathbf{F}_s$$

$$F_s = (1 + K)p_s S_{\perp}$$

$$p_s = \frac{I_s}{cR^2}$$

$\mathbf{r}$ = Vector from body CM to optical CP (m)

F_s = Solar radiation pressure force (N)

K = Surface reflectivity ($0 < K < 1$)

$S_{\perp}$ = Projected area, normal to sun vector (m^2)

I_s = Solar intensity at 1 AU (W/m^2)

c = Speed of light (m/s)

R = Distance from Sun in AU

Typical values for a spacecraft in Earth orbit:

$$r = 0.1 \text{ m}$$

$$K = 0.5$$

$$S_{\perp} = 5 \text{ m}^2$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$I_s = 1400 \text{ W/m}^2 \text{ (Solar intensity at 1 AU)}$$

$$\tau_s = 3.5 \times 10^{-6} \text{ N m}$$

Dominant effect in:

- GEO Orbits
- Deep space cruise

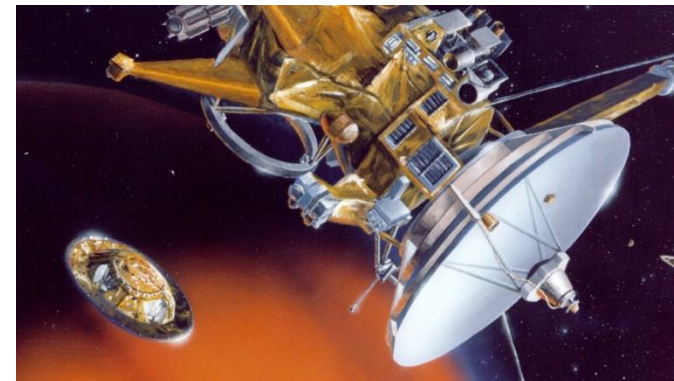
Notes: 1. The torque vector is always perpendicular to the sun line
2. Independent of altitude and velocity as long as in sunlight

From "Attitude Determination and Control", Joel Sercel, 2003

Mass Expulsion

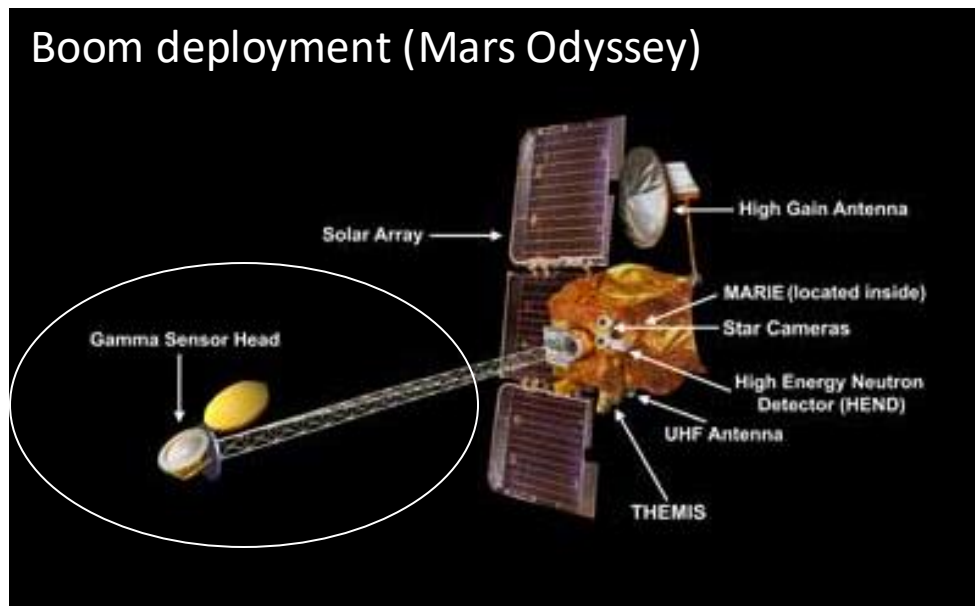
- **Mass expulsion includes a wide range of activities:**
 - Jettison: Probes, overs, expended SRMs, etc.
 - Deliberate: Thrusters, gas venting, etc.
 - Accidental: Leaks, misalignments
- **If significant, it can dominate the overall forces and result in changes to the attitude control system**
- **Mass expulsion torque:**
 - Torque (τ_{ME}) = $r_{CP} \times F_{ME}$
 - Where F_{ME} varies considerably

Cassini's release of the Huygen's probe



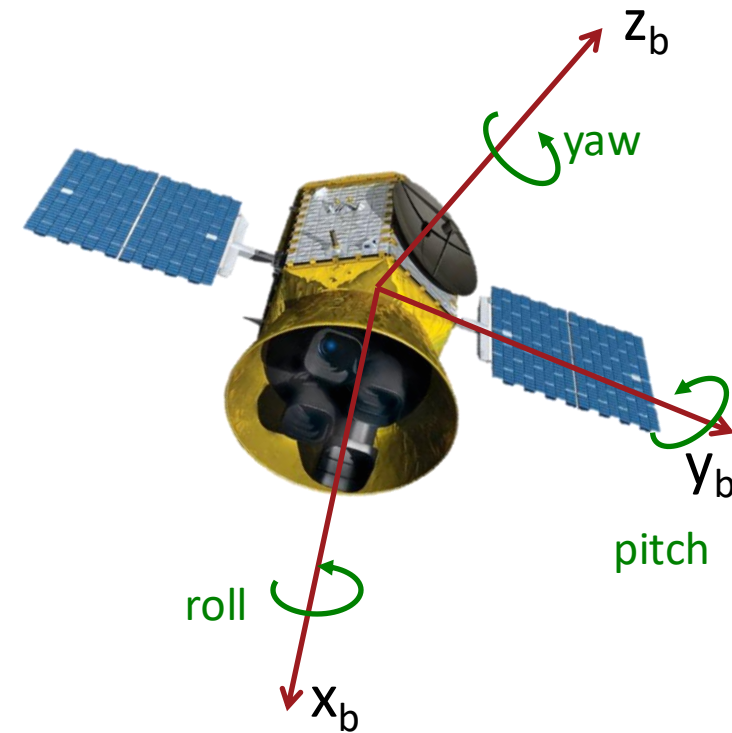
Internal Torque

- **Internal torque is typically caused by**
 - Deployments: Solar arrays, antennas, booms, instruments
 - Recurring Motions: Instrument scanning, sample retrieval, fluid flow, thermal louvers, etc.
- **It has no effect on total system angular momentum, but it still effects attitude**



Typical Pointing Concerns

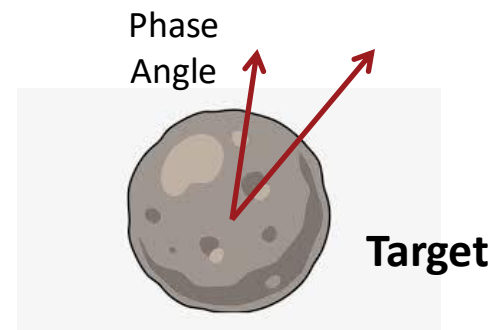
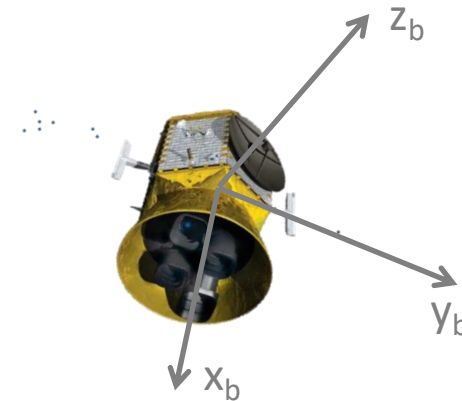
- **Payload**
 - Perform science observations
 - With the sun often within a specified phase angle range
 - Avoid Sun on sensitive instruments (eg, camera boresight)
- **Communications**
 - Avoid solar conjunction
- **Solar Arrays for Power**
 - Minimize solar incidence angle
- **Thermal**
 - Spacecraft design typically favors or excludes a particular geometry
- **Trajectory Maneuvers**
 - Pointing & stability for maneuvers
- **Spacecraft Agility**
 - How quickly can the spacecraft change directions?



**Typical Spacecraft Body-fixed
(b) Coordinate Systems**

Typical Pointing Concerns

- **Payload**
 - Perform science observations
 - With the sun often within a specified phase angle range
 - Avoid Sun on sensitive instruments (eg, camera boresight)
- **Communications**
 - Avoid solar conjunction
- **Solar Arrays for Power**
 - Minimize solar incidence angle
- **Thermal**
 - Spacecraft design typically favors or excludes a particular geometry
- **Trajectory Maneuvers**
 - Pointing & stability for maneuvers
- **Spacecraft Agility**
 - How quickly can the spacecraft change directions?



Often a driver for control, knowledge, & error

Payload Observation Architectures

Onboard ACS is typical, but other approaches can occasionally produce improved performance while maintaining cost.

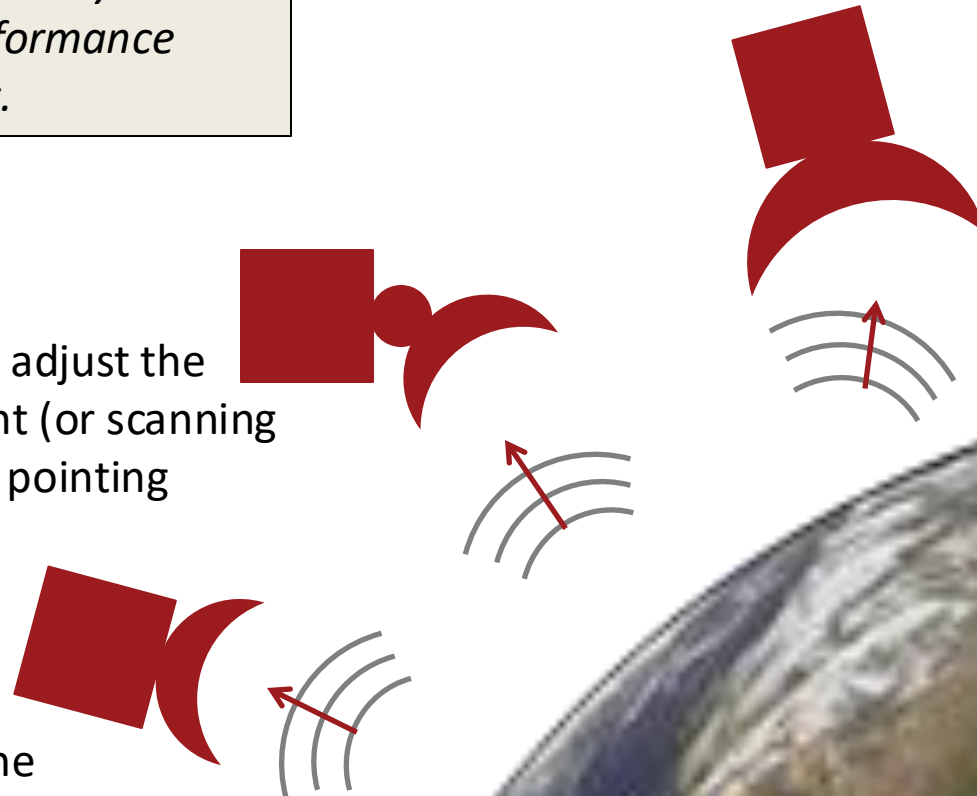
Instrument collector is oversized, such that it can internally point/filter data

Terminology

- **Nadir:** pointing towards (eg, Earth)
- **Zenith:** pointing away (eg, Earth)

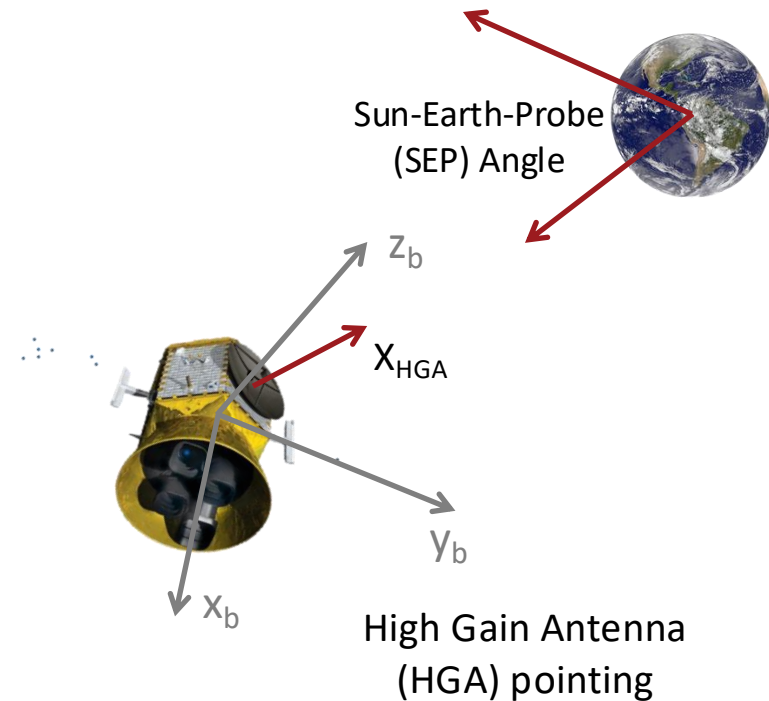
Actuators adjust the instrument (or scanning platform) pointing

Spacecraft points at the target as a rigid body (typically controlled by onboard ACS)



Typical Pointing Concerns

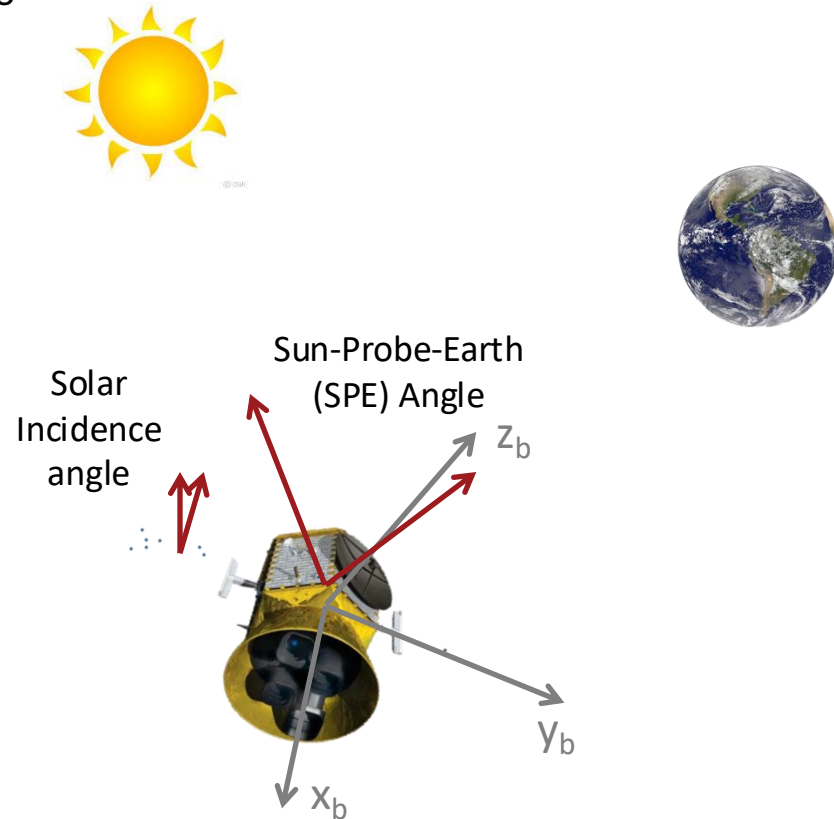
- **Payload**
 - Perform science observations
 - With the sun often within a specified phase angle range
 - Avoid Sun on sensitive instruments (eg, camera boresight)
- **Communications**
 - Avoid solar conjunction (SEP angle near 0-deg)
- **Solar Arrays for Power**
 - Minimize solar incidence angle
- **Thermal**
 - Spacecraft design typically favors or excludes a particular geometry
- **Trajectory Maneuvers**
 - Pointing & stability for maneuvers
- **Spacecraft Agility**
 - How quickly can the spacecraft change directions?



*Low gain antennas typically provide communication at any nominal attitude
High gain antennas often require 3-axis control (combined w/science & communications)*

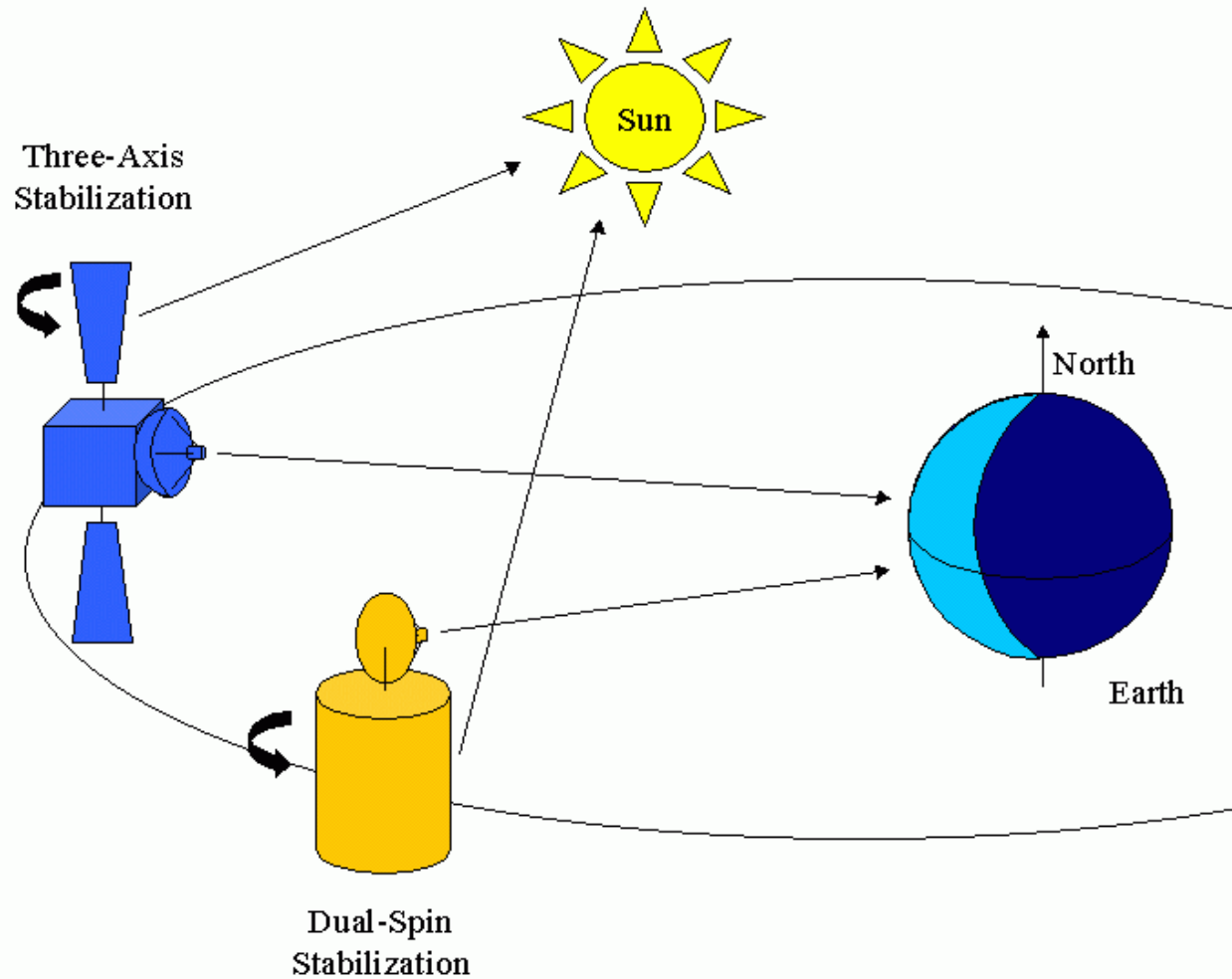
Typical Pointing Concerns

- **Payload**
 - Perform science observations
 - With the sun often within a specified phase angle range
 - Avoid Sun on sensitive instruments (eg, camera boresight)
- **Communications**
 - Avoid solar conjunction (SEP angle near 0-deg)
- **Solar Arrays for Power**
 - Minimize solar incidence angle
 - Maintain power while transmitting
- **Thermal**
 - Spacecraft design typically favors or excludes a particular geometry
- **Trajectory Maneuvers**
 - Pointing & stability for maneuvers
- **Spacecraft Agility**
 - How quickly can the spacecraft change directions?



*Arrays will often be single- or dual-axis articulated to minimize solar incidence angle
Note that arrays will occasionally be oversized to relax pointing requirements*

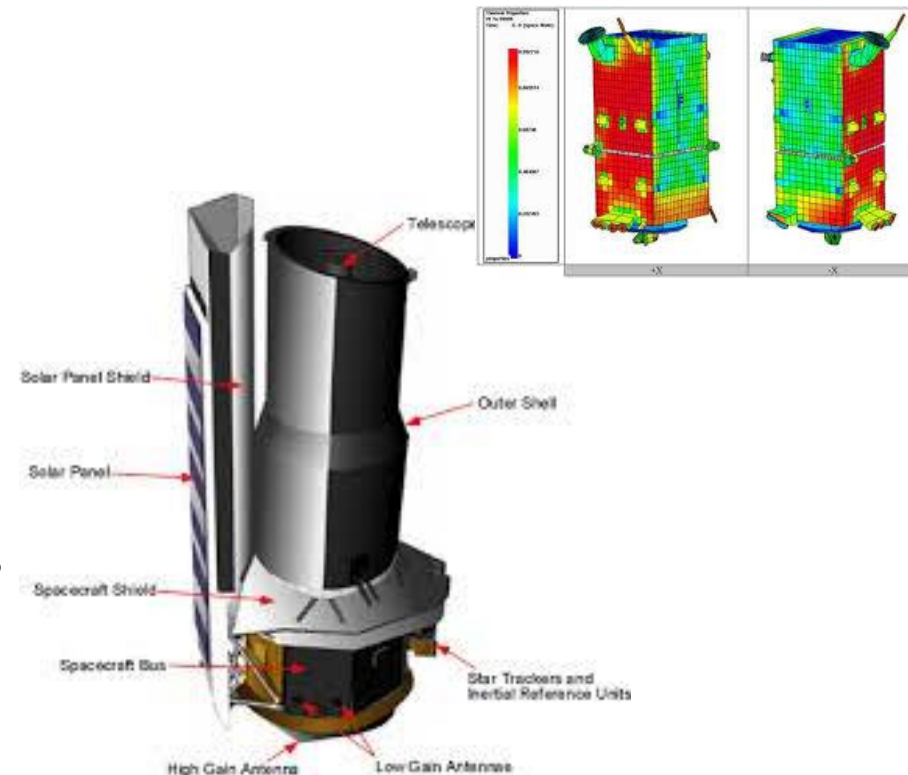
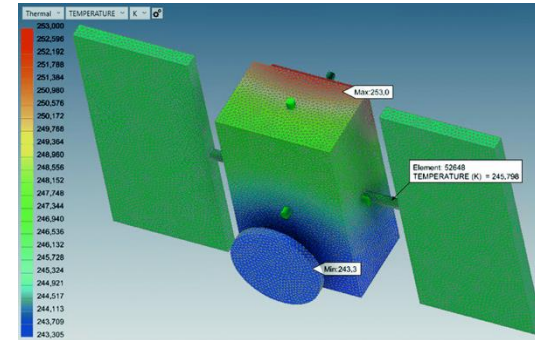
3-axis vs. Dual-Spin Stabilization



In this example, a solar array that wraps-around a spacecraft minimizes the required pointing control

Typical Pointing Concerns

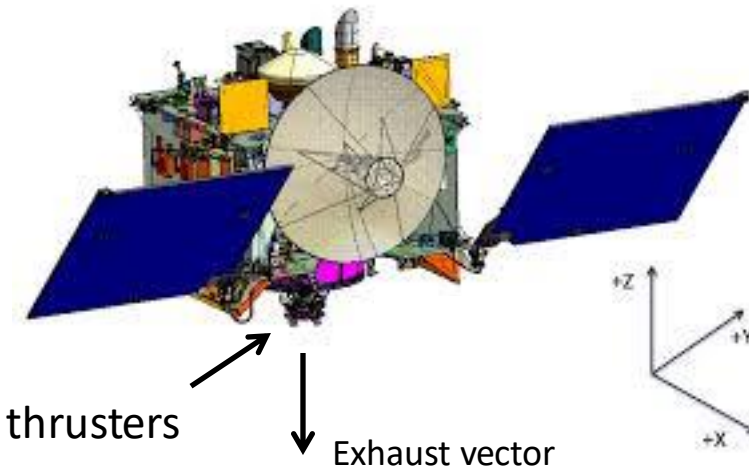
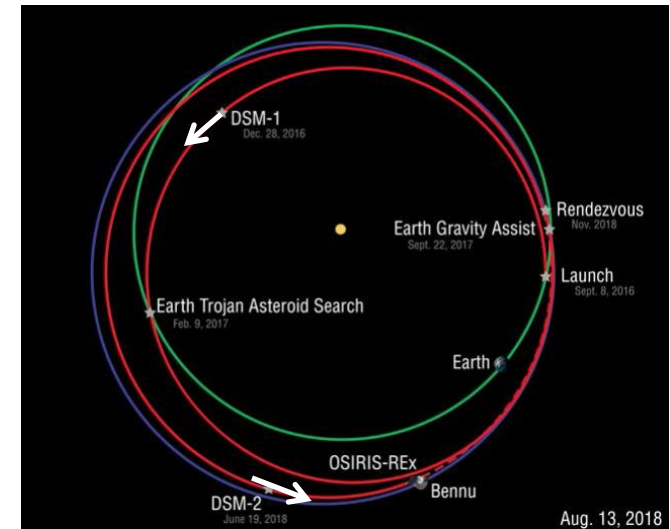
- **Payload**
 - Perform science observations
 - With the sun often within a specified phase angle range
 - Avoid Sun on sensitive instruments (eg, camera boresight)
- **Communications**
 - Avoid solar conjunction
- **Solar Arrays for Power**
 - Minimize solar incidence angle
- **Thermal**
 - Spacecraft design typically favors or excludes a particular geometry
- **Trajectory Maneuvers**
 - Pointing & stability for maneuvers
- **Spacecraft Agility**
 - How quickly can the spacecraft change directions?



Typical Pointing Concerns

- **Payload**
 - Perform science observations
 - With the sun often within a specified phase angle range
 - Avoid Sun on sensitive instruments (eg, camera boresight)
- **Communications**
 - Avoid solar conjunction
- **Solar Arrays for Power**
 - Minimize solar incidence angle
- **Thermal**
 - Spacecraft design typically favors or excludes a particular geometry
- **Trajectory Maneuvers**
 - Pointing & stability for maneuvers
- **Spacecraft Agility**
 - How quickly can the spacecraft change directions?

OSIRIS-REx Example

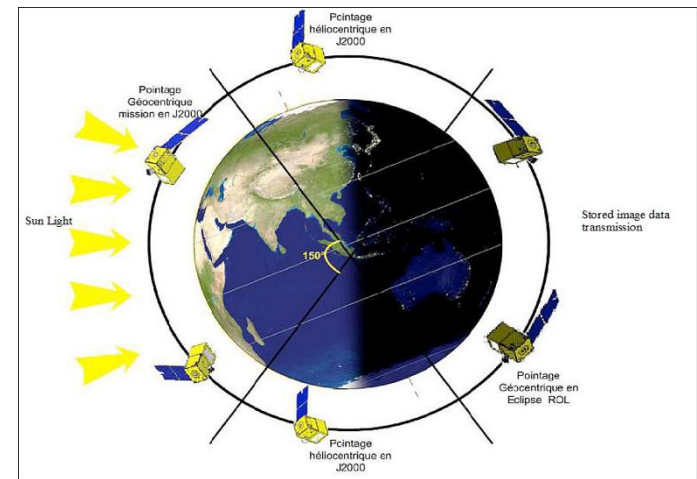


Typical Pointing Concerns

- **Payload**
 - Perform science observations
 - With the sun often within a specified phase angle range
 - Avoid Sun on sensitive instruments (eg, camera boresight)
- **Communications**
 - Avoid solar conjunction
- **Solar Arrays for Power**
 - Minimize solar incidence angle
- **Thermal**
 - Spacecraft design typically favors or excludes a particular geometry
- **Trajectory Maneuvers**
 - Pointing & stability for maneuvers
- **Spacecraft Agility**
 - How quickly can the spacecraft change attitude?

Typically driven by opportunities & threats

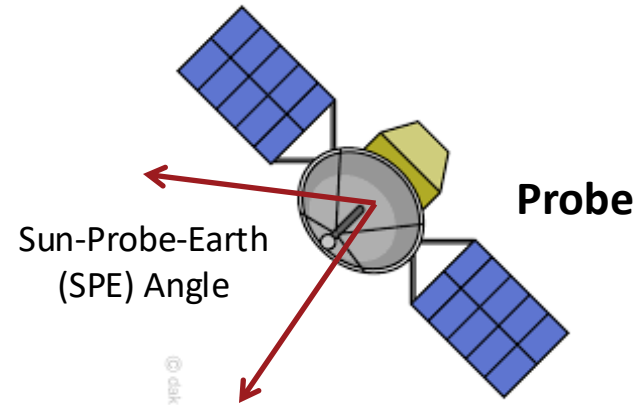
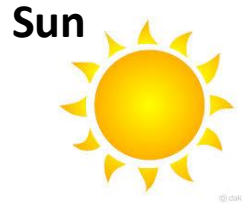
- Additional science observations?
- Science vs. communication latency?
- Attitude changes due to eclipse?



Slew Rates & Spacecraft Motion

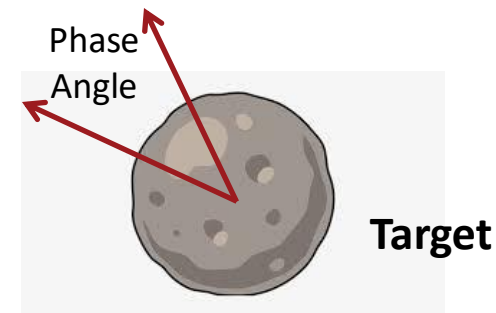
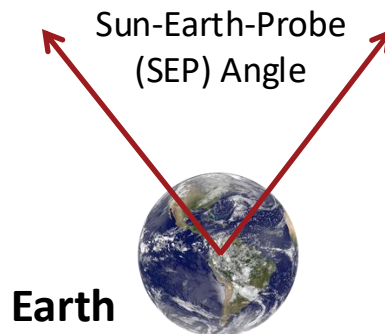
- **How fast (or slow) does a typical spacecraft turn when making attitude adjustments? Is there a typical time frame for certain attitude alterations? (few minutes? hours?)**
 - Typical spacecraft requirement is 1 deg/sec
 - Might vary between 0.25 deg/sec to 3 deg/sec
 - Note that faster slew rates require “settling time” to achieve stability requirements
 - ISS slew rate is < 0.1 deg/sec
- **How much does the typical satellite actually have to maneuver? Are they often just pointing at Earth or do they switch targets?**
 - Majority of spacecraft are for either imagery or communications (across the EM spectrum)
 - Therefore, most have telescopes, antennas, and/or detectors that require some degree of pointing (typically between 0.001-deg and 0.1-deg)
 - ACS activity is a function of the disturbance environment vs. spacecraft activity
 - Whenever spacecraft are “holding” their attitude, they are using their ACS actuators
 - Maneuvers (including slews) tend to vary greatly, for example:
 - Mars Reconnaissance Orbiter (MRO) can take dozens of pictures each day
 - Communications satellite in geo might desaturate its wheels once/week

Mission Geometry (key angles)



In concept formulation, these key angles will often drive the early design

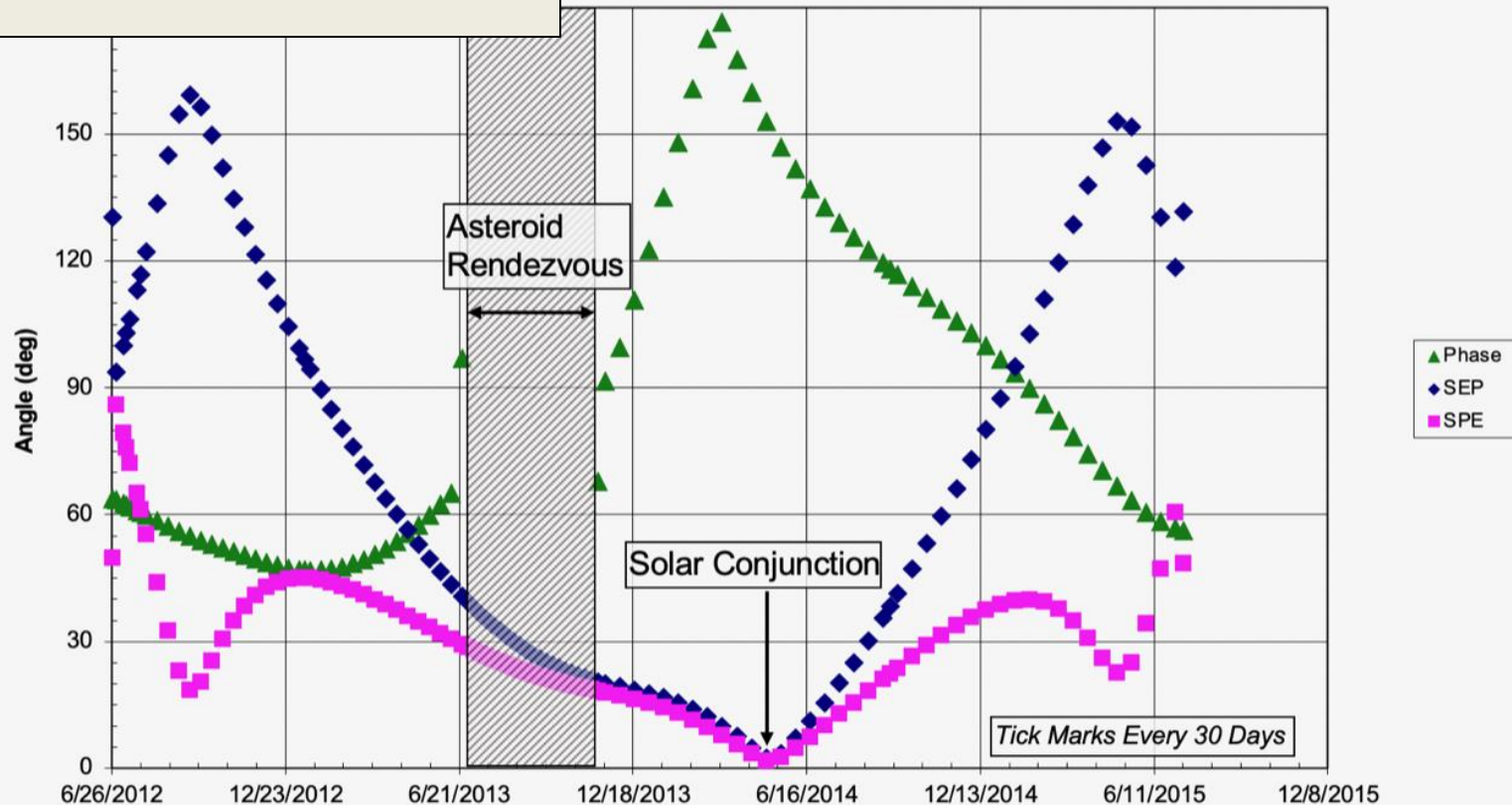
- The Mission Design & Nav. Engineer will typically product plots of these in addition to their trajectory designs and ΔV budgets



Based on the chart below...

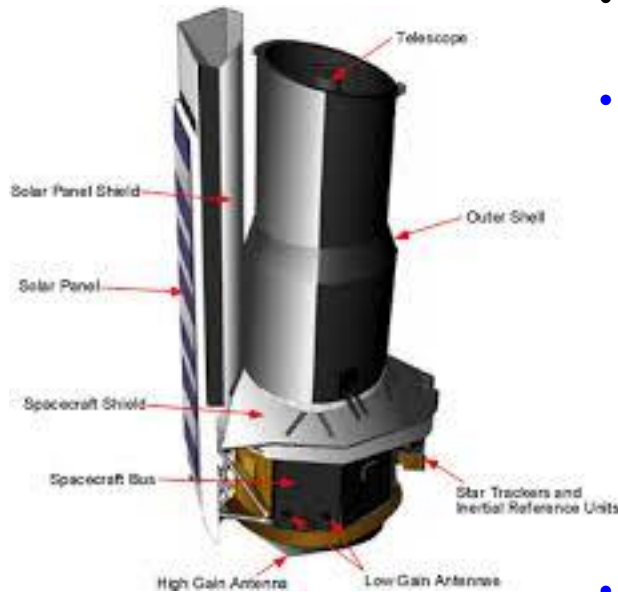
- Rendezvous is occurring at a favorable phase angle wrt science observations
- Communications blackout when SEP < 2-deg (solar conjunction)
- 0 to 45-deg SPE range suggests articulated arrays and/or antenna to maintain both power and communication

**Trajectory Geometry
Spacecraft Angles**



Thermal/ACS Relationship

- **For the thermal analysis of ACS pointing concerns, what does it mean to design a S/C that will typically favor or exclude particular geometry?**
 - While spacecraft are generally designed to be robust even in the even of attitude errors, spacecraft will still have thermal constraints.
 - For example, SIRTf must keep it's solar array on the sun both to provide power AND to keep the telescope barrel (and especially the cryogenic detector) shaded.

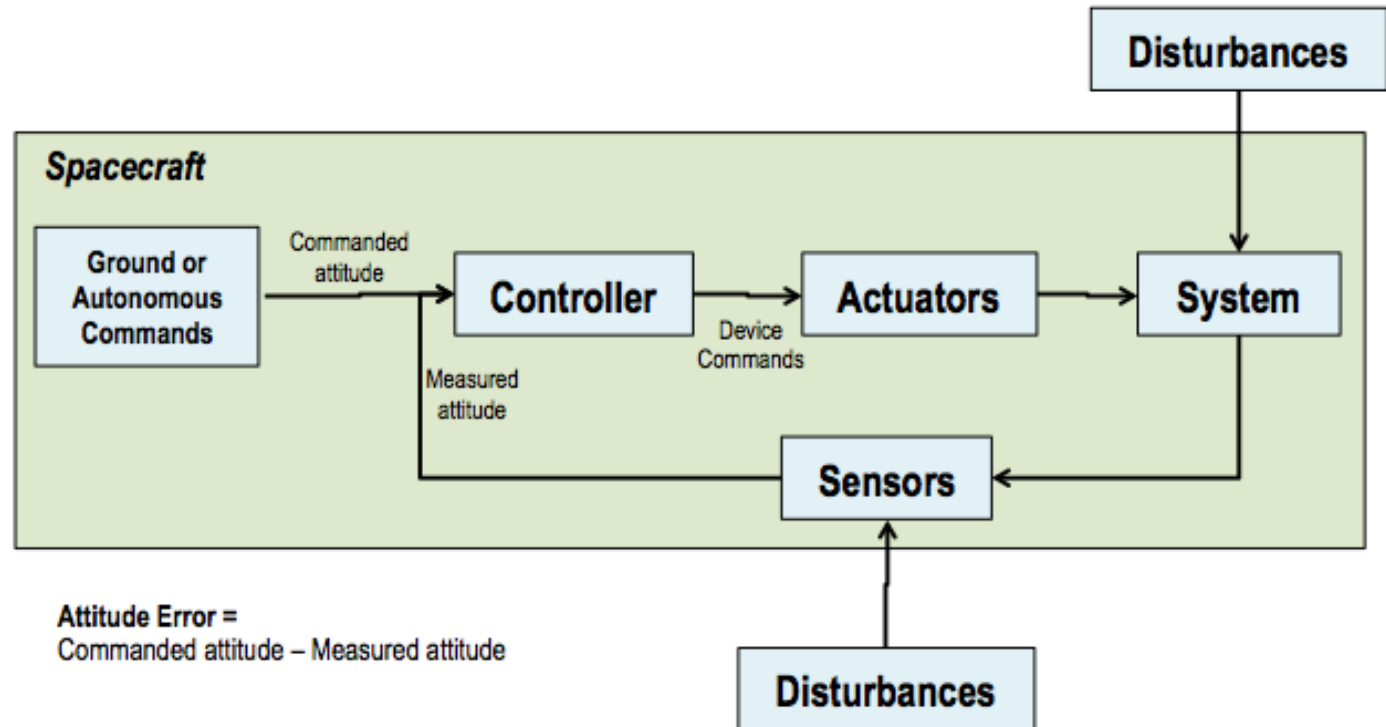


- **How do you evaluate SEP, SPE, and phase angle?**

- These angles are based on outputs from trajectory or visualization software (eg, STK)
- **Strategic Evaluation:**
 - **Phase Angle**
 - Mission Design & Nav. consider trajectory/orbital mechanics to ensure best lighting for science ops (30-60-deg incidence)
 - **Sun-Probe-Earth (SPE)**
 - Power/Telecom engineers consider solar array vs. antenna pointing during mission (close to 0-deg depending on array/antenna margin; greater flexibility with solar arrays)
 - **Sun-Earth-Probe (SEP)**
 - Mission Design & Nav. And Telecom engineers consider solar conjunction wrt communications (eg, 2-deg)
- **Tactical Evaluation**
 - What is the best attitude at any given point during the mission with respect to sun (power & thermal) and Earth (telecom)

Attitude Determination Sensors

- Sun Sensors
- Earth Sensors
- Magnetometers
- Star Trackers / Stellar Reference Units (SRUs)
- Gyroscopes / Internal Reference Units (IRUs)
- GPS Receiv



Sensor Accuracy

Cost, mass,
power, &
complexity

< \$10 M

< \$1 M

< \$100 k

Additionally, custom sensors can be built or payload instruments can be leveraged to perform similar functions.

High

- Star Trackers
- Fine sun sensors

Medium

- Digital Sun Sensors
- Earth sensors
- Gyros
- GPS Receivers

Low

- Magnetometers
- Coarse sun sensors
- IR Earth sensors

Adapted from "Attitude Determination and Control", Joel Sercel, 2003

0.1-deg

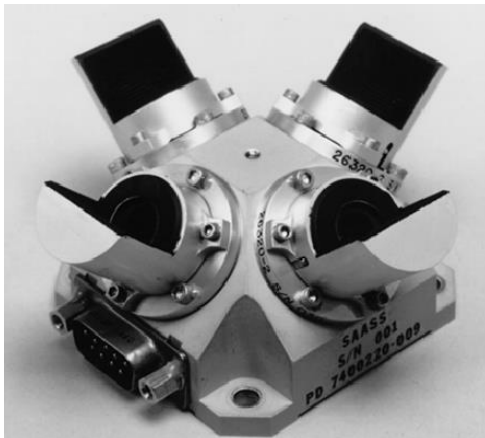
0.01-deg

0.001-deg

Accuracy

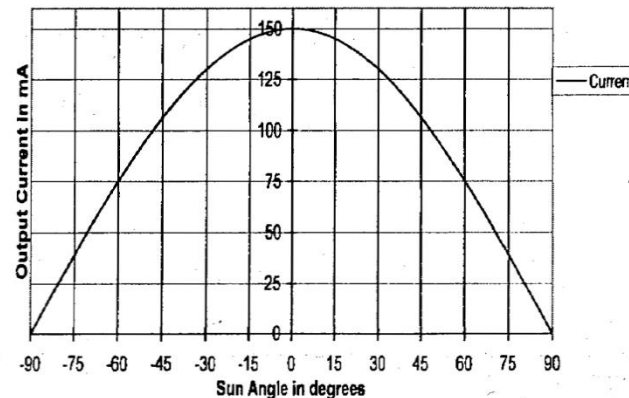
- **Used for basic attitude estimation, especially with respect fault protection**
 - For example, spacecraft with sun-pointing constraints (eg, telescopes) will have additional sun sensors to trigger safe attitude
 - Often used in orthogonally mounted pairs (4 sensors provide hemispherical coverage)

**Analog Sun
Sensor
(from Adcole)**



**Micro Sun
Sensor
(in development)**

- **Analog 4-detector pyramid (picture on left)**
 - Analog current output, digitized by s/c
 - Performance: 1.5-deg, mass = 0.12 kg



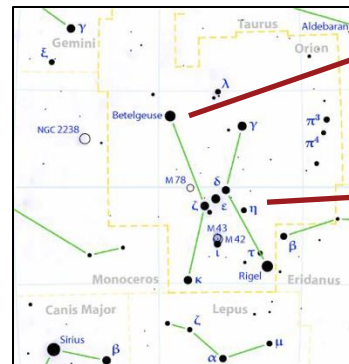
**Analog output
(converted in software
using cosine law)**

- **Fine Sun Sensor**
 - Analog current output, digitized by s/c
 - Performance: < 0.016-deg (1 arcmin)

- **Star Trackers map the positions & magnitudes of observed stars to a star catalog to provide highly accurate attitude knowledge**
 - Fixed Type (common): Scans the star field electronically (or via spacecraft motion) using a 5-20-deg FOV. It then processes, calibrates, and resolves the signal to provide quaternions.
 - Catalog typically contains 100s to 1000s of stars
 - Precision star trackers are heavy & costly.
 - Don't function well with high attitude rates (used for higher precision science missions)
 - Performance: 0.001-1 deg, Mass 1-10 kg



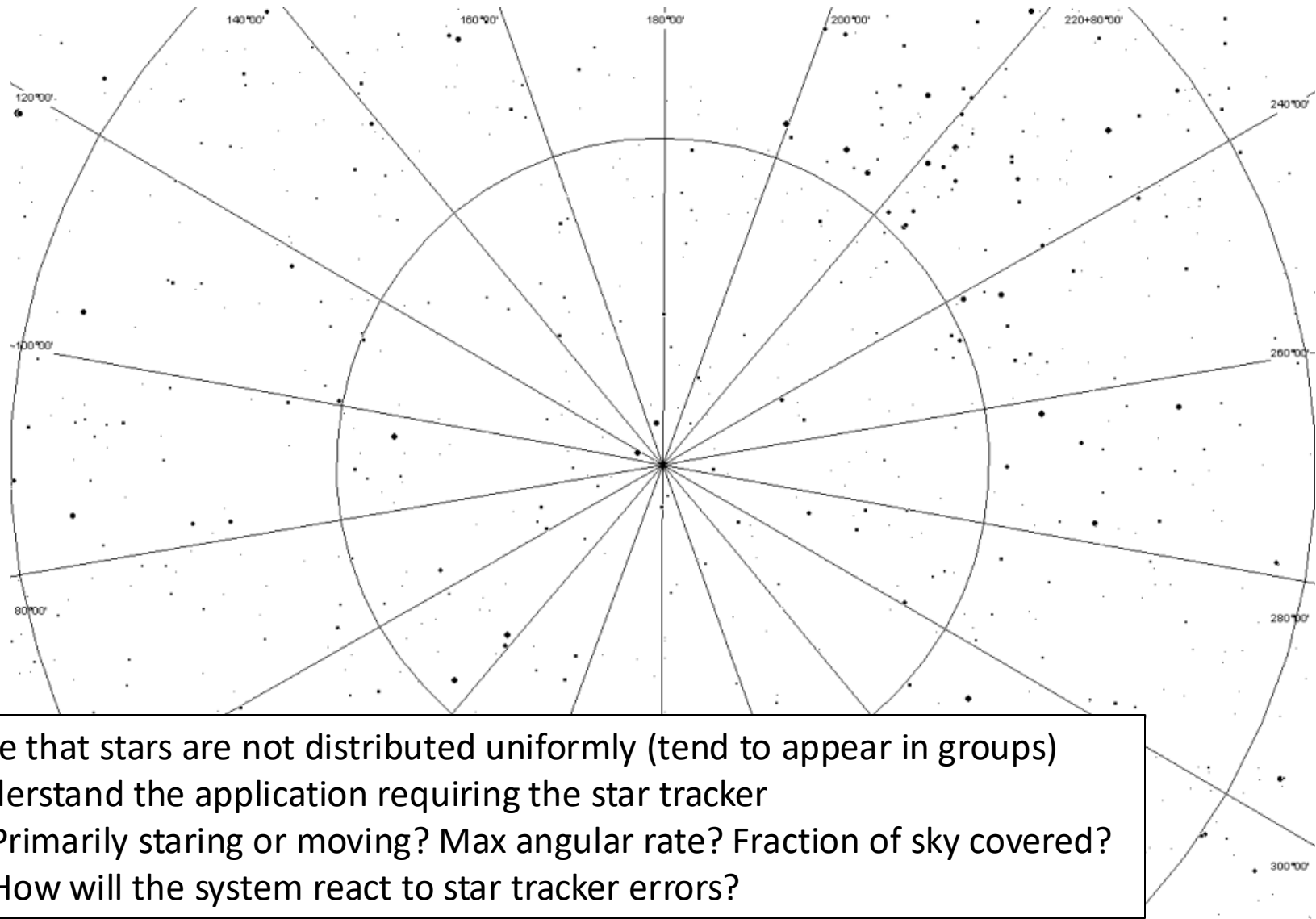
Catalog Visualization



Star Tracker Image



Star Map Example...



Note that stars are not distributed uniformly (tend to appear in groups)

Understand the application requiring the star tracker

- Primarily staring or moving? Max angular rate? Fraction of sky covered?
- How will the system react to star tracker errors?

Magnetometers

Measure 3 perpendicular components of the planet's magnetic field

Compare measurements to standard model

- Tilted center dipole is the standard used

Precision: typical $1.5^\circ \pm 0.5^\circ$

Commonly used with mag torquers (field model not reqd.)

Billingsley Magnetics TFM100S
Attitude Control Magnetometer



Radiation Tolerance: >306 kRADs
Field Measurement Range: $\pm 100,000$ nT
Accuracy: $\pm 0.5\%$ of full scale
Linearity: 0.015% of full scale
Sensitivity: 100 m V/nT
Zero Shift with Temperature: ± 0.6 nT /° Celsius
Frequency Response: - 3 dB @ > 150 Hz
Weight: 200 grams (shielded to 306 kRADs)
Size: 3.66 cm x 3.58 cm x 15.44 cm

From "Attitude Determination and Control", Joel Sercel, 2003

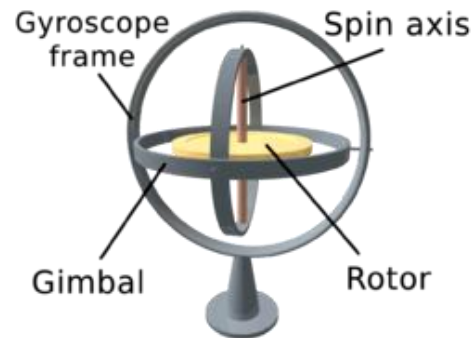
Gyroscopes (IRUs/IMUs)

- Gyros are used to maintain continuous attitude reference between updates from external references (eg, Sun, Earth, stars, etc.)
- While there are several types of gyros, the...
 - Majority of recent missions use ring-laser or fiber optic gyros, such as the LN-200 IMU
- Internal Processing
 - The gyro uses internal dynamics, environmental, and error models to produce a best estimate for the current attitude
 - Over time (eg, 0.1-deg/hr), the error will increase



LN-200 IMU
0.7 kg, 10 W
9 x 9 cm

Traditional Gyroscope



Ring-Laser Gyroscope



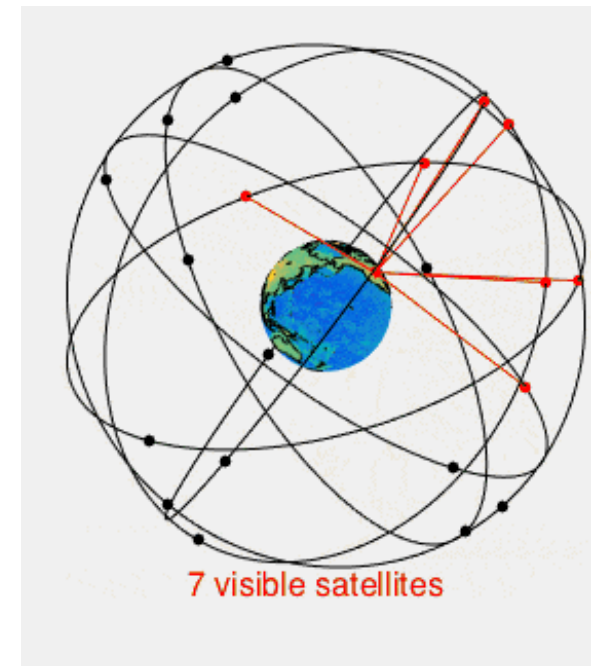
- **Using GPS for attitude knowledge has become increasingly common**
 - Spacecraft velocities to up 16,000 m/s
 - Performance varies with distance (200 km to 45,000 km)
 - LEO Performance < 0.1 deg

GD Sentinel M-Code GPS Receiver

Mass 2.5 kg



Global Positioning System



Summary of Sensors

- **Sun Sensor**
 - Simple, reliable, cheap, but also intermittent depending on sun
- **Earth Sensor**
 - Less common with narrower applications
- **Magnetometer**
 - Simple, reliable, cheap, but also requires low Earth orbit
- **Star Tracker**
 - Higher precision, narrow angle star trackers, but heavy & complex
 - Less expensive versions, but not as precise
- **Gyroscope**
 - Generally required for maintaining knowledge between attitude updates from external references
- **GPS Receiver**
 - Cheap, simple, but requires proximity to functioning GPS

- **Affect System Momentum**

- Reaction Wheels
- Momentum Wheels
- Control Moment Gyros (CMGs)

Typical Accuracy

0.0001-0.1 deg

0.1-2.0 deg

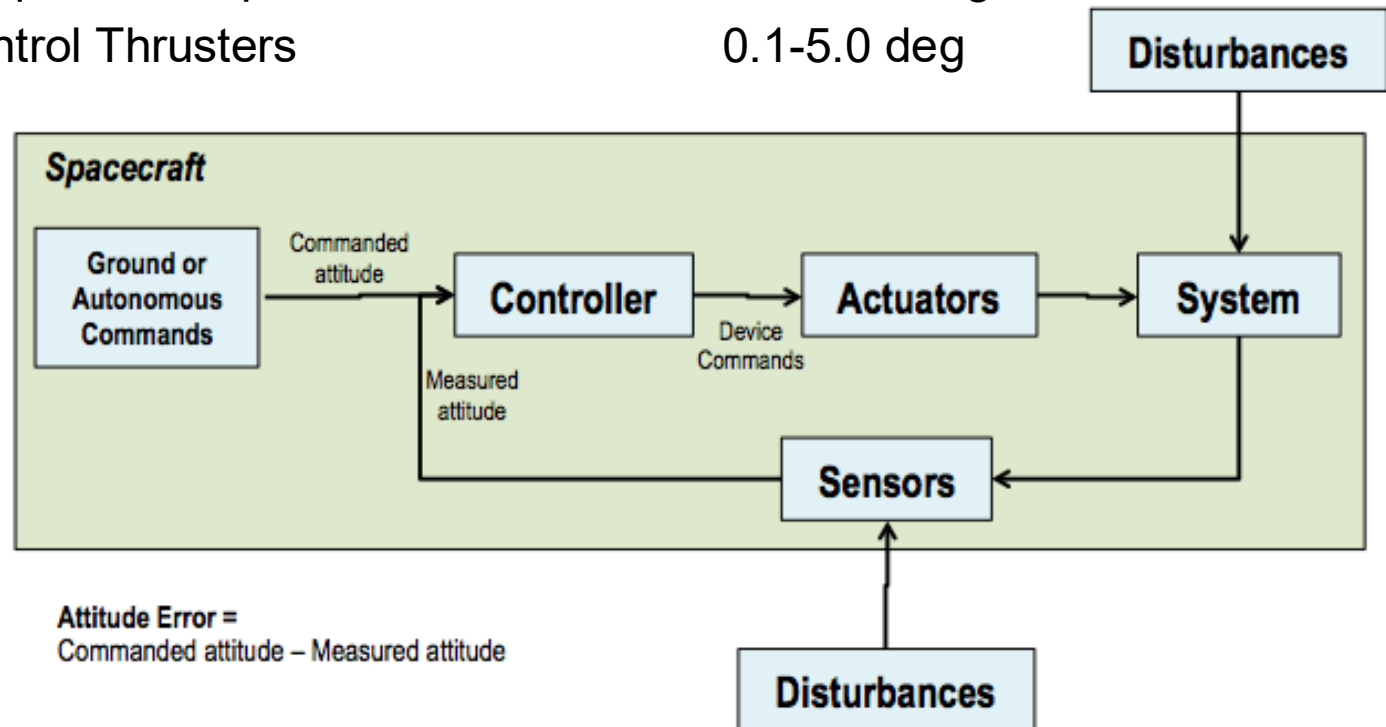
0.001-0.1 deg

- **Do Not Affect System Momentum**

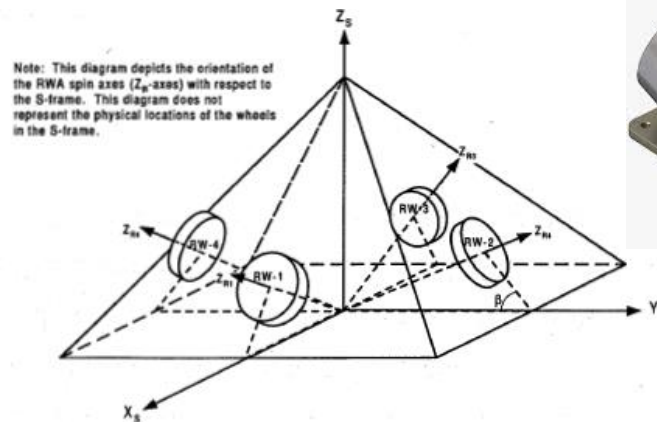
- Magnetic Torquers / Torque Rods
- Reaction Control Thrusters

1.0-10.0 deg

0.1-5.0 deg



- **Electric motor spins a wheel. The rotation is aligned with the control axes (one wheel per axis)**
 - Typical arrangement is 4 wheels in a tetrahedron for redundancy
 - Three are required (1 for each axis), so the fourth one is redundant
- **Characteristics (0.0001-0.1-deg)**
 - Low torque, high accuracy (very fast response possible, tens of Hertz)
 - Not limited by propellant, but limited by angular momentum capacity
 - Nominally operate at low speeds
 - Saturation level is defined by peak motor speed
 - Once wheels reach peak speed, required momentum dumping (eg, thruster firing) to unload
 - That is, reduce the speed without impacting the attitude



Reaction Wheel Sizing & Trade Studies

- **It wasn't clear from SMAD that any one equation was ideal for momentum wheel sizing; what determines the sizing of momentum wheels?**
 - Reaction wheels provide torque (0.01 to 1 Nm)
 - Size of reaction wheel depends on the size of the torque that is needed
 - For the quiz example, relevant torques were:
 - Magnetic torque (T_m), 2.1×10^{-5} Nm
 - Slew torque (T) = 2.9×10^{-4} Nm
 - Momentum dumps due to gravity gradient (h) = 0.039 Nms
 - Torque from RWs needs to be > than torque from disturbance to maintain attitude
 - For accumulated changes (such as momentum dumps), then
 - Slew rate (s) x torque (Nms) = momentum (Nms)
- **How long is the trade study process (typically) in determining attitude control mechanisms for a given satellite architecture?**
 - Varies significantly...
 - “Team X” design is typically done via rules of thumb & general principles (< 2-3 hrs)
 - Detailed GN&C analysis for a more complex mission can take months, including multiple simulations and reviews to finalize architecture

Momentum Wheels

- **Wheel operating at non-zero momentum to provide gyroscopic stiffness to the spacecraft**
- **Characteristics**
 - Performance: 0.1-2.0 deg
 - Effectively these are heavier reaction wheels that operate at a constant speed
 - Often used to cancel the momentum of rotating payload

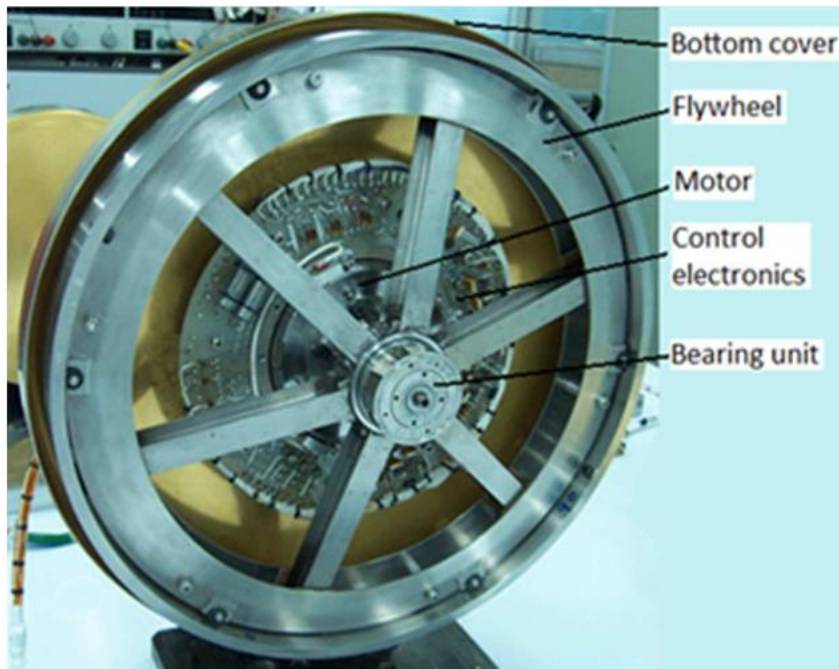
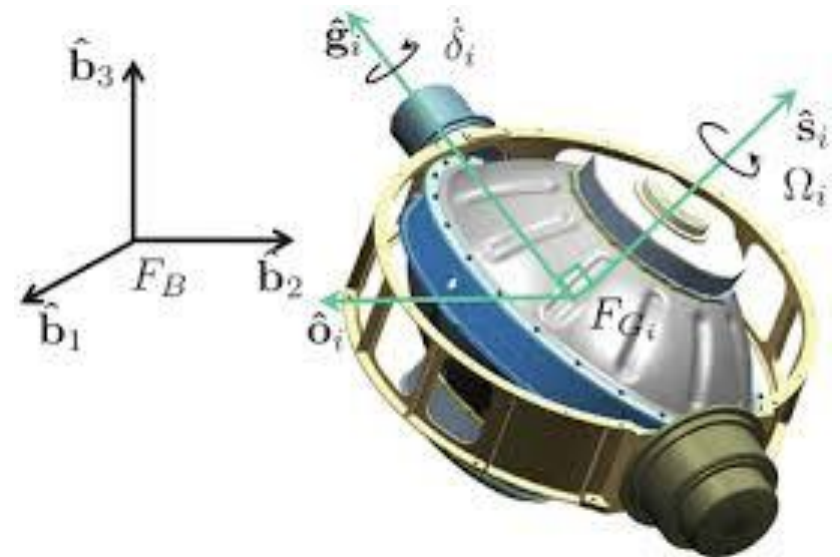


Fig 1 Momentum wheel with top cover removed [6]

Control Moment Gyros (CMGs)

- **CMG is a gimballed momentum wheel**
 - A torque is applied via the gimbal to produce a change in angular momentum, and thus a reaction torque on the body
- **Characteristics**
 - Performance: 0.001-0.1 deg
 - Momentum wheel operating at nearly steady (high) speed;
 - Higher control authority than momentum wheels (up to 100x times)
 - Relatively noisy (they induce vibrations due to the spin rate)



International Space Station (CMGs)

- **“Do they carry a surplus of gyros on the ISS now after being close to critical loss of the station or is it still not worth the weight/space resources?”**
 - No, not worth flying additional spare units beyond the 1 (of 4) installed that is redundant.
 - History
 - CMG-1 failed in 2002 and replaced in 2005 (first post-Columbia shuttle flight).
 - CMG-2 had two circuit breaker failures
 - CMG-3 failed in 2006 and replaced in 2007

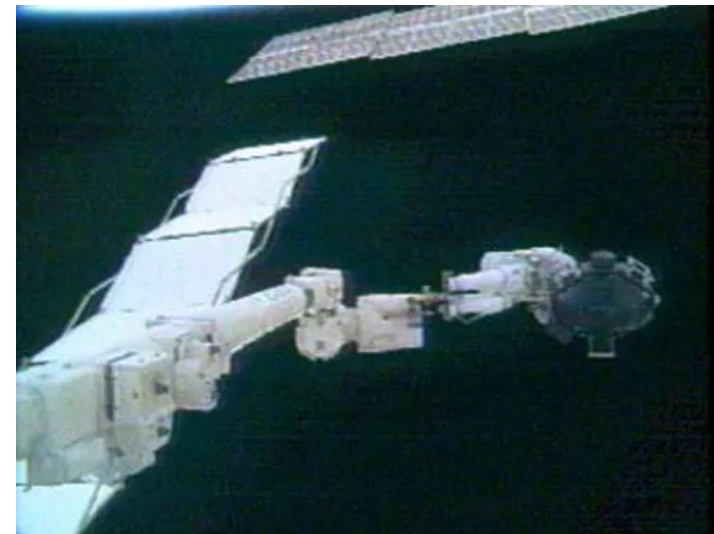
**1 of 4 CMGs during
Integration & Test**

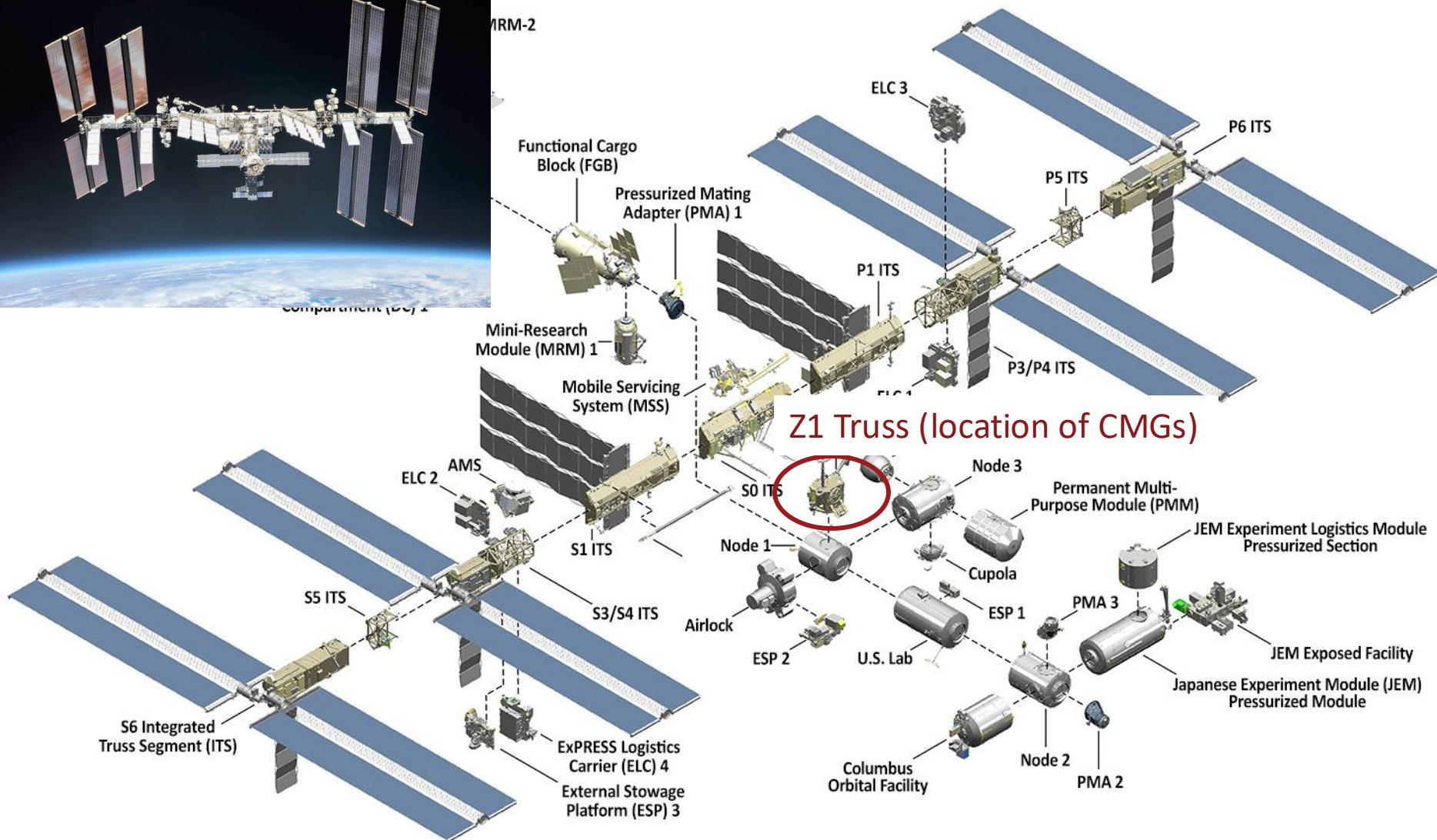


**4 CMGs installed in ISS
Z-1 Truss (as of 2000)**



**1 CMG Replaced in 2005
via a 7-hr Space Walk**

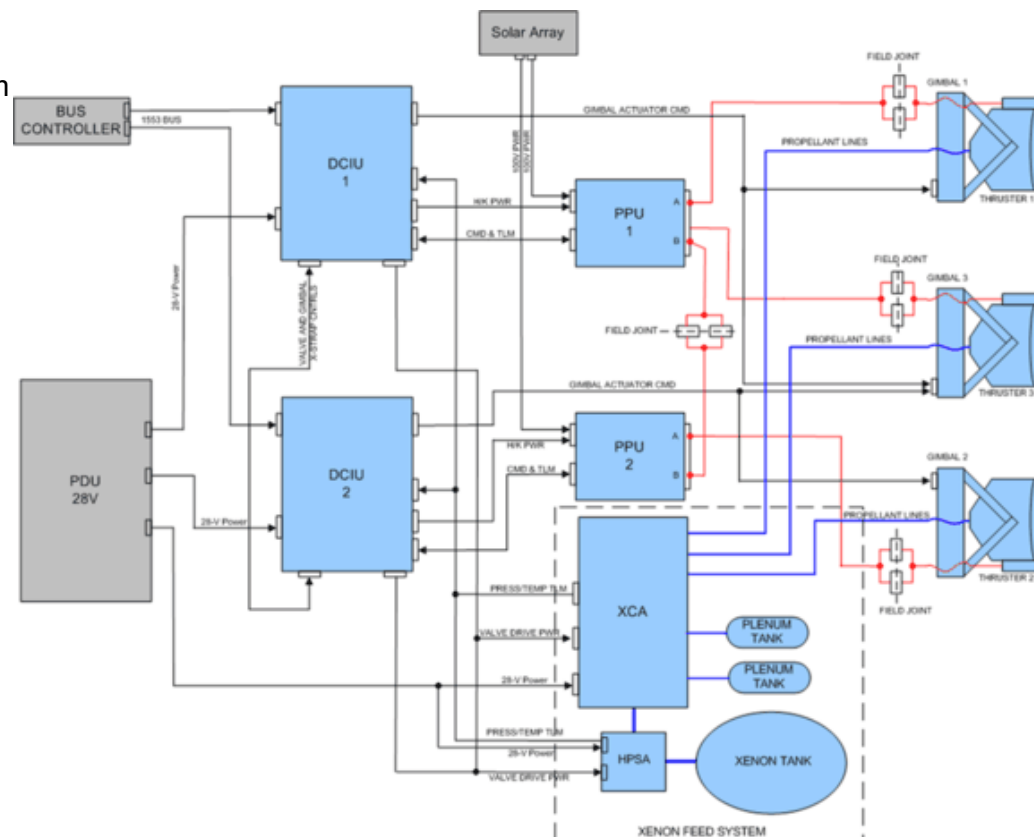




Attitude Control Redundancy

- **If part of your system controlling attitude is damaged, how can you compensate for that without losing the S/C entirely?**
 - Most spacecraft include significant redundancy for attitude control
- **Typical redundancy:**
 - Reaction Wheels or CMGs provide highest performance (w/low propellant use)
 - Typically, three wheels are required to maintain 3-axis attitude control
 - A fourth one is included, such that any three can provide full 3-axis control
 - ISS requires a minimum of 2 CMGs (uses gravity gradient as well)
 - Thrusters
 - Larger, higher-value (ie, Class A/B) missions will have redundant coupled thrusters that can provide reasonable attitude control
 - This is called “functional” redundancy. Thrusters will generally not perform as well and/or use-up greater propellant, so there will typically be some impact to the mission that is assessed
 - Or simply, by any available means...
 - Dawn used it's electric propulsion system to provide attitude control (after 3 RW failures)
 - ISS incorporates the gravity gradient to minimize attitude control requirements
 - For attitude knowledge error, onboard instruments can be pressed into service...
 - Etc.

- **Would some kind of electric propulsion thruster have enough thrust to be an effective control system?**
 - Yes. On Dawn, the 3 electric propulsion thrusters can provide 90 mN of thrust.
 - In Geostationary orbits, electric propulsion systems regularly provide attitude control given their extremely high efficiency ($> 3,000$ Isp) versus the very low disturbances
 - For a small spacecraft
 - drag = none
 - Gravity gradient = 5×10^{-9} Nm
 - Solar radiation pressure = 6×10^{-6} Nm



Magnetic Torque Rods

- **Electromagnets that are energized to provide a electromagnetic torque on the spacecraft via the local magnetic field**
- **Characteristics**
 - Performance: 1.0-10.0 deg
 - Clean way to dump momentum (with no consumables)
 - Generally consist of simple iron rods (or loops)
 - Very reliable and cheap
 - Primarily applicable to planetary/Earth orbiters with large magnetic fields.



Thrusters (1 of 2)

- **Often called a Reaction Control System (RCS) when used only for attitude control**
 - From propulsion, these are typically the smallest thrusters. Often, they functionally back-up a reaction wheels in case of a failure, but they do not provide the same precision
- **Characteristics**
 - Performance: 0.1-5.0 deg
 - Various sizes available, with different levels of performance (ie, control)
 - Propellant-limited and may introduce contamination
 - Capable of very high torque
 - Accuracy limited by thruster configuration (impulse bit)



Thrusters (2 of 2)

- **Primary Thruster Characteristics**

- Thrust-level (eg, 1-N vs. 490-N), which determines the ability to provide thrust and thus ΔV
- Efficiency (ie, Isp), which is based on both type of thruster & propellant
- Minimum impulse bit (MIB) is the smallest control torque that can be applied to the spacecraft

- **Primary Uses**

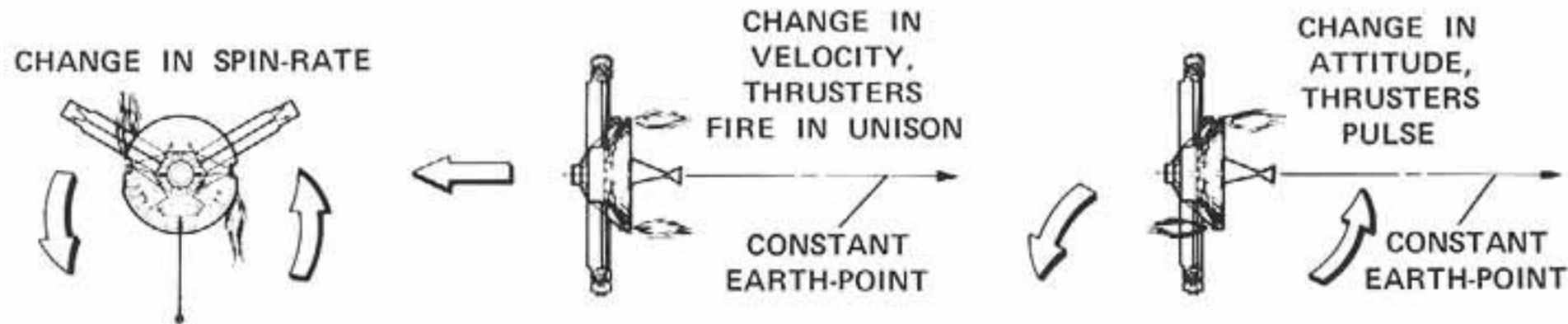
- Trajectory Correct Maneuvers (TCMs) for changing ΔV
- Orbital Maintenance
- Momentum dumps
- Attitude (or reaction) control
- Thrust vector control (TVC) that provide guidance on larger maneuvers

- **Matching Thrusters to Spacecraft**

- Consider necessary functions: large ΔV maneuvers, attitude control, etc.
- Thrusters should be evaluated for min/max burn times
 - Minimum burn time should be consistent with the minimum impulse bit
 - Maximum burn time should be consistent with the type of maneuver (eg, orbit insertion, burn time < 2-3 hrs)
- Result is often something like...
 - RCS thrusters for attitude control, momentum dumps, etc.
 - Main engine(s) for large & more efficient ΔV maneuvers
 - Other mid-size thrusters where the above thrusters aren't sufficient (eg, thrust vector control on a large ΔV burn)
- Reminder that thrust-level can be scaled with the number of thrusters

Attitude Control on Spin Stabilized S/C

Graphic from Pioneer 10 spacecraft, history.nasa.gov



- **Spin Rate**

- Two pairs of coupled thrusters act on opposite sides to change spin rate

- **Velocity**

- After the spin axis is changed to the desired orientation, the thrusters on the HGA side fire simultaneously for translational maneuver

- **Attitude**

- Pair of coupled thrusters rotate the spin-axis. Thrusters are fired in brief pulses at a precise position in the spacecraft rotation. Each pulse moves the axis a few tenths of a degree until the desired attitude is reached

• Review & Understand Design Information

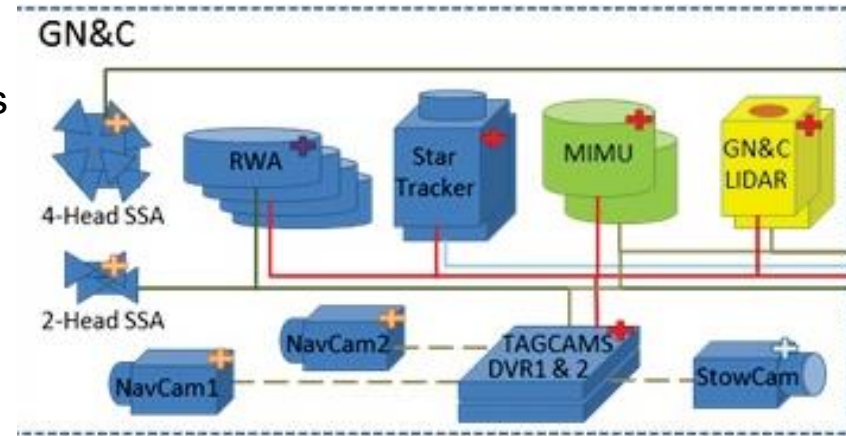
- Mission Description and/or Concept of Operations
- System and Subsystem Requirements
 - ConOps & mission geometry
 - Payload pointing requirements
 - Expected disturbances

• Create a Preliminary Design

- Identify the most likely architecture (eg, 3-axis)
- Identify likely components (eg, sun sensors, star trackers, etc.)
- Size control effectors (eg, reaction wheels, CMGs, thrusters, etc.)
 - Consider functional redundancy, system cost, etc.
 - Create end-to-end kinematic simulation, error budgets, etc.
 - Understand fault scenarios
 - There is often a trade between thrusters & reaction wheels
- Create component mass list

• Review & Iterate (w/broader team)

- Revisit other options & trades



See HW7 for additional detail.

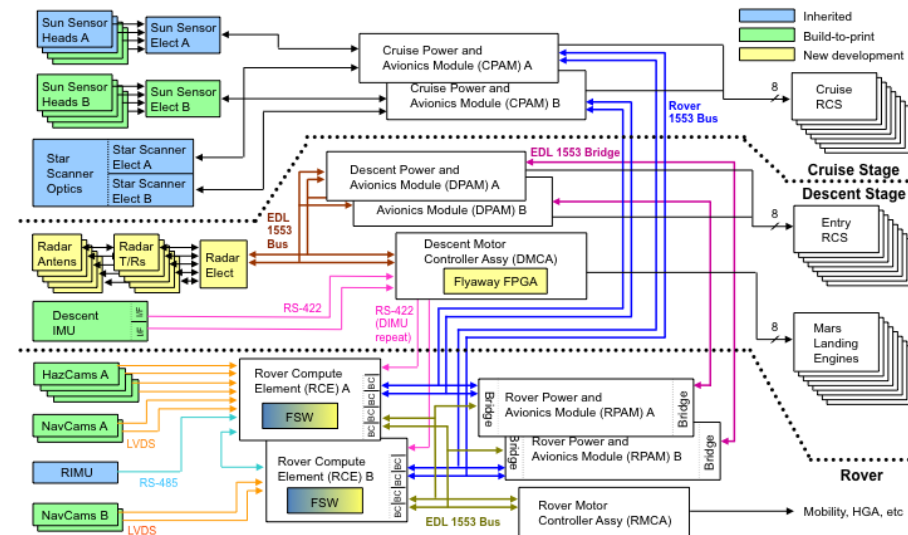
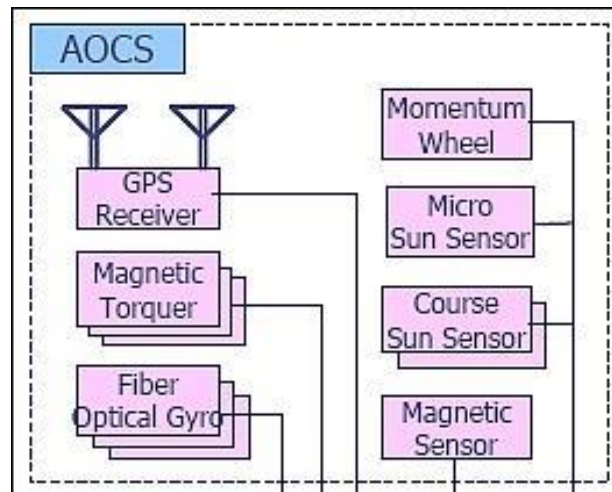
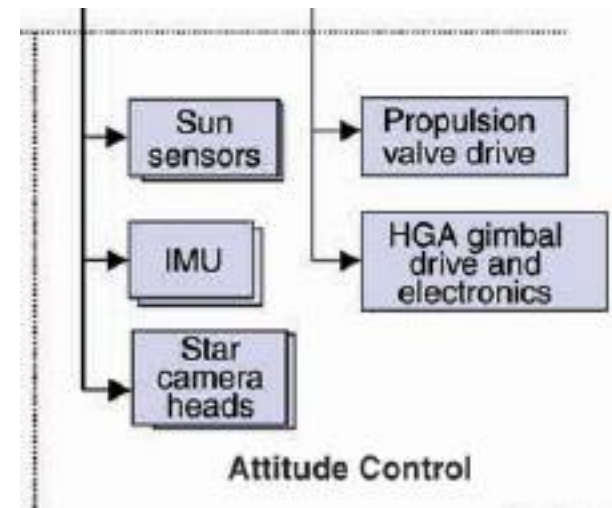
Additional Guidance

- **Three Primary Functions**
 - Control, knowledge, & stability...
- **Common Sensors Across ~All Missions**
 - Sun sensors
 - Gyros / IMUs
- **Destination Dependencies**
 - Low-Earth Orbit (LEO): Horizon sensors, GPS receivers, magnetic torquers
 - Sun < 2-3 AU: Electric propulsion for attitude control
 - Deep Space: Star trackers
 - Close Proximity Ops: Radars, LIDARs, etc.
- **Functional Dependencies**
 - Imaging s/c typically require reaction wheels, although cold gas systems are increasingly capable
 - Thrusters-only vs. reaction wheels is a common trade between cost & performance
 - Larger, more agile s/c typically need CMGs
 - Layers of redundancy increase with mission duration & budget

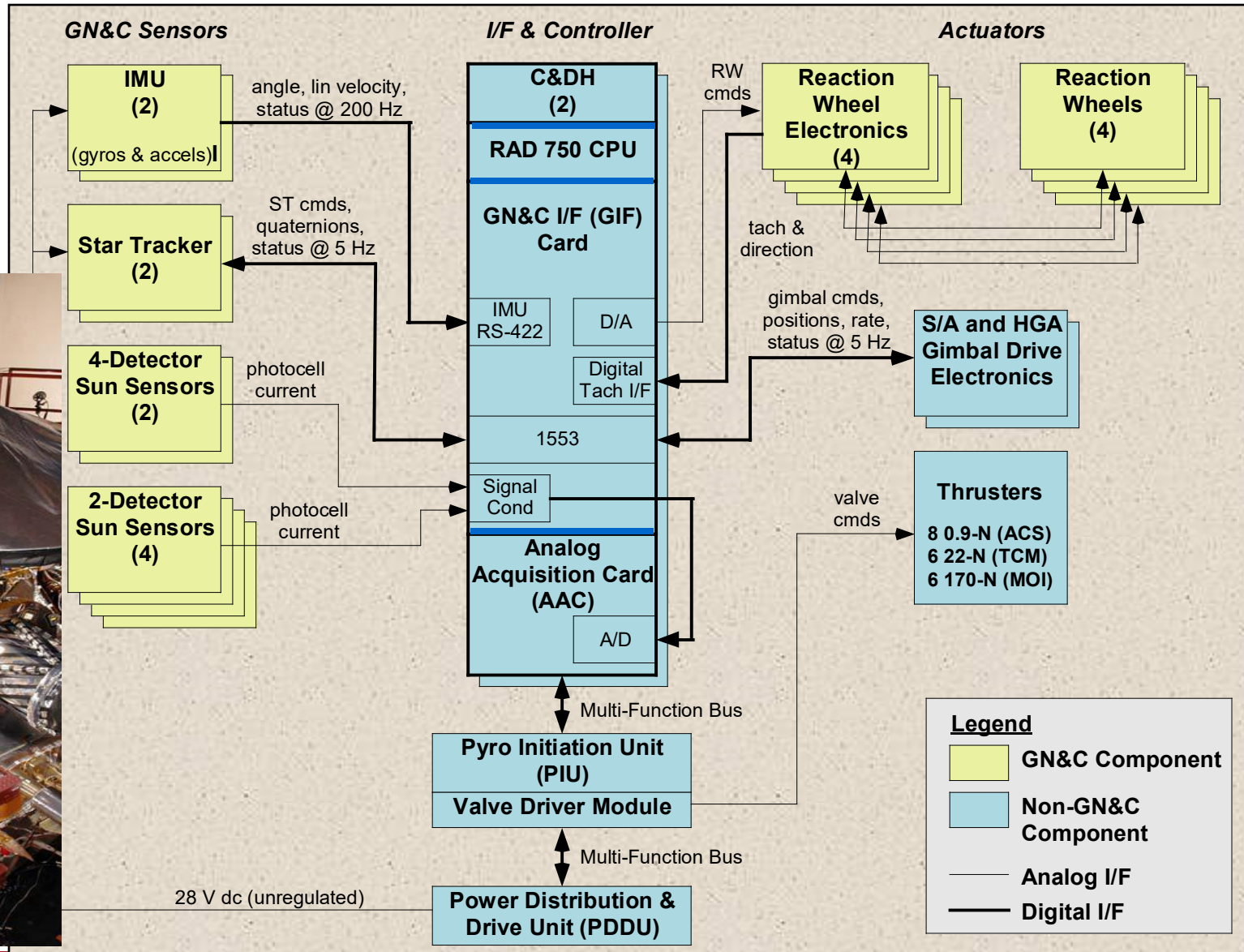
While the sizing of individual components is based on specific requirements, one can generally estimate a likely architecture based on a mission description and relative budget.

Interpreting a Block Diagram

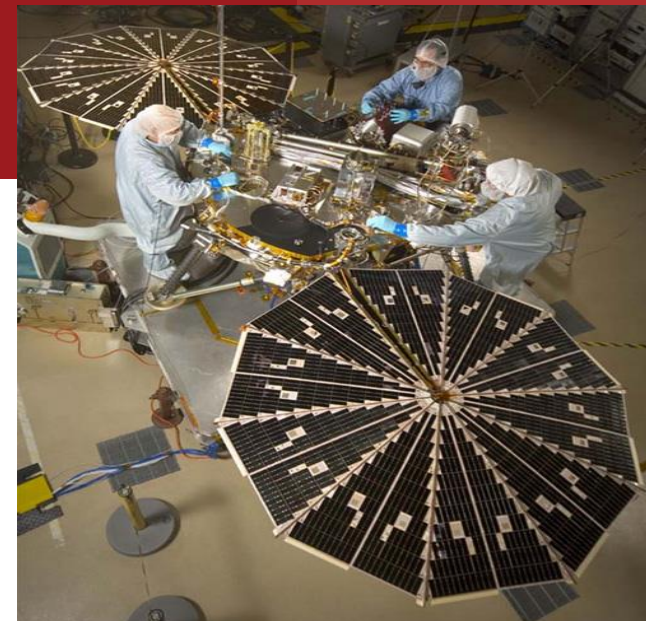
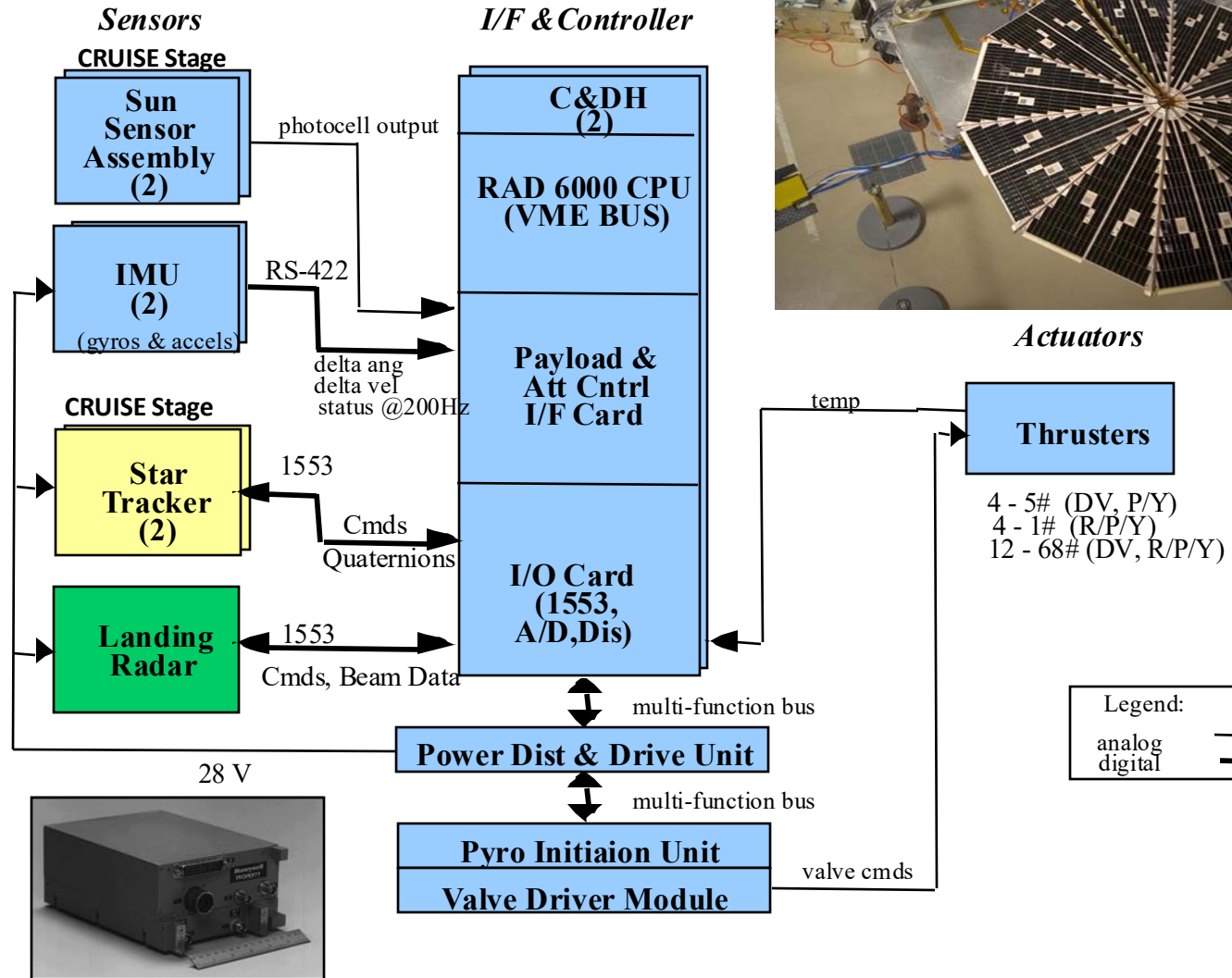
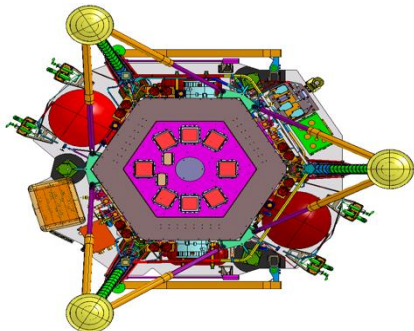
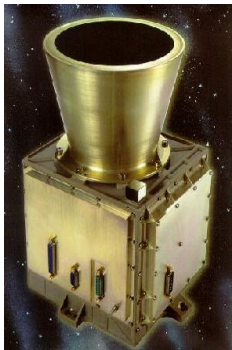
- What function do each of the GN&C components provide?
- What types of missions do they support?
- If component X fails, how does the failure propagate?
 - Assuming realtime telemetry at 1 Hz, what is the ground likely to see?
 - Is there a possibility of recovering the mission?



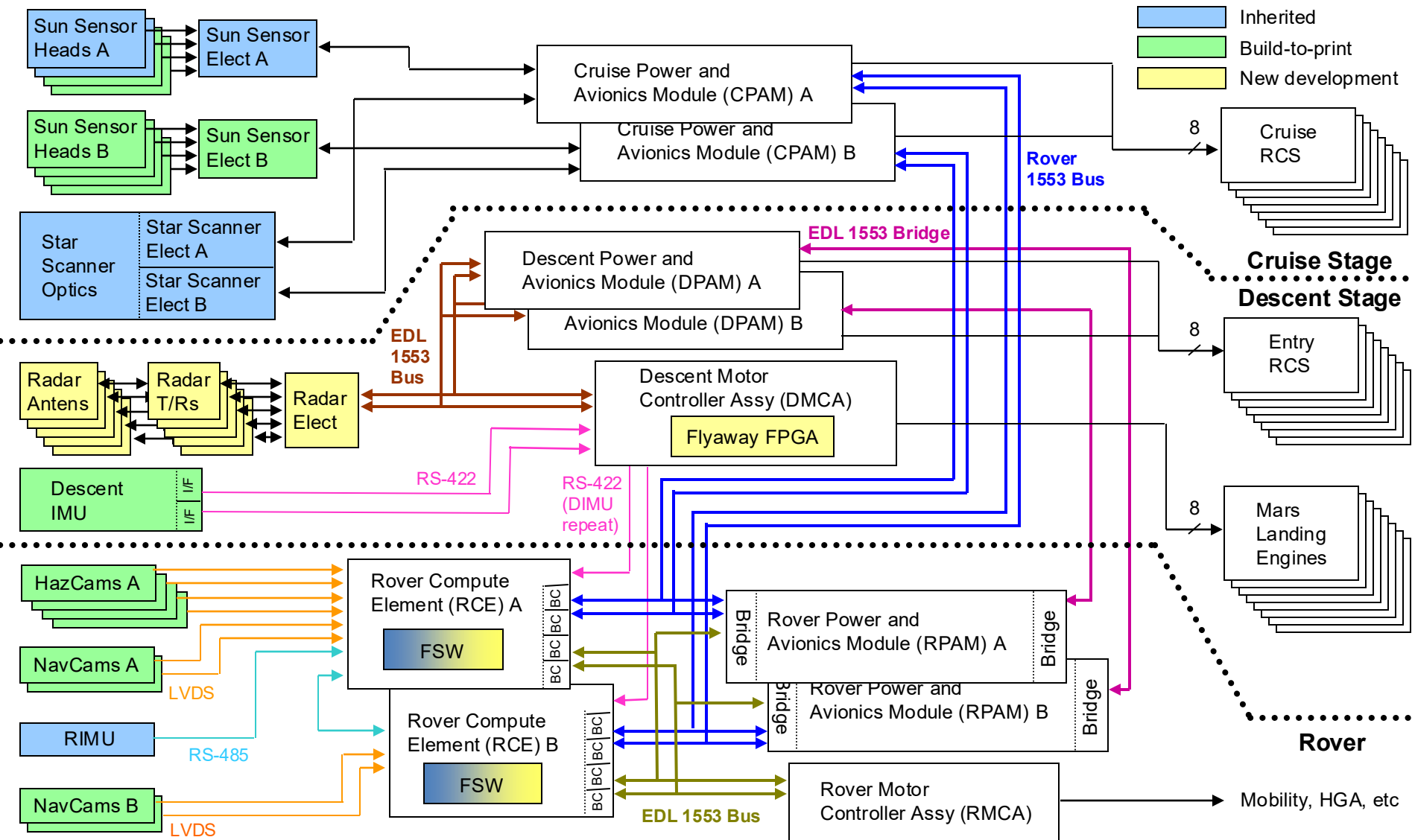
MRO GN&C Architecture



Phoenix GN&C Architecture

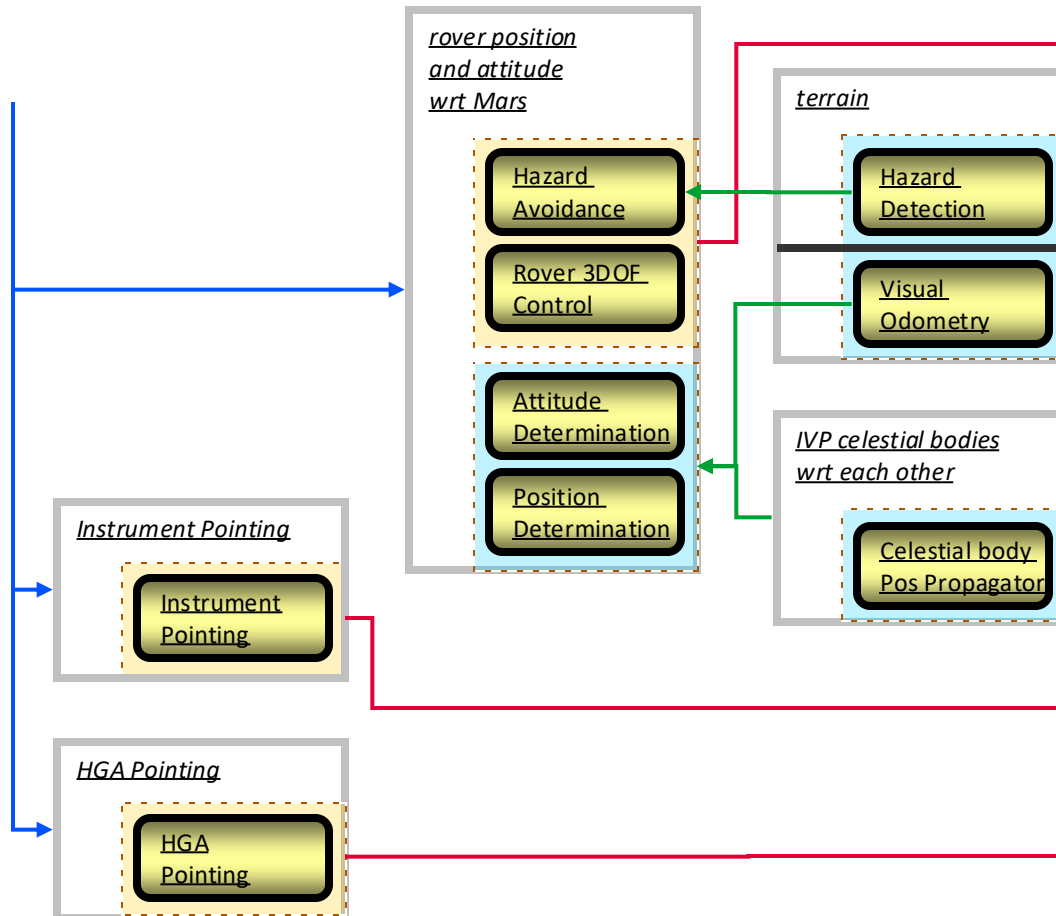


MSL GN&C Architecture

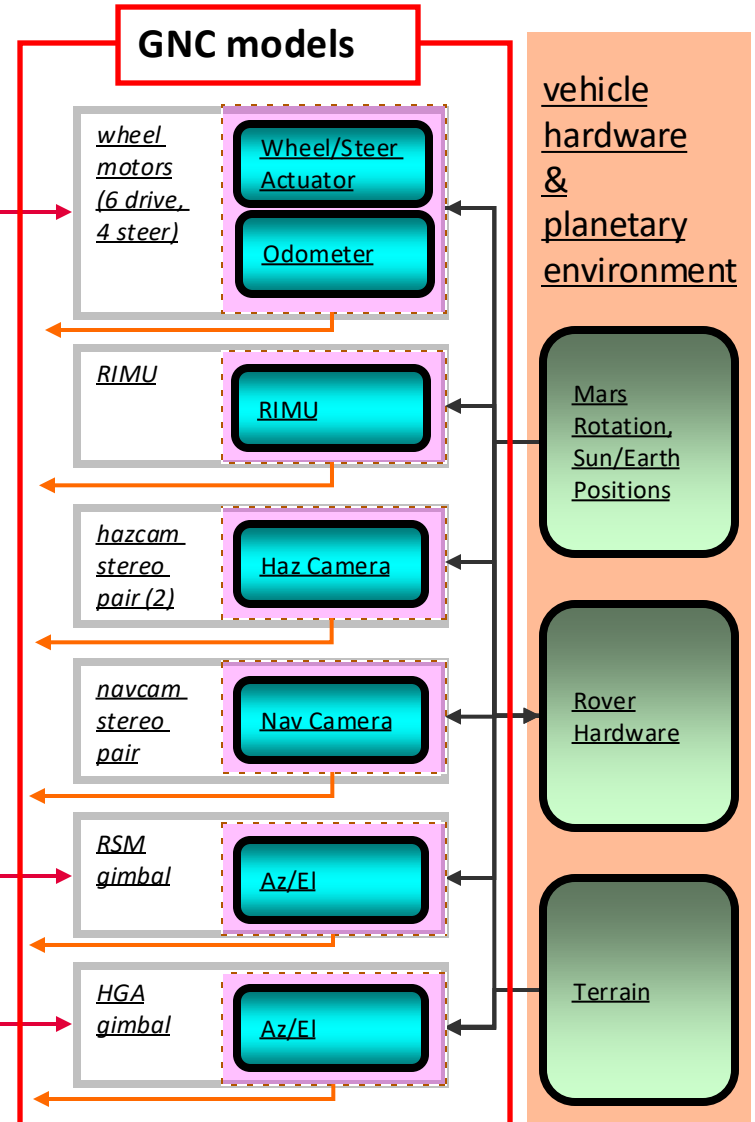


MSL GN&C Architecture (Control System)

GNC algorithms and functions



GNC models



Questions from Prior Class (1 of 3)

- **What are the most commonly used actuators and why?**
 - Magnetic torquers for LEO satellites (cheap, simple)
 - Thrusters (low precision) and reaction wheels (high precision)
- **Are there any missions that failed due to errors of GN&C actuators?**
 - Major driver of spacecraft failures & reliability, including both infant mortality & lifetime failures
 - Phasing of sensors
 - Genesis- backwards G-switch (note that phasing tests are a critical part of testing)
 - Loss of lubrication or unexpected friction
 - Dawn- reaction wheels; several failed prematurely, but Dawn was able to use electric prop.
 - ISS- failed gyros (reoccurring failures)
 - SOHO- failed gyro- (coupled with inadequate fault response nearly failed the mission)
 - In 331b, we'll look at a few significant NASA failures...
- **At what altitude does some of the drag force start to disappear?**
 - Varies with drag coefficient & solar minimum/maximum
 - At > 600 km, gravity gradient dominates over drag. At > 2,000 km disappears entirely
- **What attitude control happens at launch for the whole vehicle. Just gimbal of the main engines?**
 - Primarily the main engine gimbals

Questions from Prior Class (2 of 3)

- **Would it be possible to build a small cheap star sensor that worked from the Earth's surface assuming a cloudless night? Or would the components be far too expensive?**
 - Yes, and there are similar iPhone applications
 - Drivers are likely ensuring adequate camera sensitivity and then going through the math, but similar image processing is well-published in literature
- **Is there any reason to use reaction wheels over cold gas RCS?**
 - Greater control via reaction wheels. Cold gas thrusters are improving, but difficult to compete against 0.0001-deg RWs
- **Have there been cases where new GN&C technology needed to be developed?**
 - On MSL, the landing radar was developed for the mission (overcame issues with Phoenix radar reliability)
 - There is continued work to reduce the size of sensors & actuators (significant ongoing work on aircraft that drives space)
- **Would attitude control thrusters fall under the jurisdiction of the ADACS or the propulsion teams?**
 - Depends on the overall design
 - For a small RCS system with no other propulsion, then it's all under ACS and no propulsion subsystem required
 - Otherwise, it would typically be covered under propulsion
- **I'd like to know more about the GPS method for attitude determination. ... Would a GPS system at Mars be feasible?**
 - A GPS system functions by receiving 3+ GPS signals and then triangulating position
 - Cost of deploying a constellation at Mars is prohibitive. If this can be solved, the next issue would be power – GPS signals on Earth required ~500 W. On Mars, this translates to approximately 10 m² just for the GPS power output.

Questions from Prior Class (3 of 3)

- **On weights attached to sensors in Chart 23, what missions might require sudden, large updates of ATE versus more gradual updates?**
 - Not many, but some would be landers for terrain avoidance
- **How many total stars have been cataloged?**
 - Over a billion stars have been cataloged
 - Typically, star tracker catalogs contain thousands (eg, 10,000)
- **How was the comet sample in the Deep Impact mission relayed back to Earth? Was it just the analysis of the videos and the plume, or was there a physical sample that made it back to Earth?**
 - Only data. Instruments (including a spectrometer that can analyze the composition of the plume) transmitted data back to Earth. The probe relayed its data via the observing spacecraft.
- **How have we still been able to stay in contact with older s/c like voyager, and when are we expected to lose any communication with it and similar spacecraft?**
 - It's case-by-case.
 - All missions have a specific, funded lifetime. As the mission approaches its end-of-life, the team will typically propose a mission extension. This extension needs to include all applicable costs to maintain and communicate with the spacecraft. It is then evaluated for cost & benefit.
 - In the case Voyager, the DSN 70-meter stations continue to be maintained to support communication, along with supporting personnel, spacecraft knowledge, etc.

Part 4: Spacecraft Dynamics

• Kinematics

- Describes the motion of points, bodies (objects), and systems of bodies (groups of objects) without considering the forces that cause them to move.

• Translational Motion

- Changes in the position and velocity of an object
- For spacecraft design, this primarily corresponds with understanding position relative a gravity field and changes in ΔV (vs. thrust)

• Rotational Motion

- Changes in the attitude of an object
- For spacecraft design, this corresponds to attitude determination and control.

• Importance

- Understanding and/or modeling the spacecraft attitude provides insight into:
 - Precision of sensors & actuators required
 - Performance capability to meet payload, ascent/descent, communication, and/or other requirements
 - Amount of propellant required
 - Detailed Concept of Operations, including required spacecraft behavior and responses
 - Margin/risk assessment

No.	Quantity	Translational Motion of a particle	Rotational motion of a rigid body
1.	Inertia	Mass 'm'	M.I 'I'
2.	Displacement	$\vec{dr}, \vec{s}$	$\vec{d\theta}, \vec{\theta}$
3.	Velocity	$\vec{v} = \frac{d\vec{r}}{dt}$	$\vec{\omega} = \frac{d\vec{\theta}}{dt}$
4.	Acceleration	$\vec{a} = \frac{d\vec{v}}{dt}$	$\vec{\alpha} = \frac{d\vec{\omega}}{dt}$
5.	Momentum	$\vec{p} = m\vec{v}$	$\vec{L} = I\vec{\omega}$
6.	Cause of motion	$\vec{F}$	$\vec{\tau}$
7.	Law of motion	$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$	$\vec{\tau} = \frac{d\vec{L}}{dt} = I\vec{\alpha}$
8.	Equations of motion under uniform acceleration	$\vec{v}_f = \vec{v}_i + \vec{a}t$ $\vec{S} = \vec{v}_i t + 1/2 \vec{a} t^2$ $\vec{v}_f^2 = \vec{v}_i^2 + 2 \vec{a} \cdot \vec{s}$	$\vec{\omega}_f = \vec{\omega}_i + \vec{\alpha}t$ $\vec{\theta} = \vec{\omega}_i t + 1/2 \vec{\alpha} t^2$ $\omega_f^2 = \omega_i^2 + 2 \cdot \alpha \cdot \theta$
9.	Work done	$\vec{F} \cdot \vec{S}$	$\vec{\tau} \cdot \vec{\theta}$

<http://pinkmonkey.com/studyguides/subjects/physics/chap5/p0505103.asp>

Kinematics of Rotation

- When a rigid body rotates about a fixed axis, say, the z-axis, all the radial lines rotate the same angle
 - the same angular velocity ω in a rigid body.
- The magnitude of the velocity of any point on the rigid body is determined by R_i , the distance of the point to the axis



$$v_i = |\omega \times \mathbf{r}_i| = \omega r_i \sin \phi_i = \omega R_i$$

$$R_i = \sqrt{x_i^2 + y_i^2}$$

Comments

- Since the angular velocity varies with the radius, it is more difficult to calculate basic physical relationships, and thus the next several charts are intended to provide additional background
- I'm reusing charts from Joel Sercel's class, which should provide a good discussion found across numerous references

From ADCS Handout, Joel Sercel, 2003

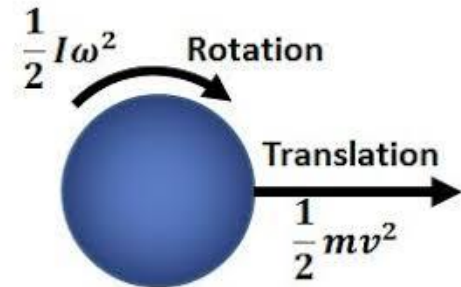
Rotational Kinetic Energy

- If the rigid body is formed by n particles, the **kinetic energy of rotation** of the rigid body is given by

$$K = \sum_{i=1}^n \frac{1}{2} m_i v_i^2 = \sum_{i=1}^n \frac{1}{2} m_i (\omega \times R_i)^2 = \frac{1}{2} \left(\sum_{i=1}^n m_i R_i^2 \right) \omega^2$$

$$= \frac{1}{2} \left(\sum_{i=1}^n m_i (x_i^2 + y_i^2) \right) \omega^2 = \frac{1}{2} I_z \omega^2$$

$$I_z = \sum_{i=1}^n m_i (x_i^2 + y_i^2)$$



Comments

- Kinetic Energy equation leads to $K = \frac{1}{2} I_z \omega^2$
- Where I_z is a constant (ie, rotational inertia) that governs how a body behaves when a force (ie, torque) is applied.
- This is similar to mass in translational physics
- I_z is the **moment of inertia** about the z-axis

From ADCS Handout, Joel Sercel, 2003

Moment of Inertia & Angular Acceleration

- Suppose a particle m can rotate about a fixed axis with radius R . Apply the Newton's 2nd law at the tangent direction, we have

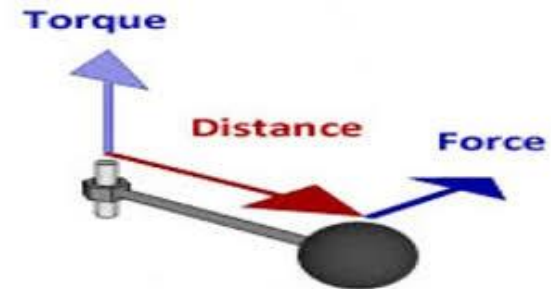
$$F_t = ma_t \Rightarrow F_t R = ma_t R = m \alpha R R = m R^2 \alpha$$

$$\Rightarrow \tau_z = I_z \alpha$$

- Notes:

- The torque τ_z and moment of inertia I_z must be with respect to the same (fixed) axis.
- This also applies to a rigid body.
- The torque must be from external forces.

6. Cause of motion	$\vec{F}$	$\vec{\tau}$
7. Law of motion	$\vec{F} = \frac{d\vec{p}}{dt} = m \vec{a}$	$\vec{\tau} = \frac{d\vec{L}}{dt} = I \vec{\alpha}$



Comments

- Moment of Inertia can now be related to Newton's 2nd Law ($F=ma$), resulting in a loose equivalent wrt torques
- α = angular acceleration (eg, radians/s²)

Calculate Moment of Intertia

- The moment of inertia for a system of discrete particles can be directly calculated from the summation.

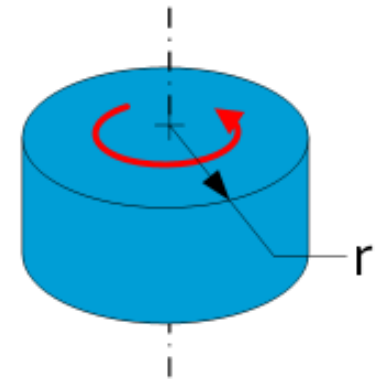
$$I_z = \sum_{i=1}^n m_i (x_i^2 + y_i^2)$$

- For a continuous rigid body and about z-axis, we have to integrate

$$I_z = \int (x^2 + y^2) dm = \int (x^2 + y^2) \rho(\mathbf{r}) dV$$

- If uniform, $\rho = m/V = \text{const}$,

$$I_z = \rho \int (x^2 + y^2) dV = \frac{m}{V} \int (x^2 + y^2) dV$$



Comments

- Note that ρ is density here
- Effectively, you are integrating x, y over the volume at a given density to calculate the moment of inertia about the z-axis
- SI units are kg m^2

From ADCS Handout, Joel Sercel, 2003

Moment of Inertia of a Ring

Moment of inertia of

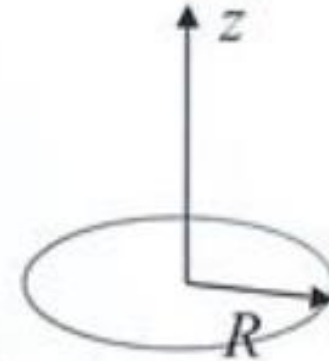
a) a uniform thin ring of radius R and mass m ,

b) a uniform disk, about its center.

Solution:

a) Linear density $\rho_l = m/2\pi R$,

$$I_z = \int (x^2 + y^2) \rho(\mathbf{r}) dV = \int_0^{2\pi} R^2 \rho_l R d\varphi = 2\pi \rho_l R^3 = mR^2$$



b) Area density $\rho_a = m/\pi R^2$. For a thin ring at r to $r + dr$, the moment of inertia from a) is

$$dI = r^2 dm = r^2 \rho_a 2\pi r dr = 2\pi \rho_a r^3 dr$$

The total moment of inertia

$$I = \int_0^R 2\pi \rho_a r^3 dr = \frac{1}{2} \pi \rho_a R^4 = \frac{1}{2} mR^2$$

Comments

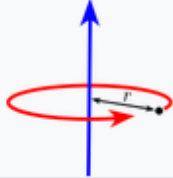
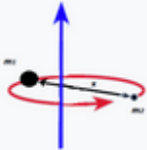
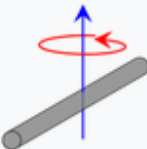
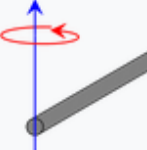
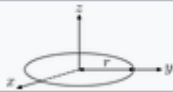
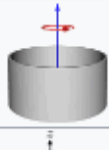
- Example for I_z of a thin 'ring'

Sources

- https://en.wikipedia.org/wiki/Moment_of_inertia
- https://en.wikipedia.org/wiki/List_of_moments_of_inertia

Comments

- Many published formulas exist for basic shapes
- This highly relevant because you can determine the moments of inertia of complex shapes by summing the individual components
- We can do this via...
 - Parallel Axis theorem
 - Perpendicular Axis theorem
- Early in formulation, the ACS engineer will estimate the s/c moment of inertia based on a general approximate of a few shapes & masses

Figure	Moment(s) of inertia
	$I = Mr^2$
	$I = \frac{m_1 m_2}{m_1 + m_2} x^2 = \mu x^2$
	$I_{center} = \frac{1}{12} mL^2$ [1]
	$I_{end} = \frac{1}{3} mL^2$ [1]
	$I_x = mr^2$ $I_x = I_y = \frac{1}{2} mr^2$
	$I_x = \frac{1}{2} mr^2$ $I_x = I_y = \frac{1}{4} mr^2$
	$I \approx mr^2$ [1]

Parallel Axis Theorem

- If I_{CM} , the moment of inertia of an object about an axis passing it CM, its moment of inertia about another axis parallel to the first one is $I = I_{CM} + Mh^2$

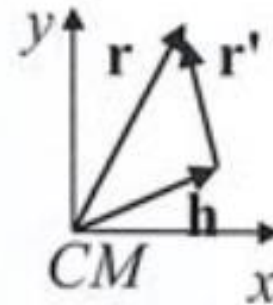
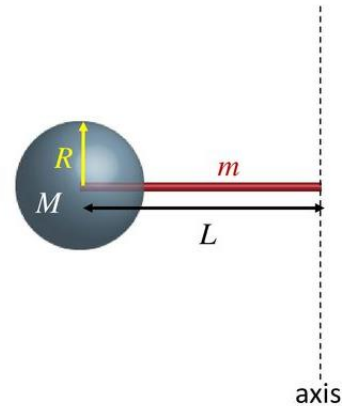
where M is the mass and h the distance between the two parallel axis.

- If we choose CM as the origin, then $h=0$. Suppose the axis is along z ,

$$I_{CM} = \int (x^2 + y^2) dm$$

- If the new axis is at $h=(a,b,z)$, $r'=r-h$

$$\begin{aligned} I &= \int (x'^2 + y'^2) dm = \int [(x-a)^2 + (y-b)^2] dm \\ &= \int [x^2 - 2ax + a^2 + y^2 - 2by + b^2] dm \\ &= \int (x^2 + y^2) dm - 2a \int x dm - 2b \int y dm + (a^2 + b^2) \int dm \\ &= I_{CM} - 2aMx_{CM} - 2bMy_{CM} + h^2 M = I_{CM} + Mh^2 \end{aligned}$$



Comments

- This formula is critical to combining moments of inertia in order to estimate more complex shapes.

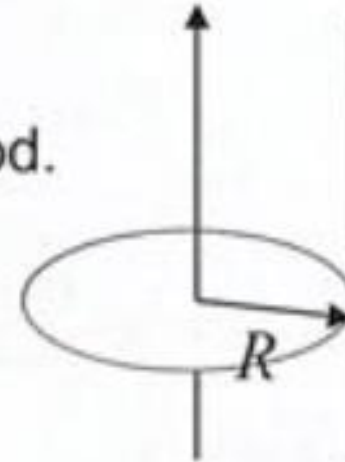
Moment of Inertia of Disk about Edge

Calculate the moment of inertia of a uniform disk about an axis perpendicular and at the edge.

Solution:
$$I_{CM} = \frac{1}{2}mR^2 \Rightarrow I = I_{CM} + mR^2 = \frac{3}{2}mR^2$$

We have obtained the moment of inertia of a uniform disk about an axis perpendicular to it and passing through its center, which is also its CM.

This formula also applies for a uniform rod.



Perpendicular Axis Theorem

- For a thin flat plate in the x-y plane:

$$I_z = I_x + I_y$$

Comments

- This is helpful for “thin” objects that are dominated by inertias about x & y

This can be seen easily by realizing for the plate $z=0$,

$$I_x = \int (y^2 + z^2) dm = \int y^2 dm, I_y = \int (z^2 + x^2) dm = \int x^2 dm$$

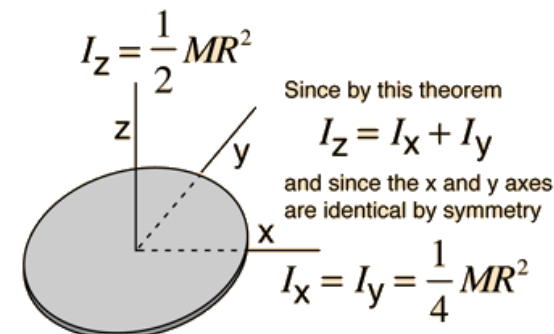
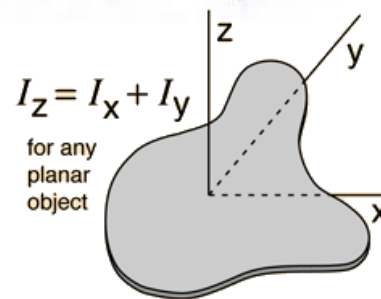
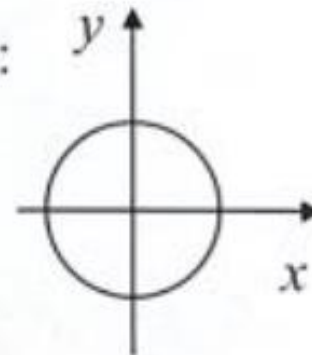
$$\Rightarrow I_x + I_y = \int (y^2 + x^2) dm = I_z$$

Application: Moment of Inertia of a Thin Disk:

For a thin disk we must have $I_x = I_y$.

From the perpendicular axis theorem,

$$I_z = MR^2/2 = I_x + I_y = 2I_x \Rightarrow I_x = MR^2/4$$



From ADCS Handout, Joel Sercel, 2003

Moment of Inertia as a Tensor

- The analogy between mass and moment of inertia is not perfect. Mass is a scalar, and linear momentum and velocity are always parallel to each other.
- A **tensor** is a physical quantity with multi-components. A vector has 3 components, and it's a first order tensor. Moment of inertia in fact has 3x3=9 components, and it's a 2nd order tensor.

$$\vec{I} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix}, \vec{h} = \vec{I} \cdot \vec{\omega} \Rightarrow \begin{pmatrix} h_x \\ h_y \\ h_z \end{pmatrix} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$

Angular rates are about each axis. This equation defines the momentum (h) about each axis

$$I_{xx} = \int (y^2 + z^2) dm, I_{xy} = \int xy dm = I_{yx}, I_{xz} = \int xz dm = I_{zx} \dots$$

- Angular momentum and angular velocity are **not parallel** to each other in general.

Comments

- This is the equivalent of moment = mass x velocity (but wrt rotation)
- Whereas in translational physics, we can just use 'mass', here we need to use a 2nd order tensor to fully define the inertia
- We need each inertial component about the axis (and their products for asymmetric shapes)

Angular Momentum

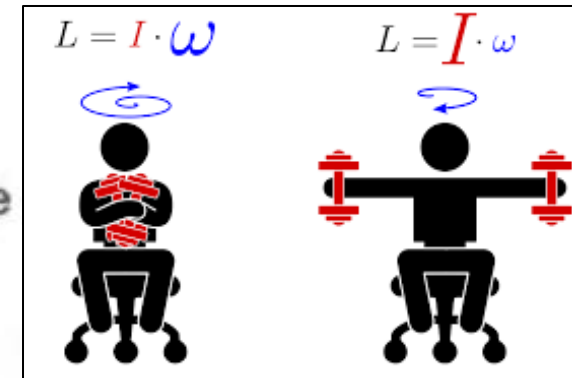
$$[I] = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix} \quad [I] \text{ is real and symmetric}$$

With this definition of the inertia matrix, the **angular momentum** of a rigid body about its center of mass can be written as:

$$\boxed{h = [I] \cdot \omega} \quad (1)$$

h = rigid body angular momentum in center of mass coordinates
 ω = angular velocity

Conservation of Angular Momentum



Comments

- This is the equivalent of our translational $h = mv$
- Previously, we used total impulse ($=Ft$) to solve for thrust
- Similarly, this will allow us to solve for the torques required to change rotational velocity
- Note: momentum often defined as 'h' or 'L'

Moments & Products of Inertia

For a collection of n point masses, define:

Moments of inertia:

$$I_{11} = \sum_{i=1}^n m_i (\rho_{i2}^2 + \rho_{i3}^2)$$

$$I_{22} = \sum_{i=1}^n m_i (\rho_{i1}^2 + \rho_{i3}^2)$$

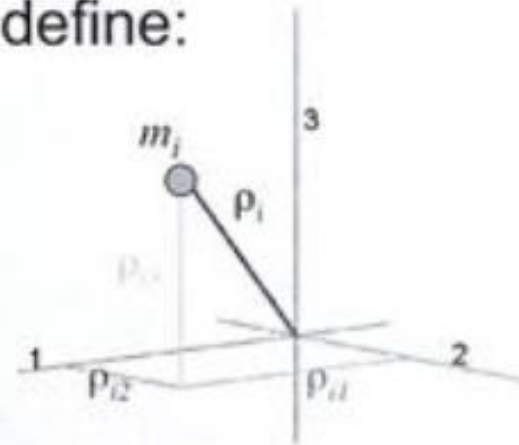
$$I_{33} = \sum_{i=1}^n m_i (\rho_{i1}^2 + \rho_{i2}^2)$$

Products of inertia:

$$I_{12} = I_{21} = - \sum_{i=1}^n m_i \rho_{i2} \rho_{i1}$$

$$I_{13} = I_{31} = - \sum_{i=1}^n m_i \rho_{i1} \rho_{i3}$$

$$I_{23} = I_{32} = - \sum_{i=1}^n m_i \rho_{i2} \rho_{i3}$$



where ρ_{in} are the components of the position vector of mass m_i in the current reference system.

Minus sign in products of inertia from conventional definition of angular momentum vector.

Principal Axes: Coordinate system in which products of inertia are all zero.

(Note: For a continuous mass distribution, the sums become integrals.)

Comments

- Note that moments of inertia (x, y, z) are always positive
- Products of inertia (xy, yz, etc.) are measures of symmetry. If an object is symmetrical across a plane (eg, XY plane), then the products associated with any axis perpendicular are zero (I_{xz} , I_{yz} = 0)

A force F_i at a position ρ_i yields a **torque**: $T = F_i \times \rho_i$

Torque from a collection of forces: $T = \sum_{i=1}^n F_i \times r_i$

Newton's law: $T = \frac{dh}{dt}$ (2)

Note: Equation (2) is only valid for:

- Rigid body
- Around the center of mass or a point fixed in space

For torque-free motion of a rigid body ($T = 0$), equation (2) implies that the angular momentum is constant. From equation (1), since the inertia matrix is constant (rigid body), the angular velocity vector ω is also constant.

Comments

- Note the restrictions above (rigid body, etc.)
- For example, if goal is to maintain constant angular moment, then we need to offset disturbances with counter torques

Euler's Moment Equations

Components of equation (2) $\boxed{\tau = \frac{d\mathbf{h}}{dt}}$:

$$\tau_1 = \dot{h}_1 + \omega_2 h_3 - \omega_3 h_2$$

$$\tau_2 = \dot{h}_2 + \omega_3 h_1 - \omega_1 h_3 \quad (3)$$

$$\tau_3 = \dot{h}_3 + \omega_1 h_2 - \omega_2 h_1$$

The set (3) is known as Euler's moment equations, and describe the general attitude motion of a rigid body. Note that the equations are not linear (ω and $\mathbf{h}$ are related by the inertia matrix).

Comments

- These equations dictate the rotational movement of a rigid body with respect to each axis
- ω is angular velocity and $\mathbf{h} = \mathbf{I} \omega$

Euler's Moment Equations

If we use principal axes, set (3) can be written as:

$$\dot{h}_1 = I_1 \dot{\omega}_1 = \tau_1 + (I_2 - I_3) \omega_2 \omega_3$$

$$\dot{h}_2 = I_2 \dot{\omega}_2 = \tau_2 + (I_3 - I_1) \omega_3 \omega_1 \quad (4)$$

$$\dot{h}_3 = I_3 \dot{\omega}_3 = \tau_3 + (I_1 - I_2) \omega_1 \omega_2$$

Comments

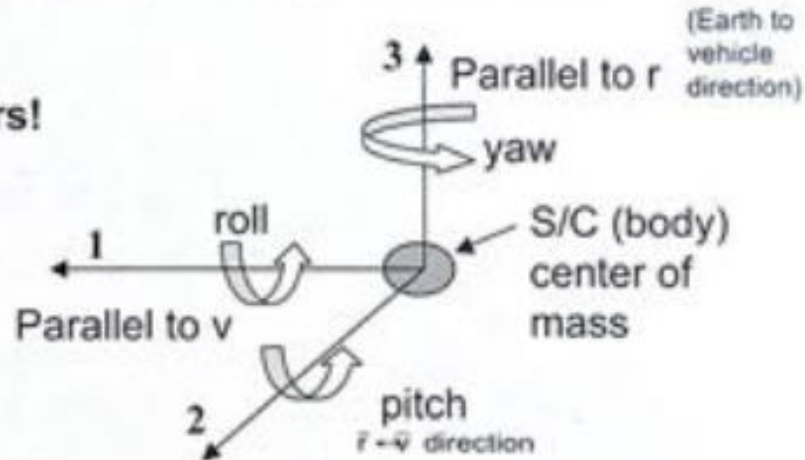
- These equations that relate the change in momentum to the torques acting on a spacecraft can then be used to create an attitude control simulation
- This will be part of HW3c

Euler's Angles

Euler angle rotation: a successive angular rotation about the three orthogonal axes of the body frame. Euler's angles are a commonly used set of angles to define the orientation of a body with respect to a reference frame.

Note: The order of the rotations matters!

- 12 possible choices
- The one in figure is commonly use for orbits around a planet
- For some choices, the Euler rotation matrix⁽¹⁾ might be singular (*locking*)



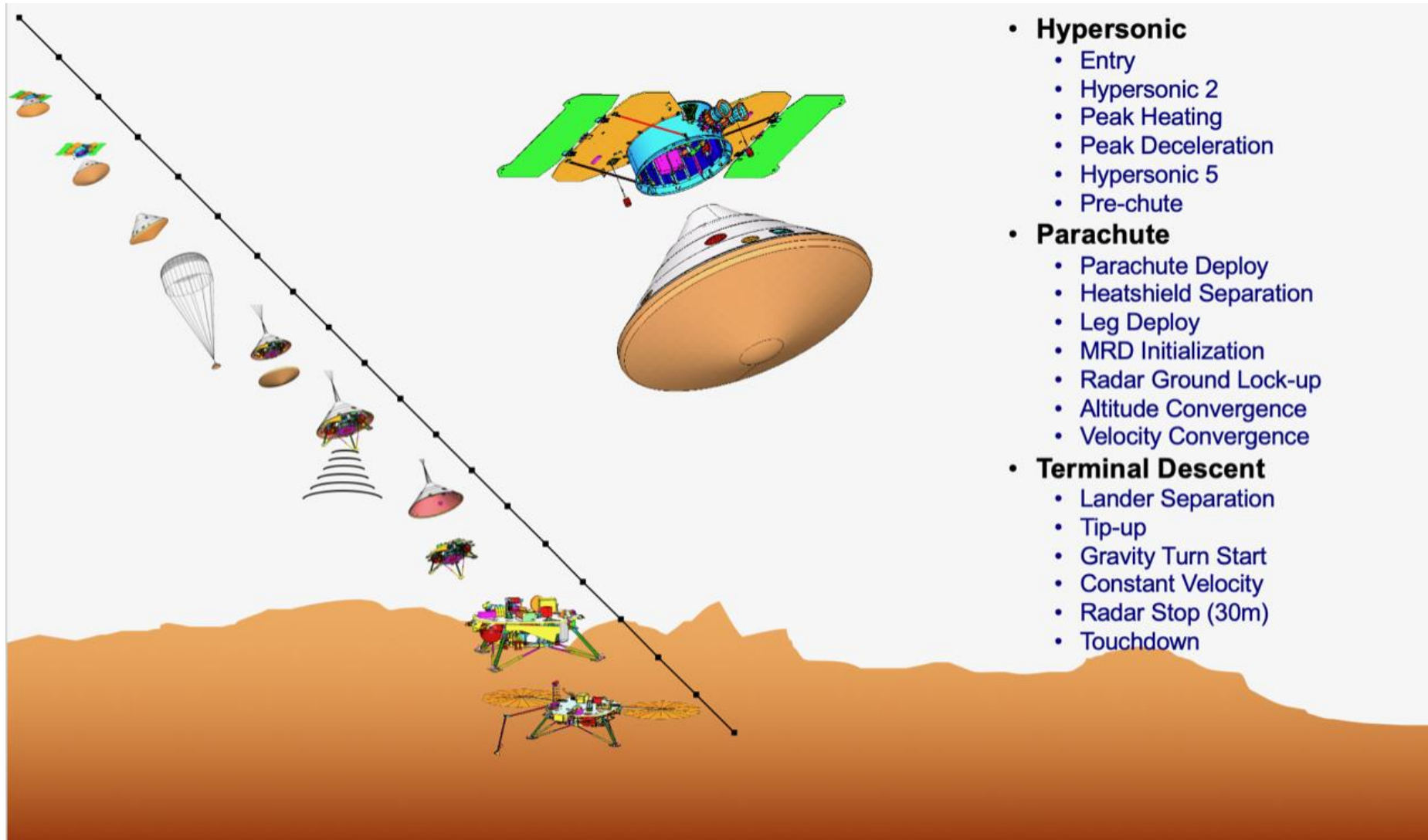
⁽¹⁾The rotation matrix $[A_{\psi\phi\chi}]$ is a linear transformation that carries the reference frame into the body frame:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{\text{Body}} = [A_{\psi\phi\chi}] \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{\text{Reference}}$$

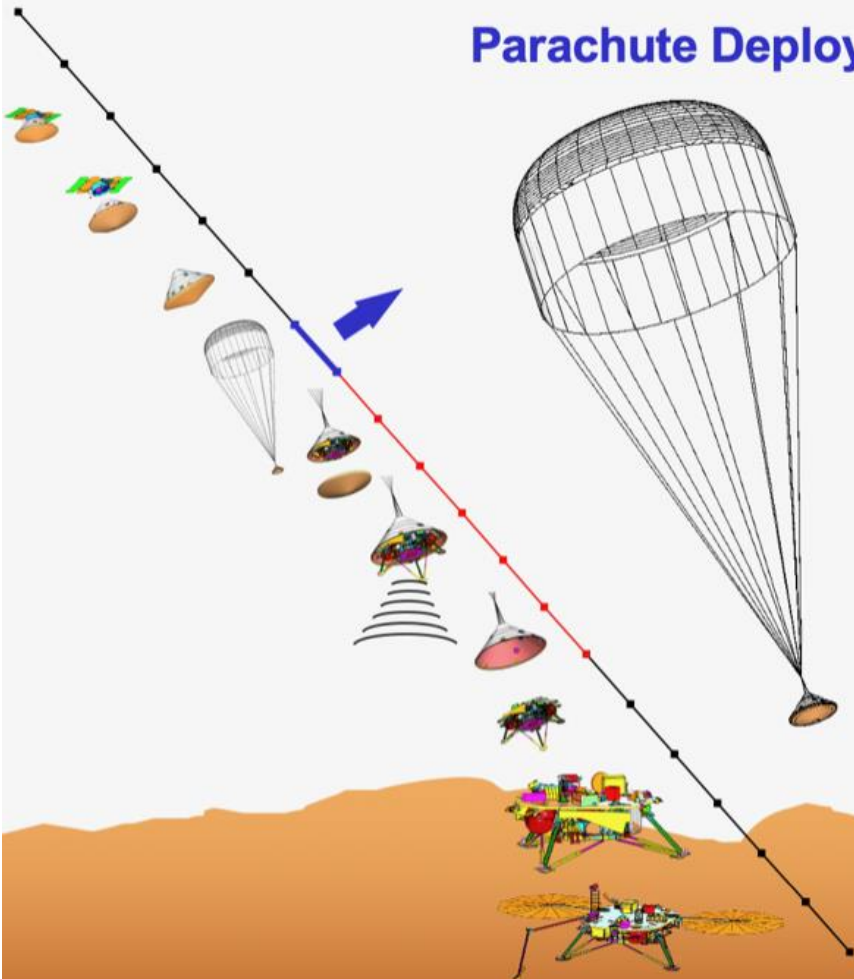
Euler angles:
(pitch, yaw, roll) =
(θ , ϕ , ψ)

Comments

- Which leads us all the way back to roll, pitch, and yaw and introduces rotation matrices.

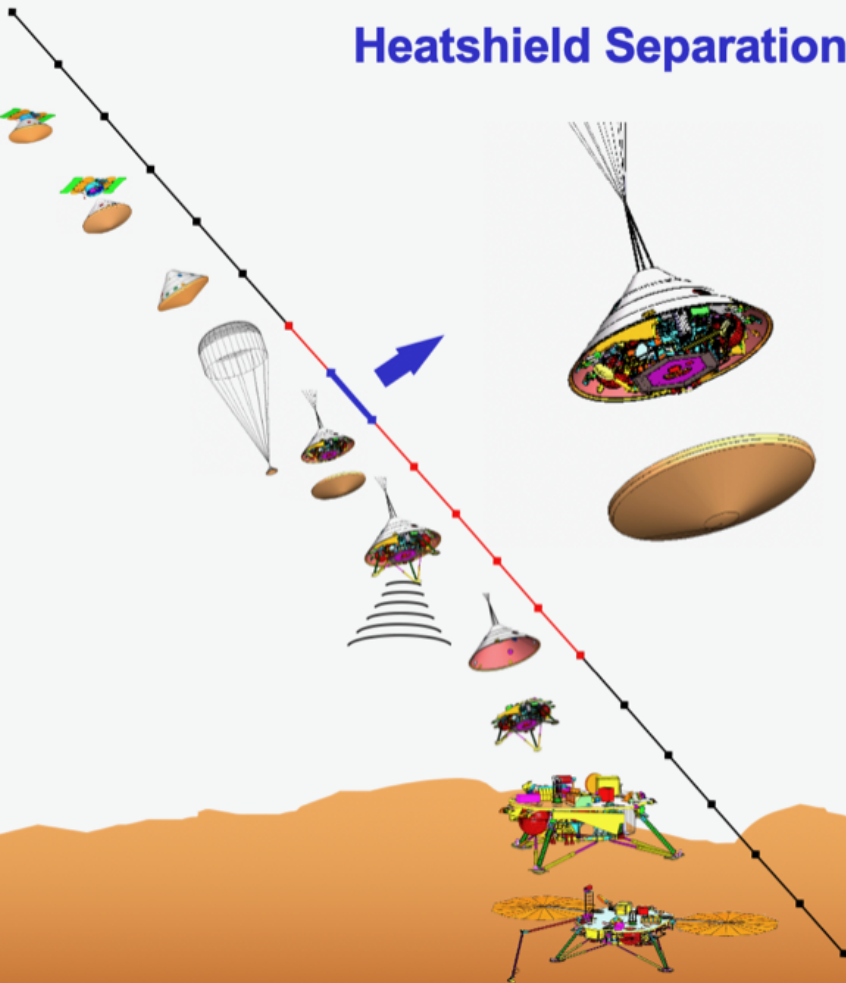


Parachute Deploy



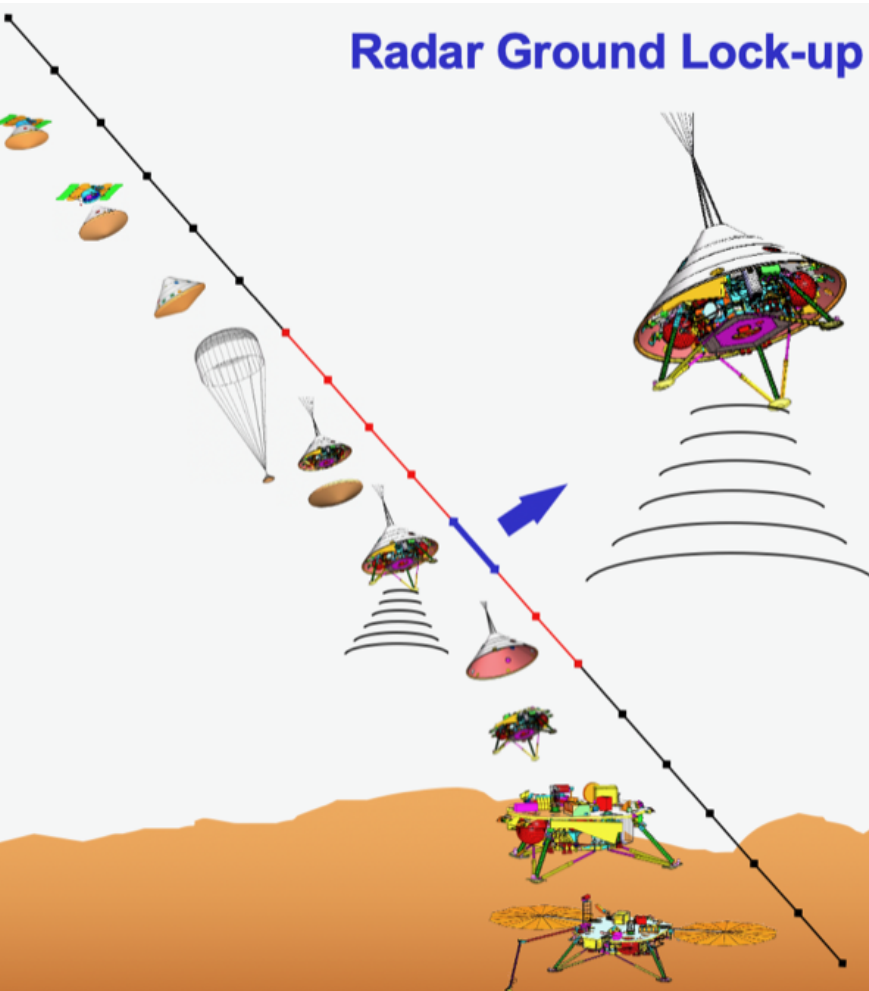
- **Hypersonic**
 - Entry
 - Hypersonic 2
 - Peak Heating
 - Peak Deceleration
 - Hypersonic 5
 - Pre-chute
- **Parachute**
 - Parachute Deploy
 - Heatshield Separation
 - Leg Deploy
 - MRD Initialization
 - Radar Ground Lock-up
 - Altitude Convergence
 - Velocity Convergence
- **Terminal Descent**
 - Lander Separation
 - Tip-up
 - Gravity Turn Start
 - Constant Velocity
 - Radar Stop (30m)
 - Touchdown

Heatshield Separation



- **Hypersonic**
 - Entry
 - Hypersonic 2
 - Peak Heating
 - Peak Deceleration
 - Hypersonic 5
 - Pre-chute
- **Parachute**
 - Parachute Deploy
 - Heatshield Separation
 - Leg Deploy
 - MRD Initialization
 - Radar Ground Lock-up
 - Altitude Convergence
 - Velocity Convergence
- **Terminal Descent**
 - Lander Separation
 - Tip-up
 - Gravity Turn Start
 - Constant Velocity
 - Radar Stop (30m)
 - Touchdown

Radar Ground Lock-up



- **Hypersonic**
 - Entry
 - Hypersonic 2
 - Peak Heating
 - Peak Deceleration
 - Hypersonic 5
 - Pre-chute
- **Parachute**
 - Parachute Deploy
 - Heatshield Separation
 - Leg Deploy
 - MRD Initialization
 - Radar Ground Lock-up
 - Altitude Convergence
 - Velocity Convergence
- **Terminal Descent**
 - Lander Separation
 - Tip-up
 - Gravity Turn Start
 - Constant Velocity
 - Radar Stop (30m)
 - Touchdown

How do you create simulations?

- **Phoenix EDL Major Sim Components (code, analysis, HW testing, etc.)**

- Simulation integrator
- Atmospheric model
- Hypersonic aerodynamics model
- Gravity model
- Dispersed hypersonic aerodynamics model
- Parachute drag model
- Parachute inflation model
- Wrist mode model (x2 types)
- Configuration aerodynamics models (x 3 configs)
- Winds model (x2 types)
- RCS propulsion model
- IMU model
- Radar model (low fidelity)
- Descent engine propulsion model
- Phoenix flight software
- ADAMS model (mechanical simulation)

Resulting Higher-Level Models

- 6-DOF Langley simulation
- 6-DOF Lockheed Martin sim
- 6-DOF Mechanical simulation
- Flowtran Lander simulation
- Flight software simulation

Validated by Tests & Peer Reviews

- Extensive literature & past missions (eg, Mars atmosphere, winds, terrain, etc.)
- Wind tunnel testing (eg, parachute)
- [Radar field testing \(added for PHX\)](#)
- Thrusters (live-fire vacuum testing)



Radar Assumptions

- **Originally, the Mars '01 (not flown) and Phoenix project had a simplified model for the radar which made the following assumptions:**
 - Infinite, sloped, flat plain
 - Noise/bias characteristics that matched the requirements
 - Very sharp, single-peak FFT velocity measurement outputs
 - Assumed perfect angle-of-arrival (radar beam canted to be an offset from this)
 - The radar will only have one target (ground only)
- **Those assumptions proved to be inadequate via field test data and development of a higher-fidelity simulation in the following ways:**
 - Terrain relief and brightness variation causes the look-vector to migrate
 - Noise/bias characteristics change with varying terrain and target brightness
 - FFT signatures are broader and show multiple peaks from nadir contamination, and wider FFT spectra peaks
 - The angle of arrival is actually biased towards nadir from the original assumed angle
 - The radar is guaranteed to see two targets (ground + heatshield)

The model also missed...

- **Several Key Items...**

- Multiple targets can confuse the search logic in the radar firmware to cause locks on false targets and ambiguities
- Never-before-discovered bugs in the radar firmware which can cause the radar to have prolonged lockup times under certain thermal & timing conditions
- Bi-modal lock-up behavior under certain conditions

- **Result**

- Significant time required just prior to launch and during cruise to build, test, and incorporate a high-fidelity radar model into the Phoenix EDL simulation
- This fixed several anomalies that had the potential to fail the mission. But, it also uncovered new anomalies....

How do you create simulations?

- **Phoenix EDL Major Sim Components (code, analysis, HW testing, etc.)**

- Simulation integrator
- Atmospheric model
- Hypersonic aerodynamics model
- Gravity model
- Dispersed hypersonic aerodynamics model
- Parachute drag model
- Parachute inflation model
- Wrist mode model (x2 types)
- Configuration aerodynamics models (x 3 configs)
- Winds model (x2 types)
- RCS propulsion model
- IMU model
- Radar model (low & added **high fidelity**)
- Descent engine propulsion model
- Phoenix flight software
- ADAMS model (mechanical simulation)

Resulting Higher-Level Models

- 6-DOF Langley simulation
- 6-DOF Lockheed Martin sim
- 6-DOF Mechanical simulation
- Flowtran Lander simulation
- Flight software simulation

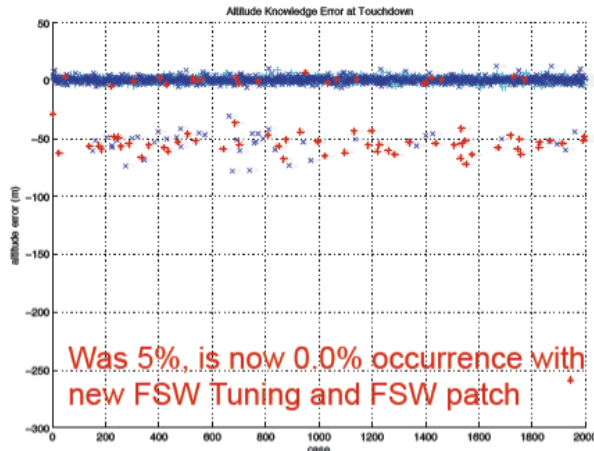
Validated by Tests & Peer Reviews

- Extensive literature & past missions (eg, Mars atmosphere, winds, terrain, etc.)
- Wind tunnel testing (eg, parachute)
- **Radar field testing**
- Thrusters (live-fire vacuum testing)

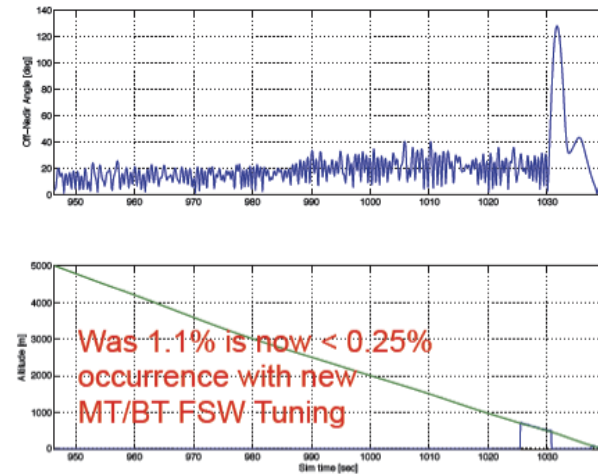


Radar Data from Simulation

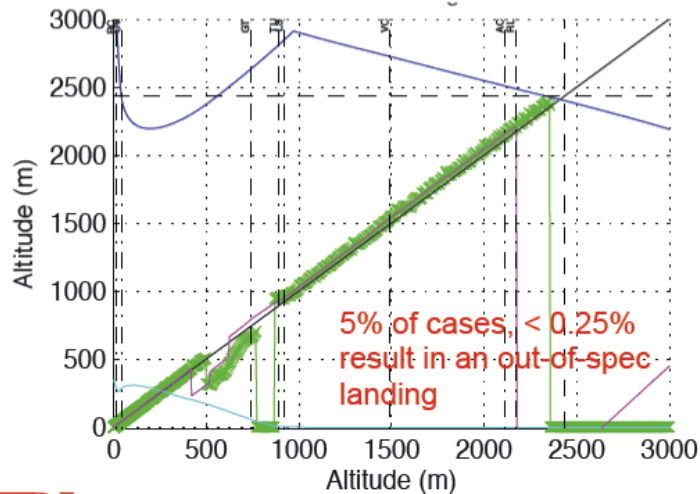
Nav State Jump at Low Altitude due to Radar Error



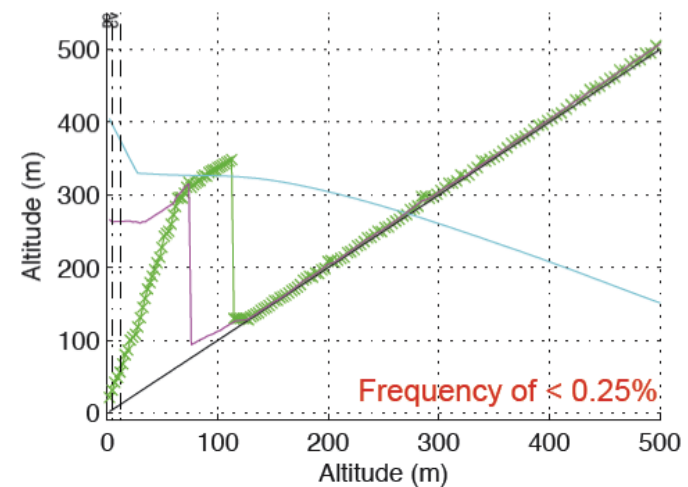
Late- or Non-Radar-Lockup (wrist-mode-related)



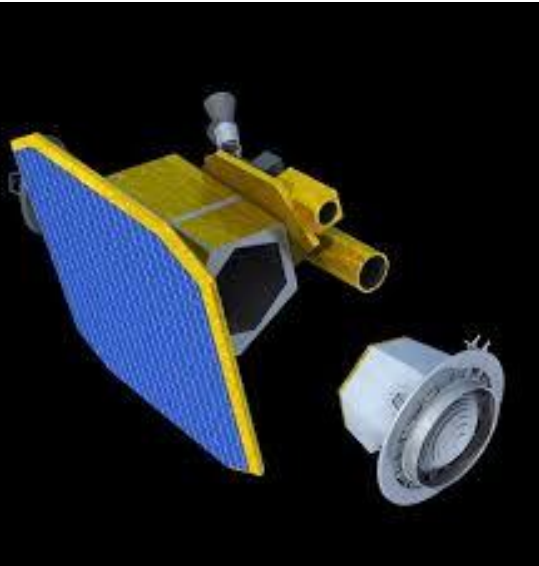
Barker Code Side Lobe (most likely sim artifact)



Radar Slant-Range Lockup (non-sep-related)

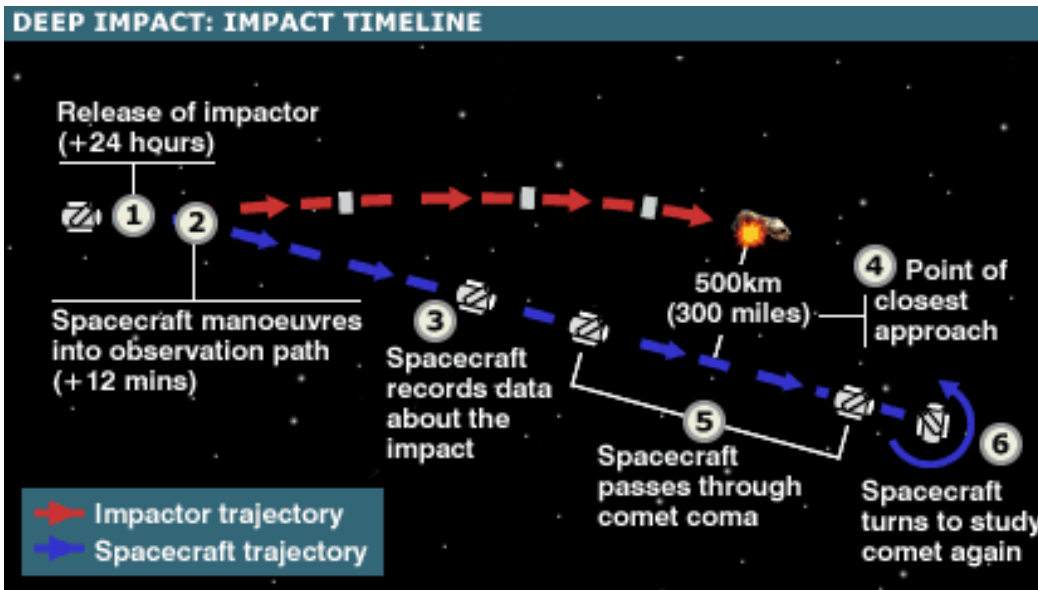


Deep Impact Star Tracker Anomaly

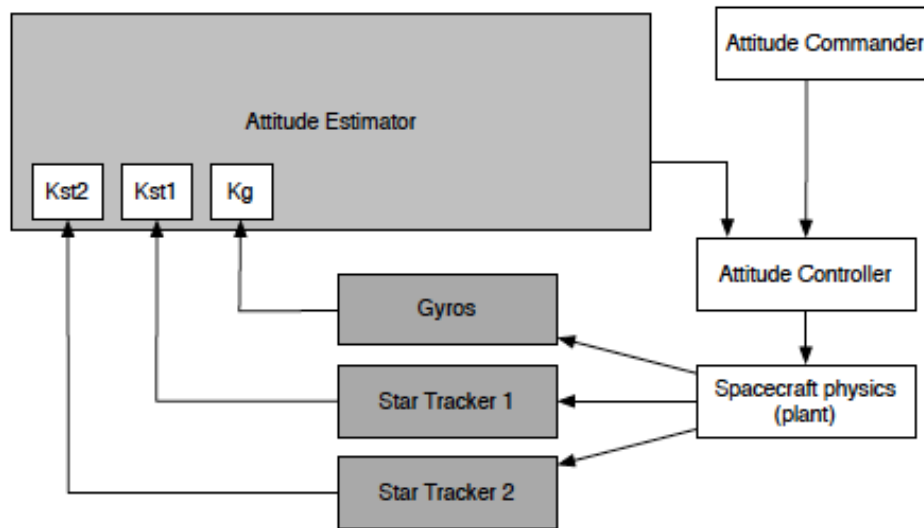


• Deep Impact

- NASA spacecraft designed to study the interior composition of a comet by releasing an impactor into the comet
- Launched in 2005, it resulted in a successful comet impact
- However, it almost never made it...



Deep Impact Attitude Estimation



• Attitude Estimation

- Combines weighted measurement from multiple sensors into a single attitude estimate
- Typically uses include gyro(s) for rate data and star tracker(s) for celestial attitude data
- Able to propagate for limited times in the absence of star tracker data (given an initial reference)
- Weights chosen to match strengths of sensor
 - Low weights indicate that the sensor cannot abruptly change the estimate as fast as other sensors
 - Conversely, high weights allow a sensor to make sudden, large updates

• Star Tracker (Primary & Redundant)

- Matches star images to a catalog to produce a full 3-axis attitude with respect to a standard CS reference (eg, J2000)
- Can quickly determine attitude without any initial knowledge
- Can't sustain lock under fast rotations
- Requires clear view of the sky (can be dazzled by bright objects near by or in FOV)
- Needs adequate star density

• Gyros

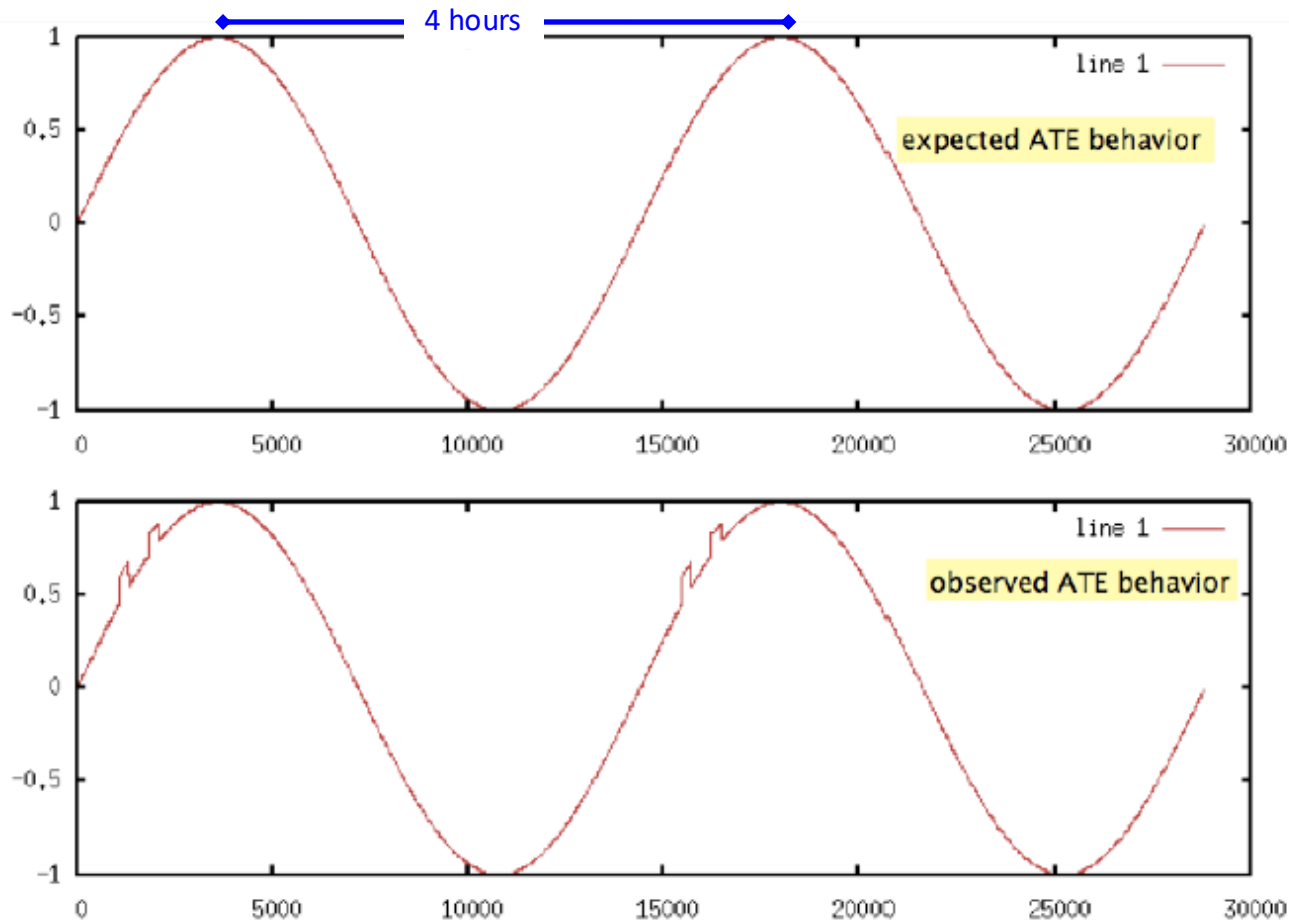
- Provide angular rates, which can be integrated over time to obtain angular position relative to a given reference
- Generally operate in “relative mode”, relative to an assumed reference and propagating changes in attitude without having absolute knowledge
- Able function under relatively “fast” rotations whereas other sensors may not
- Low signal-to-noise ratio (ie, less accurate)
- Generally, there is bias “or drift” rate over long periods of integration without an absolute reference

- **After launch, Deep Impact enters a planned “safe-mode.”**
 - In safe mode, it uses gyros and sun sensors, holding +Y axis on the sun and rotating once every 4 hours (“sunline” ACS state)
 - Star trackers are mounted on the backside, facing somewhat anti-sun. They are not used at the beginning of the mission
- **Attitude Estimator (ATE) is integrating gyro data from a “wake-up” frame. This frame is defined as the s/c frame at the instant that the s/c is commanded to sunline mode**
- **After post-launch orbit check-out and initial spacecraft health assessment, the star trackers are turned on by ground command**
- **Nominally, ATE will accept data from the star trackers and will slowly correct its estimate to reflect the data coming from the STs.**
 - The ATE mixes in gyro data with ST data with much greater weight placed on the gyro data. Over time, the STs will ‘win’ but in the short time the current ATE estimate wins. This is considered a “slow drip” correction.
- **The nominal plan is that after the ATE has converged, the ground will command the spacecraft to go from safe-mode to 3-axis control**



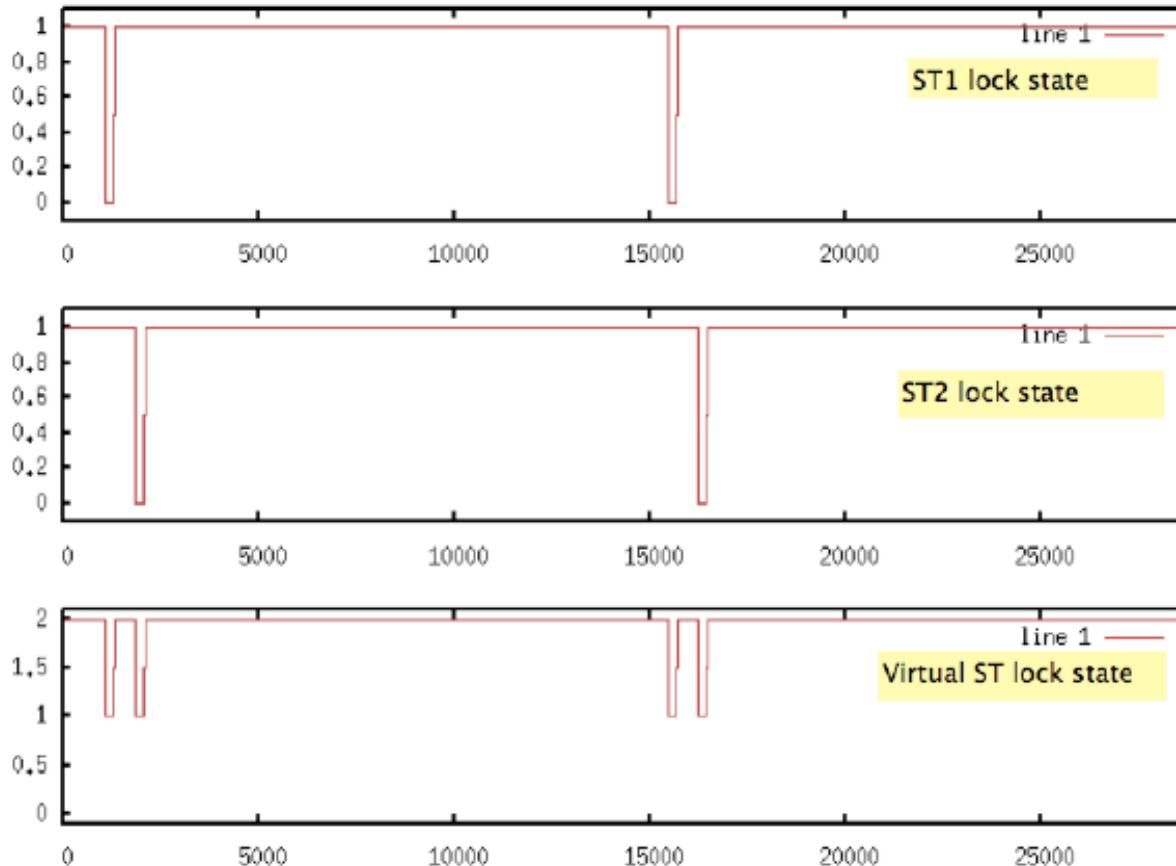
Attitude Estimation

- Twice every 4 hours, the ATE estimate jumps up & down discontinuously. The jumps are 11-12 min apart
 - Note that 1 full rotation = 4 hours



Star Tracker “Lock” State

- Coincident with the jumps are star tracker outages...
- Interval between jumps corresponds to 19-deg (relative alignment of star trackers), suggesting a systemic issue...
- Star tracker data alone suggests the sun is significantly (~90-deg) off from expectations. Given sun sensor & solar array data, this suggests possible ST alignment issue
- Data suggests this issue is occurring when star trackers are pointed towards celestial true north



Possible Issues

- Star tracker alignment
- ACS/ATE software
- Earth in FOV
- “Dark spot” in sky

Documentation Discovery...

- Going through documentation, an error in the coordinate frame transformation was discovered...
 - A 90-deg rotation to translate from mechanical from to spacecraft frame was dropped...
 - All three star trackers (including impactor) had incorrect alignments

Subject Title	CAGE Code	Project Title
Attitude Determination and Control Design Description Document, Revision B		DEEP IMPACT

$$q_{J2000_2_tracker_mech} = q_{CT633_OUT} \otimes \begin{bmatrix} 0 \\ 0 \\ -1 \\ \frac{\sqrt{2}}{2} \\ 1 \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

Where q_{CT633_OUT} is the quaternion in the CT - 633 output data stream.
(See S99.41830.S/W.008B, "Software Command and Data Interface for CT - 633 Star Tracker," Section 2.6, page 14.)

Red term is correction that rotates ST output from the ST mechanical frame to spacecraft frame. This term was omitted in the alignment parameter load to the FSW.

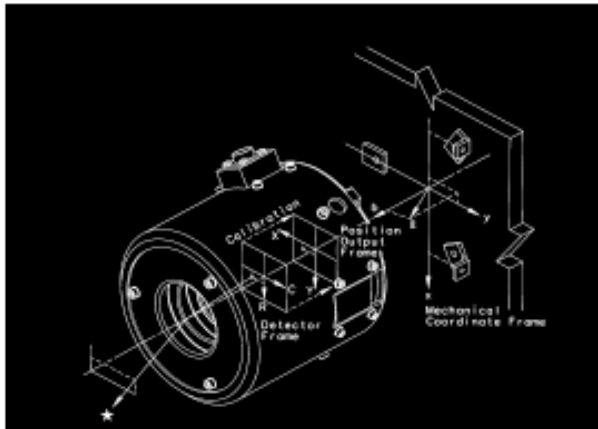


Figure 6-1. The Ball CT-633 star tracker provides stellar attitude estimates relating the J2000 Earth Centered Inertial and tracker mechanical coordinate frames. Features on the mounting interface define the mechanical frame. Note, the DI light shade is different than shown here.

Spacecraft Body to Tracker Mechanical Transformation	Notes
$T_{scTracker} = \begin{bmatrix} 0.906308 & 0.000000 & 0.422618 \\ -0.130596 & -0.951057 & 0.280065 \\ 0.401934 & -0.309017 & -0.861950 \end{bmatrix}$ $V_{tracker} = T_{scTracker} \cdot V_{sc}$	Closest to SIM ~18 degrees up from inst platform, 20 degrees off HRI anti-normal, toward SIM
$T_{scTracker} = \begin{bmatrix} 0.707107 & 0.000000 & 0.707107 \\ -0.218508 & -0.951057 & 0.218508 \\ 0.672499 & -0.309017 & -0.672499 \end{bmatrix}$	Farthest from SIM. ~18 degrees up from inst platform, zero degrees off HRI anti-normal.

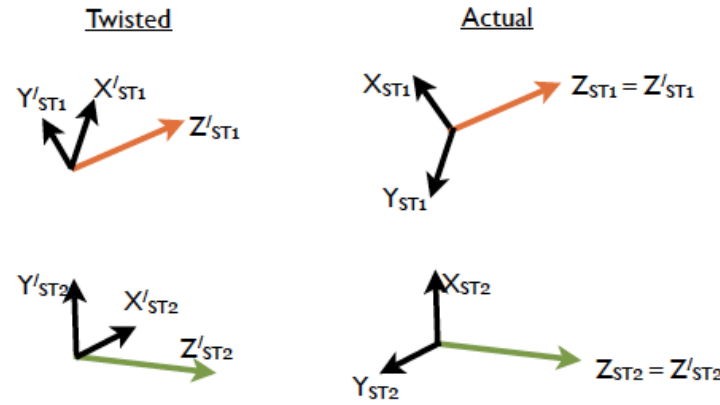
Table 6-2. Nominal Impactor Star Tracker Orientation

Impactor Body to Tracker Mechanical Transformation	Notes
$T_{scTracker} = \begin{bmatrix} 0.905984 & -0.165391 & 0.389666 \\ -0.423313 & -0.353974 & 0.833972 \\ 0.0 & -0.920515 & -0.390706 \end{bmatrix}$ $V_{tracker} = T_{scTracker} \cdot V_{sc}$	Boresight is in the Impactor Y-Z plane, 23 degrees off -Y toward -Z. Tracker X is rolled off Impactor body X by 25.04 degrees in a right-handed sense about tracker

These were the alignments loaded to the s/c. Note the text indicates "Mechanical Frame" not "spacecraft frame." The correction term on page 20 (shown above) was dropped when the alignment parameters were loaded for launch.

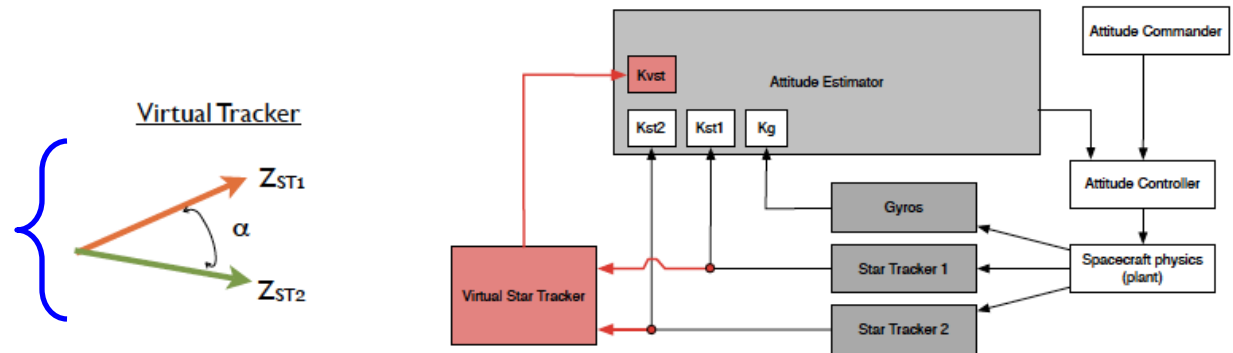
B. Buchholtz, 04-002.)

Virtual Star Tracker

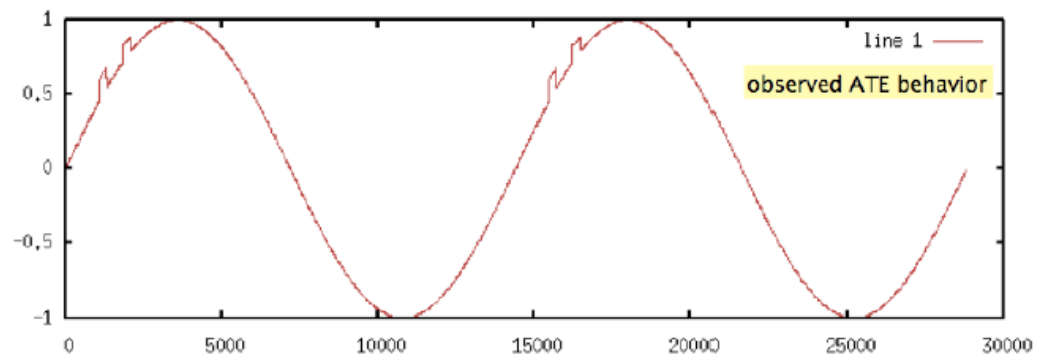


Virtual Star Tracker

When both star trackers on and in-lock, the attitude algorithm only uses the boresight vectors of the actual trackers (which are correct)



However, when only 1 star tracker is in-lock, the whole quaternion from the star tracker is used for the attitude, which includes the 90-deg error



Another Discovery...

- Star Catalogs contain large numbers of stars, indexed, and sorted by declination
- While these star trackers had significant heritage, additional stars were added to the catalog to ensure a high density given the mission in deep space
- However, when these stars were added, there was a corresponding parameter that set the length of the list – and it wasn't increased
- Therefore, the list was cut-off at declinations of greater than 80-deg.
- Thus, the failure to lock attitude near the celestial North Pole

Star Catalog (representation)

- **Keep It Simple**
 - “Virtual Star Tracker” seemed like a good idea, but it obscured the problem...
 - Fortunately, safe mode wasn't relying on ATE for attitude estimate
 - This allowed the anomaly to be de-bugged while in sunline mode (via coarse sun sensors)
- **Phasing Tests Are Critical...**
 - A star tracker phasing test “should” have identified the mis-alignment.
 - Unclear whether it occurred or properly verified
- **Explain All the Observables**
 - Multiple errors occasionally occur and require significant exploration even after some/most of the issues have been explained
 - This is the benefit of using a fishbone diagram and driving to closure
- **Beware of Heritage**
 - Initially, the team was very skeptical that there was a star tracker issue given the significant heritage. Finally, after significant convincing, the team investigated and discovered the incorrect parameter

Note that the alignment error was found on all three star trackers, including the one on the impactor. If it had been limited to the impactor, the issue may not have been caught until there was insufficient time to fix prior to the comet flyby.